Electronic Companion to 'Close to the cap: who plays the whole match across nine sports'

Contents

EC1 Key terms . . . 2
EC2 The rulebooks . . . 2
EC3 Panels and certification . . . 9
EC4 The exposure identity . . . 11
EC5 The identity in full . . . 12
EC5.1 The proof . . . 12
EC5.2 The measurement . . . 13
EC5.3 Football's gap, counted . . . 17
EC5.4 Brazil . . . 19
EC5.5 Rugby union . . . 20
EC5.6 Baseball, where substitutes replace substitutes . . . 21
EC5.7 Football's own exception . . . 22
EC5.8 Month by month . . . 22
EC6 Who plays the whole match . . . 23
EC7 Who plays the whole match, by position . . . 26
EC8 A cap on one player: cricket's bowling quota . . . 29
EC9 The other elements of the decision . . . 30
EC10 The rotational recount . . . 31
EC11 The count in full . . . 31
EC12 The window in full . . . 42
EC12.1 The form, and what it rules out . . . 43
EC12.2 The cells . . . 44
EC12.3 The lines, with the covariance . . . 45
EC12.4 The spread . . . 48
EC12.5 Bundling . . . 50
EC12.6 The first change: injury and position . . . 53
EC12.7 The 90th minute . . . 54
EC13 The model of stoppages . . . 55

EC13.1 Against the fitted seasons' own tables 59
EC13.2 The count of changes and the stoppage minutes . . . 60
EC13.3 How evenly a team spaces its stoppages 62
EC13.4 The score 64
EC13.5 A test on matches not yet read 67
EC14 Register of pre-specified predictions 69

EC1 Key terms

Every term of the main text is defined where it is first used, and Table EC1 gathers them in the order the main text meets them.

EC2 The rulebooks

A panel is the set of records read for one analysis. Table EC2 lists every rule the panels are read under, with the seasons it covers and its reference. The rules are read edition by edition, and consecutive seasons with the same value share one row. Where no edition was read, the source column names the release that announced the rule. Football's limit in each league-season is the competition's choice between three and five, read from its own lineups. The move to five was meant to protect players in a crowded calendar, in which muscle injuries rise as matches come closer together (Bengtsson et al., 2013). The Laws of 2026–27 let senior national teams use eight substitutes in friendly matches, and more by agreement (The International Football Association Board, 2026*b*). The same Laws give a substituted player 10 seconds to leave the field. Below the elite game, football's Laws also allow return substitutes (The International Football Association Board, 2017). A season is labelled by the year it starts, and a calendar-year season, such as Super Rugby's, by its year.

Table EC2: The rules by sport, season and reference or source. Consecutive seasons of one value share a row; a league in brackets has a rule of its own.

Sport	Rule	Value	Seasons	Reference or source
Association football	players on the field	11	2000–	Law 3.1
Association football	substitution limit	3	2000–2019	match sheets
Association football	substitution limit	5	2020–	match sheets
Association football	stoppages for changes in play	unlimited	2000–2019	Law 3.2
Association football	stoppages for changes in play	3	2020–2021	IFAB release, 2020

Table EC2, continued

Sport	Rule	Value	Seasons	Reference or source
Association football	stoppages for changes in play	3	2022–	Law 3.2
Association football	added in extra time	0	2000–2017	Law 3.2; Laws, Introduction
Association football	added in extra time	1	2018–	Law 3.2
Association football	a replaced player may return	no	2000–	Law 3.2; Law 3.3
Association football	temporary replacements	none	2000–	Law 3.3
Association football	concussion replacement	none	2000–2019	Law 3.2
Association football	concussion replacement	trial, at the competition's option	2020–2023	Laws, Introduction
Association football	concussion replacement	permanent, at the competition's option	2024–	IFAB release, 2024
Association football	a player sent off replaced	before the start only	2000–	Law 3.6
Association football (GB1)	concussion protocol	trial: up to two a team, and as many for the opponent	2020–2023	Premier League release, 2021
Association football	concussion protocol	permanent: one a team, and one more for the opponent	2024–	IFAB laws, 2026
Rugby union	players on the field	15	2000–	Law 3.1
Rugby union	substitution limit	conditional	2000–2013	Law 3.4; Law 3.15
Rugby union	substitution limit	8	2014–	Law 3.4(a); Law 3.4(b)
Rugby union	a replaced player may return	limited	2000–	Law 3.7; Law 3.12
Rugby union	temporary replacements	blood	2000–2013	Law 3.10
Rugby union	temporary replacements	blood and head injury	2014–	Law 3.10; Law Amendment Trial (Temporary replacement - head injury assessment)
Rugby union	concussion replacement	none	2000–2013	Law 3.10
Rugby union	concussion replacement	temporary, for a head injury assessment	2014–	Law 3.10 Law Amendment Trial (Temporary replacement - head injury assessment)
Rugby union	a player sent off replaced	no	2000–2024	Law 3.6; Law 10.7
Rugby union	a player sent off replaced	after 20 minutes	2025–	Law 9.30 (elite rugby)

Table EC2, continued

Sport	Rule	Value	Seasons	Reference or source
Rugby union (Pro D2)	a replaced player may return	up to four, for an injured team-mate	2018–2025	LNR release, 2018
Rugby union (Top 14)	a replaced player may return	up to four, for an injured team-mate	2018–2025	LNR release, 2018
Rugby union (Pro D2)	concussion replacement	head injury assessment and blue card	2018–2025	LNR release, 2018
Rugby union (Top 14)	concussion replacement	head injury assessment and blue card	2018–2025	LNR release, 2018
Gaelic football	players on the field	15	2000–	Rule 2.1, Rules of Specification
Gaelic football	substitution limit	5	2000–2013	Rule 2.4(i), Rules of Specification
Gaelic football	substitution limit	6	2014–2019	Rule 2.4(i), Rules of Specification
Gaelic football	substitution limit	5	2020–2025	Rule 2.4(i), Rules of Specification
Gaelic football	added in extra time	3	2000–2025	Rule 2.4(ii), Rules of Specification
Gaelic football	a replaced player may return	limited	2000–	Rule 1.5(i) Blood Rule, Rules of Control
Gaelic football	temporary replacements	blood	2000–2020	Rule 1.5(i) Blood Rule, Rules of Control
Gaelic football	temporary replacements	blood and head injury	2021–	Rule 1.5(b)(ii), Rules of Control
Gaelic football	a player sent off replaced	no	2000–2013	Rule 2.4(i), Rules of Specification
Gaelic football	a player sent off replaced	black card only	2014–2019	Rule 2.4(iv) Exception (ii), Rules of Specification
Gaelic football	a player sent off replaced	no	2020–	Rule 2.4(iv), Rules of Specification
Hurling	players on the field	15	2000–	Rule 2.1, Rules of Specification
Hurling	substitution limit	5	2000–2025	Rule 2.4(i), Rules of Specification
Hurling	added in extra time	3	2000–2025	Rule 2.4(ii), Rules of Specification
Hurling	a replaced player may return	limited	2000–	Rule 1.5(i) Blood Rule, Rules of Control
Hurling	temporary replacements	blood	2000–2020	Rule 1.5(i) Blood Rule, Rules of Control
Hurling	temporary replacements	blood and head injury	2021–	Rule 1.5(b)(ii), Rules of Control
Hurling	a player sent off replaced	no	2000–	Rule 2.4(i), Rules of Specification
Baseball	players on the field	9	2000–	Rule 1.01 (2011)
Baseball	substitution limit	unlimited	2000–	Rule 3.03 (2011)
Baseball	a replaced player may return	no	2000–	Rule 3.03 (2011)
Baseball	a player sent off replaced	yes	2000–	Rule 9.01(d); Rule 3.03 (2011)

Table EC2, continued

Sport	Rule	Value	Seasons	Reference or source
Baseball	pitchers on the roster, at most	13	2022–	MLR 2(b)(2)(A) (2021)
Baseball	batters a pitcher must face	1	2000–2019	Rule 3.05(b) (2011)
Baseball	batters a pitcher must face	3	2020–	Rule 5.10(g) (2020)
Baseball	designated hitter	each league's option	2000–2021	Rule 6.10 (2011)
Baseball	designated hitter	universal	2022–	Rule 5.11(a) (2024)
Rugby league	players on the field	13	2000–	Section 4.1
Rugby league	substitution limit	10	2008–2015	NRL release, 2014 to NRL release, 2015
Rugby league	substitution limit	8	2016–2025	NRL release, 2015 to NRL release, 2025
Rugby league	a replaced player may return	yes	2000–	Notes on Laws 2013, 1 Interchange (Foul Play); Section 4.2(a)
Rugby league	temporary replacements	none	2000–	Section 4.2(b)
Rugby league	concussion replacement	free, for a head injury assessment	2014–	NRL release, 2014
Rugby league	an eighteenth player	yes	2021–	NRL release, 2021
Rugby league	a player sent off replaced	no	2000–	Section 16, Notes 6; Section 4, Notes 1
Australian rules	players on the field	18	1899–	AFL page, 1899 entry
Australian rules	substitution limit	unlimited	2000–2013	AFL release, 2013
Australian rules	substitution limit	120	2014–2015	AFL release, 2014
Australian rules	substitution limit	90	2016–2020	AFL release, 2016
Australian rules	substitution limit	75	2021–	AFL release, 2021
Australian rules	players on the bench	4	2011–2015	AFL page, 2011 entry
Australian rules	players on the bench	4	2016–2020	AFL release, 2016
Australian rules	players on the bench	5	2021–2025	AFL release, 2021
Australian rules	a replaced player may return	yes	1978–	AFL page, 1978 entry
Australian rules	temporary replacements	blood and head injury	2014–	Regulation 11.11(c)(iii)(A)-(B)
Australian rules	concussion replacement	none	2000–2020	AFL release, 2020
Australian rules	concussion replacement	medical substitute	2021–2022	AFL page, 2021 entry
Australian rules	concussion replacement	tactical substitute	2023–2025	AFL release, 2023
Australian rules	changes at the breaks counted	no	2014–	Reg. 12.9(c)(ii)(C)
Australian rules	a player sent off replaced	no player is sent off	2000–	Law 20.1
Ice hockey	players on the field	6	2000–	Rule 400

Table EC2, continued

Sport	Rule	Value	Seasons	Reference or source
Ice hockey	substitution limit	unlimited	2000–	match sheets
Ice hockey	players a team may dress	20	2000–	Rule 5.1
Ice hockey	a replaced player may return	yes	2000–	Rule 74.1
Ice hockey	a player sent off replaced	yes	2000–	Rule 23.1; Rule 21.2
Basketball (EURO)	players on the field	5	2000–	Art. 4.2.2
Basketball (EURO)	substitution limit	unlimited	2000–	Art. 19.2.1
Basketball (EURO)	players a team may dress	12	2000–	Art. 4.2.1
Basketball (EURO)	a replaced player may return	yes	2000–	Art. 19.2.4
Basketball (EURO)	temporary replacements	treatment	2000–	Art. 5.3
Basketball (EURO)	a player sent off replaced	yes	2000–	Art. 19.3.6
Rugby union (Top 14)	players on the bench	8	2008–	match sheets
Rugby union (Pro D2)	players on the bench	8	2008–	match sheets
Rugby union (Premiership)	players on the bench	8	2009–	match sheets
Rugby union (United Rugby Championship)	players on the bench	8	2010–	match sheets
Rugby union (Super Rugby)	players on the bench	8	2014–	match sheets

Rugby union. Law 3 carries two front-row exceptions that no lineup records (World Rugby, 2023). When a front-row player is suspended or sent off and no one on the field can take the place, another player leaves so that a front-row replacement can come on (Laws 3.19 and 3.20). A player replaced for tactical reasons may return to replace an injured front-row player (Law 3.33). Both move a player outside the plain count, so they sit in rugby union's gap and cannot be separated from it. So do temporary replacements for bleeding, which referees' reports once counted at 34 per 100 scheduled matches of Scottish club rugby (Sharp et al., 2001), and for head injury assessments, during which a possible concussion is assessed off the field (Fuller et al., 2015). The bench of eight came to each league in its own season, read from its lineups. It came in 2008 to the Top 14 and the Pro D2, France's first two divisions, and in 2009 to the Premiership, England's top league. It came in 2010 to the United Rugby Championship, the league of clubs from Ireland, Italy, Scotland, South Africa and Wales, and in 2014 to Super Rugby, the league of the southern hemisphere. The Laws made

Table EC1: Key terms, in the order the main text meets them.

Term	Meaning
Substitution limit	The number of players, or of interchange events, a rulebook lets a team bring on in a match; the cap for short.
One-way sport	A sport in which a replaced player cannot return, apart from temporary replacements: football, rugby union, Gaelic football, hurling and baseball.
Re-entry sport	A sport in which a replaced player may come back on: rugby league, Australian rules, ice hockey and basketball.
Identity	An equation that holds by counting alone: here, the starters withdrawn equal the substitutions, corrected by five named terms.
Whole match	On the field from the start to the final whistle.
Chain	A substitution that takes off a player who had come on earlier, not a starter.
Withdrawal rate	The share of a team's starters recorded as leaving before the end of the match.
Substitution rate	A team's substitutions per player on the field.
Late entry	A change at minute 90 or in the second half's added time, listed at minute 90, so that its starter seems to finish.
Departure without replacement	A starter who leaves and is not replaced, as after a red card.
League-regime	One competition under one substitution rule, such as a league under five substitutions.
Gap	The substitution rate less the withdrawal rate. The identity says what it is made of.
Poisson count	The number of events that arrive independently at a steady rate. Its variance equals its mean.
Binomial count on the cap	The number of a team's places on the bench used when each is used independently with one chance. Its variance is $1 - \mu/c$ of its mean.
Censored normal	A narrow count of changes, drawn from a normal distribution and cut off at the limit.
Cell	The average minute of the k-th of n changes, over the teams that make exactly n.
Window	The stretch of the match in which a team's changes fall.
Stoppage	A pause in play in which a team makes one or more changes, read as the changes it books in one minute, or under Law 3.2 in one minute and the next.

eight general in 2014. From 2018–19 to 2025–26 the French leagues also let up to four tactically replaced players a team return in place of a team-mate who was injured, bleeding, blue-carded or under a head injury assessment. The dispensation ends after 2025–26 (Serevin, 2026), so 2026–27 is the first French season under the plain one-way rule. A return takes a starter off with no replacement recorded, so returns would pull the gap down. On the old lineups the French leagues' gap moved by -0.0002 across 2018–19 against the other leagues' (95 per cent interval -0.0011 to 0.0007, with the other leagues' seasons as the comparison). That allows at most 0.017 returns per team-match. No lineup names more than eight replacements, and the count of changes takes no step at 2018–19 (largest $|z| = 1.0$). From 2025–26 a player sent off may be replaced after 20 minutes, the orange card of the French leagues (Ligue Nationale de Rugby, 2025), and World Rugby made that replacement full law from 1 July 2026 (World Rugby, 2026). Replacing a player sent off puts a replacement on for a starter already recorded short, which lifts the gap against a sending-off that is not replaced. Under the full law fewer starters sent off go unreplaced, so the departures without replacement D shrink and rugby union's gap rises.

Rugby league. From 2014 a player removed for a head injury assessment who returns within 15 minutes uses no interchange, and the rule was used 167 times in its first season (Gardner et al., 2016). From 2021 a team may activate an 18th player after failed head injury assessments or a match-ending injury from foul play. Neither is counted against the eight interchanges.

Australian rules. The cap counts rotations during play. Changes at quarter time, half time and three-quarter time are not included, and neither is the substitute. The feed's count is the capped one: 0.2 per cent of team-games or fewer exceed the cap in any cap era, from 2,518 team-games. Its rotations per team-game agree with the AFL's published average for the first nine rounds of 2016, 85.3 against 85.3 (Schmook, 2016). For 2014 and 2015 the feed gives 109.2 and 109.2, about one per cent above the AFL's 107.9 for each season, which the published figure does not explain.

Baseball. The designated hitter is a batter who bats in the pitcher's place. With the designated hitter the lineup names 10 players, 9 who bat and a starting pitcher, which is the n_f the identity uses. The designated hitter became universal in 2022, and a club may still decline it. Since 2022 a starting pitcher may also be named the designated hitter and, once relieved, stay on as the hitter (Rule 5.11(b) of the 2024 rules), which leaves nine

players where the identity counts ten. None of the 4,858 team-games of 2024 names its starting pitcher in its batting order. The rules were renumbered in 2015, so each baseball rule in Table EC2 names the edition it was read in. A pitcher must face at least three batters from 2020 (Major League Baseball, 2020). A roster may carry at most 13 pitchers, a limit adopted for 2020 and first enforced on 20 June 2022 (Associated Press, 2022). Relief pitchers are counted on nine-inning team-games. Counts over every game run slightly higher, 3.43 in 2021 and 3.30 in 2022 (Jaffe, 2022), against 3.29 and 3.17 here.

EC3 Panels and certification

A season enters a panel only if its records are complete and the pattern of changes its teams make shows the rulebook's limit, and we call this check the certification. A season that fails is refused whole. In the football panels four league-seasons were refused, because the substitutions their teams made fit neither limit (Table EC3). All fall in the 2020 and 2021 seasons, when the five-substitution rule was being adopted, and none belongs to the leagues of Table EC7. The three-substitution panels also leave out every match played under the temporary five-substitution amendment of 8 May to 31 July 2020. The count and timing panels leave out the 2019–20 season altogether. Its matches under the amendment were kept apart and read once, as the held-out test of the model of stoppages (Section EC13). Read as five-substitution seasons, the refused seasons used 65 to 70 per cent of the cap. A panel that reads its cap from the lineups cannot admit them, so the statement that teams use most of their cap is read on the seasons that show their cap. Table EC4 gives the panel behind each result of the main text.

Table EC3: The football league-seasons the certification refused.

League	Season	Team-matches	Substitutions per team-match
BE1	2020	612	3.23
BE1	2021	610	3.31
RU1	2021	480	3.51
SC1	2020	396	3.25

Table EC4: The panel behind each result of the main text.

Result	Sports	Panel	Left out
Who plays the whole match	football, rugby union, Gaelic football, hurling, rugby league, Australian rules	the starters of the paired records' eight European leagues, on the field at the final whistle; five rugby union leagues, bench of eight; the All-Ireland finals; the NRL; the AFL's appearances	sides without a complete lineup, NRL sides with faulty minutes
Who plays the whole match, by position	football, rugby league, baseball, Australian rules, ice hockey, basketball	1,454,066 player-games	players without recorded minutes
The gap	football, rugby union, Gaelic football, hurling, baseball	18 football league-regimes; five rugby union leagues on the old lineups; five and nine finals; MLB 2024	sides without a complete lineup or with a contradictory record; refused finals
The check of the records	football	42,303 team-matches in nine leagues	the 2020 amendment window
The monthly gap	football	the eight European leagues under five substitutions (event records); seven women's leagues (appearances)	months with fewer than thirty matches
The exposure check	rugby league, Australian rules, ice hockey	10 panels; the NHL's 2023 regulation games	games with overtime in the NHL
The count	football, rugby union, basketball, baseball	29 football leagues with at least 500 five-substitution team-matches (59,982); rugby union's eight-bench sides; EuroLeague and EuroCup; MLB	the 2019–20 season; refused league-seasons
The timing	football, rugby union	the same football team-matches with one to five changes; rugby union's eight-bench sides	the 2019–20 season
The model of stoppages	football	fitted on the timing panel; tested on 1,270 held-out team-matches of 2019–20 in eight leagues	unrecorded matches

Gaelic football and hurling. The panels hold All-Ireland senior finals. Of the finals read, 16 in Gaelic football and 10 in hurling were refused for missing or disagreeing lineups. A further eight and 12 team-finals were refused because their pages carry no substitution entries and so parse as zero. The Gaelic football finals kept are those of 2002, 2011, 2012, 2015 and 2024, and the 2015 final was played under six substitutions. The hurling finals kept are those of 2003, 2007, 2008, 2010, 2015, 2021, 2022, 2024 and

2025.

Rugby union. The lineups come from one source in two formats. Up to 2024–25 each lineup lists the replacements who entered, with each one's entry minute and the minute each starter was replaced. From 2025–26 the lineups list the whole bench and give every player's minutes. A league-season is read as the new format when at least 10 per cent of its sides list a replacement who never entered, and every 2025–26 season is. A season's label is read from its match dates, because the source files Super Rugby's seasons under the year before. On the 2025–26 Top 14 lineups, six play-off matches, 12 sides, record no replacement minutes for either side, and they are set aside from the paired reading and counted (Table EC5). A half ends when the ball next goes dead, so recorded team minutes run over 80 and the two sides of one match do not reconcile. Of the starters, 0.51 are recorded at exactly the full match, against 0.026 recorded just short of it. Of the bench places, 0.054 are recorded at exactly zero.

Table EC5: The Top 14 sheets of 2025–26 set aside: play-off matches whose sheets record no replacement minutes, with the starters each side's sheet records short of 80 minutes.

Date	Sides with no replacement minutes	Starters short, home	Starters short, away
2026-06-13	2	7	6
2026-06-14	2	0	0
2026-06-14	2	0	0
2026-06-19	2	0	0
2026-06-20	2	0	0
2026-06-27	2	0	0

Rugby league. In the seasons under 10 interchanges, the NRL's minutes carry a fault. On some sides no starter is recorded at the full 80 minutes, because the starters who never left are recorded a few minutes short. Those sides cannot say who finished, so they are set aside and counted, never repaired.

EC4 The exposure identity

A team's recorded minutes should sum to the players on the field times the length of the match. Table EC6 gives the ratio of the two in every re-entry panel. The check is read on each panel's mean and spread. A panel passes when its mean ratio lies within 0.005 of one and its standard deviation is below 0.02. Read this way it passes the NRL's 2015 season, although 88 of

that season's sides were set aside for faulty minutes (Table EC16). So the check misses a fault that leaves a panel's mean intact. The AFL's time on ground comes from the league's own interchange records, so its sum may hold by construction. In ice hockey the panel holds the 1,040 of the season's 1,312 games that were decided in regulation. Overtime adds time on ice, at three skaters a side for up to five minutes, so a season with it can sum to more than six places for 60 minutes. Over a season with overtime the sum has been put at 0.994 of that total, and the gap put down to overtime and penalties (Donskov et al., 2019; McKay et al., 2014). In regulation only time with fewer than six on the ice removes a place. The shortfall tracks each team's penalty time, and penalty time does not account for all of it. In regulation the sum is 0.984 of the total, 1.6 per cent short, and its shortfall correlates with each team's penalty time at $r = 0.51$. Penalty minutes are only a proxy for time below six, because a misconduct penalty removes no one. The box scores do not give time below six. A goalkeeper pulled for an extra attacker moves minutes from the goalkeeper to the skaters and leaves the sum of six places unchanged. Shift reports have been summed against official time on ice before, with no discrepancy in the last two full seasons checked (Knodell, 2025).

Table EC6: The exposure identity. The ratio is the team-game's summed minutes over $n_f t_M$, averaged over team-games; for ice hockey it is summed time on ice over six players for sixty minutes.

Panel	Season	Team-games	Named	$\sum t_i/(n_f t_M)$	s.d.
Australian rules, men	2013	414	22.00	1.0000	0.0012
Australian rules, men	2014	414	22.00	0.9999	0.0013
Australian rules, men	2017	414	22.00	0.9997	0.0027
Australian rules, men	2018	414	22.00	0.9998	0.0025
Australian rules, men	2019	414	22.00	0.9997	0.0045
Australian rules, men	2022	414	22.56	1.0000	0.0009
Australian rules, men	2023	432	22.99	1.0000	0.0010
Australian rules, women	2023	198	21.00	1.0000	0.0012
Rugby league, men	2015	402	17.00	1.0014	0.0163
Rugby league, men	2016	402	17.00	0.9994	0.0024
Ice hockey, men	2023	2,080	20.00	0.9837	–

EC5 The identity in full

EC5.1 The proof

The Proposition of the main text follows from counting the players each substitution takes off. Each substitution takes off exactly one player. The $S - C - L - X$ substitutions that are not chains, not credited with the full match and not unlisted each take off a starter recorded short of the full

match. The only other starters recorded short are the D sent off and the U who left otherwise. So $n_f s_w = S - C - L - X + D + U$, which is the identity, and $C, L, X \geq 0$ with $S \leq n_{\text{eff}}$ gives the bound.

EC5.2 The measurement

Table EC7 reads the two rates on the same team-matches in every league-regime, which is one competition under one substitution rule. Rules have been read for what they bound before, for example the order of kicks in a penalty shoot-out (Apesteguia & Palacios-Huerta, 2010; Kocher et al., 2012). In rugby union, bonus points raised a measure of how evenly matched the teams are, mainly by raising the highest value the measure could take (Avila-Cano et al., 2023). Probability models in this journal have related scoring rates to the uncertainty of outcomes (Scarf et al., 2022) and derived the order in which teams score (Hamada & Tanaka, 2021). We describe a gap as small when its point estimate lies within 0.03 of zero, which is 3 per cent of the team, the unit of the figures. That is 0.33 of a starter per team-match in football and 0.45 in rugby union. The margin was set on football's league-regimes after their gaps had been read, the largest at 0.028, so football's gaps describe its records and do not test the margin. Outside football the margin tests little. Rugby union's old lineups give the substitutions and the withdrawals from one list of replacements, and the Gaelic reports list each change as one player for another, so a gap near zero there is close to built in. Against substitution rates of 0.25 to 0.52, the tolerance is 6 to 12 per cent of the rate. The intervals resample matches, each with both its sides, and resampling club-seasons instead changes no verdict. Every football league-regime's gap is below the margin (Figure EC1). The largest gap is that of Turkey's top league under five substitutions, 0.0281, and its 90 per cent interval reaches 0.0306, so the equivalence test fails there alone.

On the old lineups of rugby union's five leagues no league-season lies more than 0.0052 from zero. The source began in 2025–26 to list every player's minutes, and on these newest lineups the gap rose in every league, to between 0.0099 and 0.0297 (Figure EC2). The Top 14's minutes on the new lineups sum to more than 15×80 on 299 team-matches. A half ends when the ball next goes dead, so a sum above 15×80 does not by itself show a fault, and only official timed reports could say which format is right. The same season let a team replace a player sent off (Ligue Nationale de Rugby, 2025), which moves the gap up. It was also the last in which the French leagues let a replaced player return for an injured team-mate (Serevin, 2026), which moves it down. On the old lineups the French returns left no measurable

trace, a change of −0.0002 against the other leagues across 2018–19. Gaelic football and hurling rest on five and nine finals, and with so few units a bootstrap only restates the count, so no equivalence test is read on them. Gaelic football's gap is exactly zero in nine of its 10 team-finals, and hurling's in 17 of 18.

Table EC7: The two rates on paired records. Each row reads the substitutions and the withdrawals of the same n team-matches, or team-finals in the Gaelic games: S substitutions per team-match and a withdrawal rate s_w. The gap is $S/n_f - s_w$, with bootstrap intervals over matches. The equivalence test of Schuirmann (1987) asks the 90 per cent interval to lie inside ±0.03, and a dagger marks the one row that fails it. In the Gaelic rows every team-final but one has a gap of exactly zero, so they give the count instead of an interval and carry no equivalence test. The chain share C/S is counted where the record names the player replaced. In brackets it is the gap's share of S/n_f, the chain share plus late entries and unlisted players, less departures without replacement.

Competition	n	S	s_w	Gap	95% interval	90% interval	C/S (%)
Association football, five substitutions (calibration)							
Turkey	3,251	4.32	0.3648	0.0281	0.0253, 0.0309	0.0257, 0.0306†	0.59
England	2,280	4.04	0.3411	0.0264	0.0239, 0.0287	0.0242, 0.0285	0.65
Portugal	2,826	4.54	0.3928	0.0202	0.0178, 0.0227	0.0182, 0.0223	0.48
Germany	2,970	4.42	0.3838	0.0185	0.0161, 0.0207	0.0165, 0.0203	0.66
Netherlands	2,902	4.15	0.3637	0.0139	0.0121, 0.0158	0.0124, 0.0156	0.55
Brazil	757	4.79	0.4213	0.0139	0.0100, 0.0180	0.0107, 0.0173	0.86
France	3,333	4.22	0.3708	0.0132	0.0110, 0.0153	0.0113, 0.0151	0.50
Spain	3,553	4.55	0.4019	0.0120	0.0102, 0.0137	0.0105, 0.0134	0.52
Italy	3,699	4.53	0.4010	0.0112	0.0096, 0.0129	0.0098, 0.0126	0.72
Association football, three substitutions (calibration)							
England	3,492	2.75	0.2377	0.0121	0.0108, 0.0135	0.0110, 0.0133	0.54
Turkey	1,690	2.93	0.2559	0.0105	0.0080, 0.0129	0.0085, 0.0126	0.61
Germany	1,667	2.90	0.2543	0.0094	0.0076, 0.0115	0.0078, 0.0112	0.50
Netherlands	1,679	2.72	0.2398	0.0072	0.0053, 0.0094	0.0056, 0.0089	0.22
Portugal	1,458	2.94	0.2618	0.0055	0.0031, 0.0080	0.0036, 0.0075	0.28
France	2,035	2.86	0.2558	0.0038	0.0018, 0.0056	0.0021, 0.0053	0.33
Spain	1,909	2.92	0.2626	0.0028	0.0011, 0.0044	0.0014, 0.0041	0.16
Italy	2,048	2.94	0.2652	0.0020	0.0002, 0.0037	0.0005, 0.0035	0.43
Brazil	754	2.96	0.2691	−0.0001	−0.0028, 0.0025	−0.0024, 0.0022	0.31
Rugby union and the Gaelic games (test)							
Five leagues	25,028	7.19	0.4771	0.0022	0.0020, 0.0024	0.0020, 0.0023	[0.5]
Top 14, 2025–26	364	7.82	0.5007	0.0205	0.0163, 0.0249	0.0170, 0.0244	[3.9]
Gaelic football	10	3.90	0.2533	0.0067	9 of 10 zero	exact	2.56
Hurling	18	4.39	0.2889	0.0037	17 of 18 zero	exact	1.27
Baseball (test)							
MLB	4,858	4.71	0.2262	0.2444	0.2405, 0.2480	0.2412, 0.2475	≥45 [52]

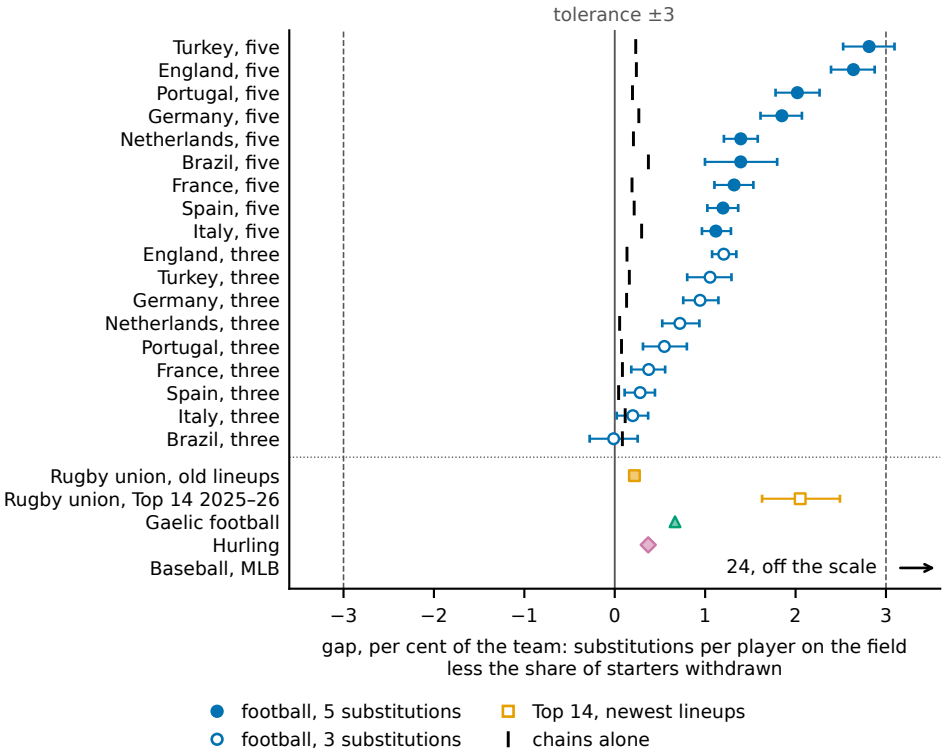


Figure EC1: Every gap but baseball's lies within the tolerance, and football's chains are small. The gap between the substitution rate and the withdrawal rate in every league-regime, in per cent of the team, with its 95 per cent interval; the dashed lines are the tolerance, set on football. The markers are Figure 1's: football's filled points are under five substitutions and its open points under three, and the open square is the Top 14's newest lineups. The small bar on each football row is the chains alone, C/n_f, and a dotted rule separates football from the other codes. Turkey's 95 per cent interval under five substitutions reaches past the tolerance, and so does its 90 per cent interval, so it is the one row that fails the equivalence test (Table EC7). Rugby union's interval on the old lineups is about as wide as its marker. Rugby union's old lineups and the Gaelic finals give both counts from one list, so their small gaps are close to built in, and they are drawn lighter. The Gaelic rows are exact counts and carry no interval, and baseball lies off the scale. Panel b of Figure 2 of the main text draws football's gap counted part by part.

Alt text: A dot plot of the gap for twenty-three rows. Every football and rugby union row lies between two dashed lines at plus and minus the tolerance, and each football row carries a small bar near zero for its chains. The two lighter Gaelic rows are points just above zero, and baseball is an arrow off the right edge.

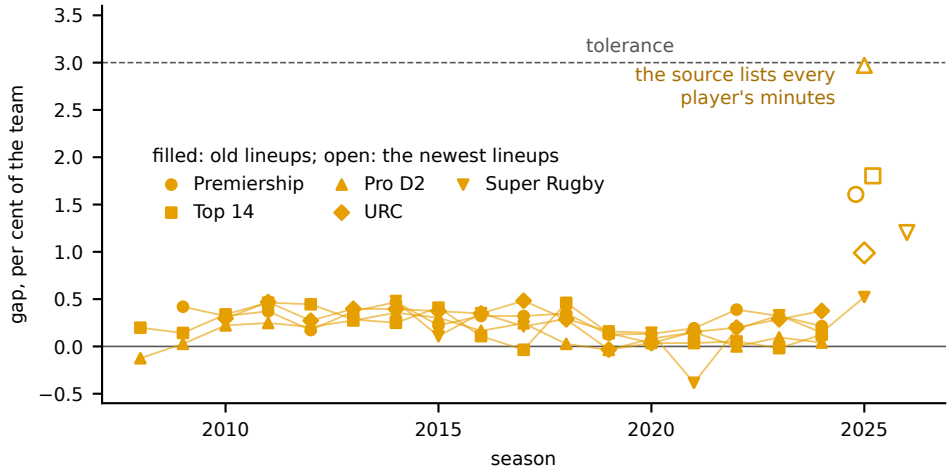


Figure EC2: Rugby union's gap stays close to zero season after season on the source's old lineups, and rises in every league once the source lists every player's minutes. The gap of each league and season, in per cent of the team; open markers are the newest lineups of 2025–26, or 2026 in Super Rugby, and the dashed line is the tolerance. A season is labelled by the year it starts, Super Rugby's by its calendar year; URC is the United Rugby Championship; the newest Premiership and Top 14 points are nudged apart so that both show. The Top 14's newest point counts all 366 sides read, the play-off sides that record no replacement minutes among them, as Table EC11 does; on its 364 paired sides, drawn in Figure EC1 and in Figure 1 of the main text, its gap is 2.1 per cent.

Alt text: A line chart of rugby union's gap by season for five leagues, each with its own marker. On the old lineups every league stays within about half a point of zero from 2008 to 2025. In the newest season every league's open marker sits higher, between about one and three points, the highest at the dashed tolerance line.

EC5.3 Football's gap, counted

Table EC8 gives the counted parts of football's gap in every league-regime. Each substitution event names the player who left and the player who came on, and each card says whether it sent the player off. When several players change at once, a provider may infer who replaced whom, for instance from the order of the swap (Stats Perform, n.d.). The chain count C counts outgoing players who had come on earlier, whoever replaced them. The gap is exactly $(C + L + X - D - U)/n_f$. Here X counts the substitutions whose outgoing player is on neither list, and these records hold none. The term U counts the other starters recorded short with no change or card, at most 0.013 per team-match. Late entries run from 0.08 to 0.38 per team-match, and players sent off pull the other way, at 0.05 to 0.11. With every change in added time recorded at its minute, L would be zero and the gap would be $(C + X - D - U)/n_f$. Under five substitutions that shrinks every league's gap, to at most 0.7 per cent of the team from 1.1 to 2.8 per cent as recorded, which holds by construction once L is set to zero. Panel b of Figure 2 of the main text draws the counted parts instead. Under three it would not shrink them all: the gaps would read -0.0087 to -0.0029, because there the late entries had partly offset the departures without replacement. So under three substitutions a correct record can show a gap near zero, and the gap alone cannot tell it from a record that drops its late entries. Adding back the starters sent off per player, which appearance tables usually carry, leaves the chains and the late entries less the few other starters recorded short, $(C + L + X - U)/n_f$. On every league-regime here that sum is 0.8 to 3.6 per cent of the team, under both rules (Table EC10). A side is set aside when the event records carry no lineup for it, 335 sides, or when its record does not carry 11 starters with minutes, 1,668 more. A side is also set aside when its record contradicts itself, 14 sides: a player taken off twice, brought on twice, or named going off and coming on in one substitution. One of them is a match suspended and resumed with new players. With those set aside, the counted parts add up to the gap on every team-match, as the counting requires. The events are those of the snapshot the other football analyses read, and it carries a lineup for at least 75 per cent of each league's games in every season. The gap's interval resamples matches, and the column Club resamples club-seasons instead, which moves no verdict. The audit flags a team-match whose recording terms, $L + X$, are not zero, 8 to 27 per cent of team-matches, each for a change whose starter is credited with the full match. Table EC9 follows the audit through the Premier League's seasons, across the arrival of exact added time in 2023–24 (Premier League, 2023). Late entries rose there from 0.26 per team-match in

2022–23 to 0.36 in 2023–24, because a longer added time puts more changes where the source's clock has stopped. Per substitution, late entries rose in seven of the nine leagues. In Italy they held, moving by less than 5 per cent of their level, from 0.033 to 0.034, and in the Netherlands they fell, from 0.054 to 0.046. The three- and five-substitution seasons are different years, and a difference-in-differences study of four leagues has read the five-substitution rule's effect on match outcomes (Zhang et al., 2025). A flag cannot see a change missing from both lists, because the substitutions and the withdrawals then drop it together. Brazil's dropped entries showed in the league's gap, not in its flags (Section EC5.4). Arithmetic checks of published statistics find values that cannot occur (Nuijten et al., 2016; Brown & Heathers, 2017), and public event data are screened for impossible sequences of events (Pappalardo et al., 2019). This check finds something milder, a convention by which the source lists added time at minute 90. In official statistics a record that fails an equality it must meet, an edit rule, is deduced to hold an error, and rounding is a common cause (Scholtus, 2011). Listing added time at minute 90 is such a rounding of the clock. Live event coding by trained operators has been found highly reliable (Liu et al., 2013), and the flags here come from the listing convention, not from the coding. Injury histories from the same source show fewer injuries than other sources, except severe ones (Hoenig et al., 2022). Exploratory checks of records have been proposed in this journal as a tool for auditability (Cooke & Huber, 2026). Baseball's rules prove a box score, the game's table of statistics, by making its columns balance (Major League Baseball, 2026, Rule 9.03(c)). A larger dataset does not cure an error that repeats in every match (Bradley et al., 2021; Lazer et al., 2014). Many teams of analysts given one dataset reach widely different answers (Silberzahn et al., 2018; Botvinik-Nezer et al., 2020; Breznau et al., 2022), and an exact identity gives every analyst the same check. Season by season, the Premier League's flags rose from 13 per cent of team-matches in 2017–18 to 26 per cent in 2023–24. A football analyst has read Serie A's 9.37 substitutions a game as 42.6 per cent of the starters taken off (Segar, 2024), which is the identity with its correction terms set to zero.

Table EC8: Football's gap, counted on the event records and Brazil's match reports. Read is the per cent of sides whose line-up and changes are read whole, and n the team-matches. Per team-match: S substitutions, C chains, L changes booked at minute 90, D starters dismissed and not replaced, and U other starters recorded short. No substitution takes off a player on neither list, so $X = 0$, and the gap is exactly $(C + L - D - U)/n_f$. Club is the upper end of the gap's 90 per cent interval when club-seasons, not matches, are resampled, and Flag the per cent of team-matches whose recording terms are not zero; Read and Flag are in per cent.

League	Seasons	Read	n	S	C	L	D	U	Gap	Club	Flag
Five substitutions											
Turkey	2020–24	87	3,251	4.32	0.026	0.382	0.085	0.013	0.0281	0.0310	27
England	2022–24	100	2,280	4.04	0.026	0.310	0.046	0.000	0.0264	0.0292	23
Portugal	2020–24	92	2,826	4.54	0.022	0.300	0.097	0.002	0.0202	0.0229	21
Germany	2020–24	97	2,970	4.42	0.029	0.234	0.053	0.007	0.0185	0.0213	18
Netherlands	2020–24	95	2,902	4.15	0.023	0.193	0.062	0.000	0.0139	0.0161	15
Brazil	2024	100	757	4.79	0.041	0.194	0.082	0.000	0.0139	0.0174	15
France	2020–24	95	3,333	4.22	0.021	0.219	0.089	0.005	0.0132	0.0156	17
Spain	2020–24	94	3,553	4.55	0.024	0.184	0.076	0.000	0.0120	0.0138	14
Italy	2020–24	97	3,699	4.53	0.033	0.153	0.063	0.000	0.0112	0.0130	12
Three substitutions											
England	2017–21	97	3,492	2.75	0.015	0.165	0.047	0.000	0.0121	0.0137	15
Turkey	2017–19	100	1,690	2.93	0.018	0.207	0.109	0.000	0.0105	0.0138	19
Germany	2017–19	100	1,667	2.90	0.014	0.143	0.053	0.000	0.0094	0.0118	13
Netherlands	2017–19	99	1,679	2.72	0.006	0.146	0.073	0.000	0.0072	0.0093	14
Portugal	2017–19	88	1,458	2.94	0.008	0.156	0.104	0.000	0.0055	0.0081	15
France	2017–19	98	2,035	2.86	0.009	0.131	0.099	0.000	0.0038	0.0060	12
Spain	2017–19	93	1,909	2.92	0.005	0.098	0.072	0.000	0.0028	0.0044	9
Italy	2017–19	100	2,048	2.94	0.013	0.096	0.087	0.000	0.0020	0.0036	9
Brazil	2019	100	754	2.96	0.009	0.081	0.092	0.000	−0.0001	0.0032	8

Table EC9: The audit by season in the Premier League: changes booked at minute 90 per team-match (L), the per cent of team-matches the audit flags, and the gap. Exact added time came in with 2023–24.

Season	Limit	Team-matches	L	Flagged (%)	Gap
2017–18	three	759	0.133	13	0.0093
2018–19	three	748	0.175	16	0.0129
2019–20	three	557	0.171	16	0.0114
2020–21	three	731	0.159	15	0.0118
2021–22	three	697	0.191	17	0.0149
2022–23	five	760	0.255	19	0.0222
2023–24	five	760	0.355	26	0.0305
2024–25	five	760	0.320	23	0.0264

EC5.4 Brazil

The match reports of Brazil's top league are event records, like Europe's. Every substitution names the player who came on, the player who went off and the minute, and every sending-off names the player and the minute.

The rows of Table EC8 count them under the European rule, on the 2019 season under three substitutions and the 2024 season under five. Counted chains are 0.31 and 0.86 per cent of substitutions. A first reading used an appearance panel built from the same reports, which drops every entry at the 90th minute or later. A change in added time then leaves neither a substitute nor a withdrawal, so that panel counts $C - D - U$ where every other row counts $C + L + X - D - U$. Read on it, the same seasons gave gaps of -0.0077 and -0.0043, and the audit is what found the panel's rule. Read from the event records, the gaps are -0.0001 and 0.0139. The gap alone would have flagged the event records of the three-substitution season too, since their gap is near zero although they are correct. Adding back the starters sent off per player separates the two readings: 0.06 and 0.31 per cent of the team on the appearance panel, against 0.8 to 3.6 on every counted record.

Table EC10: The aggregate audit rule with the starters sent off added back, in per cent of the team. The gap plus the starters sent off per player estimates the chains and the late bookings, $(C + L)/n_f$, less the few other starters recorded short. On the counted records of every league-regime it is clearly positive, and on Brazil's first reading, an appearance panel that drops every entry at minute 90 or later, it falls near the chains alone.

League	Rule	Record	Gap	Sent off per player	Sum
BRA1	five	counted	1.39	0.74	2.14
ES1	five	counted	1.20	0.69	1.89
FR1	five	counted	1.32	0.81	2.13
GB1	five	counted	2.64	0.42	3.06
IT1	five	counted	1.12	0.57	1.69
L1	five	counted	1.85	0.48	2.33
NL1	five	counted	1.39	0.56	1.96
PO1	five	counted	2.02	0.88	2.90
TR1	five	counted	2.81	0.77	3.58
BRA1	three	counted	−0.01	0.83	0.82
ES1	three	counted	0.28	0.65	0.93
FR1	three	counted	0.38	0.90	1.28
GB1	three	counted	1.21	0.43	1.63
IT1	three	counted	0.20	0.79	0.99
L1	three	counted	0.94	0.49	1.43
NL1	three	counted	0.72	0.66	1.38
PO1	three	counted	0.55	0.95	1.50
TR1	three	counted	1.05	0.99	2.04
BRA1	three	appearance panel	−0.77	0.83	0.06
BRA1	five	appearance panel	−0.43	0.74	0.31

EC5.5 Rugby union

Table EC11 gives rugby union's two rates and their gap league by league, on each format of the source's lineups. Pooled over the five leagues the

old lineups give a gap of 0.0022, with a 95 per cent confidence interval of 0.0020 to 0.0024. On the old lineups every league's gap lies between 0.0014 and 0.0030, and on at most 5 per cent of a league's team-matches do the replacements outnumber the starters recorded short. On the new lineups of 2025–26 that share is 16 to 38 per cent. On those new lineups the gap rose in every league to 1.0 to 3.0 per cent of the team, so the source's two formats disagree (Figure EC2). The Top 14's new lineups give a gap of 0.0205, with a 95 per cent interval of 0.0163 to 0.0249, on the 364 team-matches that pair both sides. Table EC11 reads the same season on 366 sides: those of the 364 paired team-matches and the two sides of the play-off match of 13 June 2026, which record starters short but no replacement minutes (Table EC5). Its Top 14 row gives 0.0180. On them 99 team-matches record more replacements than starters short, 122 replacements in all, and 10 record one starter short beyond their replacements. Only 55 of the 364 sum their minutes to exactly 15×80.

Table EC11: Rugby union's identity on the team sheets of five leagues, every season with a bench of eight, by the source's format: the old sheets list only the replacements who entered, the new sheets of 2025–26 the whole bench. URC is the United Rugby Championship. A season is labelled by its start year, or by its year in Super Rugby. The interval resamples matches. Worst is the largest league-season gap, and More the per cent of team-matches on which the replacements outnumber the starters recorded short.

League	Sheets	Seasons	Team-matches	Gap	95% interval	Worst	More
Five leagues	old	2008–25	25,028	0.0022	0.0020, 0.0024	0.0052	4
Premiership	old	2009–24	4,136	0.0029	0.0025, 0.0035	0.0047	5
Pro D2	old	2008–24	8,149	0.0014	0.0010, 0.0017	0.0036	4
Super Rugby	old	2014–25	2,424	0.0026	0.0019, 0.0033	0.0052	5
Top 14	old	2008–24	6,224	0.0020	0.0016, 0.0025	0.0046	4
URC	old	2010–24	4,095	0.0030	0.0025, 0.0035	0.0048	5
Five leagues	new	2025–26	824	0.0167	0.0133, 0.0201	0.0172	25
Premiership	new	2025	112	0.0161	0.0095, 0.0232	0.0161	22
Pro D2	new	2025	92	0.0297	0.0217, 0.0384	0.0297	38
Super Rugby	new	2026	72	0.0120	0.0056, 0.0204	0.0120	19
Top 14	new	2025	366	0.0180	0.0109, 0.0239	0.0180	27
URC	new	2025	182	0.0099	0.0051, 0.0147	0.0099	16

EC5.6 Baseball, where substitutes replace substitutes

Baseball is strictly one-way, because a player removed from the game cannot return. Across 4,858 team-games of the 2024 season its gap is 0.24, 8.1 times the tolerance. A count that does not depend on the gap bounds its chains. Under the designated hitter every relief pitcher after a team's first enters for a pitcher who had come on as a substitute. So a team-game has at least 2.13 chains, at least 45 per cent of its substitutions. Relief pitchers

per nine-inning team-game are many, 2.75 in 2008, 3.23 in 2019 and 3.12 in 2024. Baseball keeps the letter of the one-way rule more strictly than any sport here, and it still has a large gap. So a small gap needs few chains as well as a ban on re-entry.

EC5.7 Football's own exception

Football's Laws let a competition allow concussion substitutes outside the normal limit, and the protocol has changed (Table EC2). Before 2020–21 there was none. From 2020–21 to 2023–24 each competition could adopt a trial protocol, and the Premier League and the Football Association allowed from 6 February 2021 up to two concussion substitutes a team, the opponent as many (Premier League, 2021), while the Premier League still used three substitutions. Since 1 July 2024 the Laws carry one protocol (The International Football Association Board, 2024): a team may use one concussion substitute, and its opponent may then use one more substitute, in one more stoppage (The International Football Association Board, 2026*a*). So a team may lawfully pass its normal limit by two under the permanent protocol, and by four under England's trial. A prediction, entered in the register only after its test had been written, counted the team-matches that use more substitutions than their season's coded limit. Across 108 coded league-seasons and 58,545 team-matches, 343 exceed their limit, 0.6 per cent of them, which moves s_w by 0.0006. The source marks no concussion substitute, so the counts against each protocol's most are bounds. Of these team-matches, 46 across 13 league-seasons sit more than one above their limit, four more than two and two more than four. In England under the trial, 14 of 1,090 team-matches under three substitutions exceed the limit, and none exceeds the trial's most. Leaving England's trial seasons out, the rate above the limit is 3.1 times higher under three substitutions than under five, and Ukraine's and Scotland's 2021–22 seasons hold 55 per cent of the excess under three. So the excess sits in a few league-seasons, where competition formats and the coding of the limit explain it better than concussion does.

EC5.8 Month by month

The months are counted from each league-season's first match, month one being the calendar month of that match, and each month pools the leagues of a family, the men's leagues or the women's. The men's family is the eight European leagues with paired records under five substitutions, and the women's is seven leagues. A month's substitution rate is its substitutes

who played over twice its matches and over n_f, and its withdrawal rate is the share of its starters recorded short of 90 minutes. So the two rates are compared in the month, not in the team-match. That is what lets the women's records, which name the player and the match but carry no club, enter at all. The men's worst monthly gap is 0.016 across 11 months, and the women's 0.003 across nine. An appearance panel can drop a change listed at minute 90, as Brazil's did, and that pulls the gap toward zero. So the men's months are also read from the event records, team-match by team-match, on the five-substitution sides of the eight leagues (Table EC12). Read that way the largest monthly gap is 0.023 across 11 months, larger than the appearances give and still below the margin. The women's records carry no events, so their reading stays the appearances', and by the same reasoning their months may understate their gap.

Table EC12: The gap month by month in the eight European leagues under five substitutions, read from the event records: a team-match's substitutions from its substitution events and its withdrawals from its starters recorded under 90 minutes. Month one is the calendar month of a league-season's first match.

Month	Matches	Team-matches	S/n_f	s_w	Gap
1	1,052	1,989	0.4012	0.3869	0.0143
2	1,222	2,315	0.4085	0.3954	0.0130
3	1,314	2,491	0.4006	0.3883	0.0123
4	1,117	2,125	0.3922	0.3738	0.0184
5	1,326	2,517	0.3878	0.3679	0.0199
6	1,364	2,632	0.3824	0.3618	0.0206
7	1,413	2,762	0.3955	0.3785	0.0170
8	1,286	2,506	0.3954	0.3766	0.0188
9	1,596	3,114	0.4004	0.3814	0.0190
10	1,152	2,269	0.4059	0.3864	0.0195
11	47	94	0.4207	0.3975	0.0232

EC6 Who plays the whole match

The prediction, entered in the register only after its test had been written, ranked every sport under each of its limits by the limit per player on the field. If the share that finishes fell monotonically in that ranking, the law would be restated on it. If not, the law would be restricted to the sports it was first measured in. The ranking was not monotone, so the second branch binds (Spearman -0.10 on starters, $p = 0.87$, across five pairings of a sport and a limit). Figure 5 of the main text draws these shares, football's read on the field at the final whistle rather than on the record's minutes, and Table EC15 gives every pairing. The ordering fails as well when the re-entry sports are placed at their bench per player, 0.31 in rugby league and

0.22 to 0.28 in Australian rules. A practitioner series had counted the NRL's share before this paper measured it, 47 per cent of player-games reaching 80 minutes in 2021 (LeagueEyeTest, 2021). Studies of match demands compare whole-match and interchanged players as two groups (Black & Gabbett, 2014).

The main text reads a starter as playing the whole match when the starter is on the field from the start to the final whistle, and each source is read on its own terms. Football's starters are read from the event records, rugby union's and the Gaelic finals' from the lineups, which name each starter replaced. The NRL's are read on the full 80 minutes recorded, baseball's on every out, the AFL's and the NHL's on the game's own clock, and the EuroLeague's on every minute of a game without overtime. Football's records credit a starter replaced in added time with the full match. So the paired records' eight European leagues are read both ways (Table EC13). On the field at the final whistle are 73 per cent of the starters under three substitutions and 60 per cent under five, against 75 and 62 per cent recorded at the full match. The ranking above was registered on the record's minutes, so it reads them, and the shares below are the record's.

Rugby union's share has three readings. The main text gives 52.3 per cent on the old lineups of the five leagues and 49.9 per cent on the newest, each one less the withdrawal rate of Table EC7. The newest reading takes the 364 sides of the 2025–26 Top 14 that pair both sides of a match. Table EC15 gives 51.3 per cent, the starters recorded at the full match on all 376 sides of the same season. The 12 play-off sides set aside from the pairing record 167 of their 180 starters at the full match, and they lift the share from 49.9 to 51.3 per cent. So under a bench of eight about half the starters play the whole match, and which of the source's formats is right decides which side of one half it is.

Table EC13: Football's starters who play the whole match, recorded at 90 minutes and on the field at the final whistle, in the eight European leagues of the paired records. A starter is on the field at the final whistle when the record gives 90 minutes and no substitution took the starter off and no card sent the starter off. The last column is the difference: the starters recorded at 90 minutes who left before the end.

Rule	Starters	Read	Recorded (%)	On the field (%)	Left late (%)
three	goalkeepers	16,198	98.9	98.8	0.1
three	outfield starters	162,309	72.3	70.8	1.6
five	goalkeepers	25,972	98.7	98.5	0.1
five	outfield starters	259,925	58.4	55.8	2.5
three	every starter, mean of 8 leagues	175,758	74.6	73.1	1.5
five	every starter, mean of 8 leagues	272,954	62.3	59.9	2.4

In football the bound leaves at least $1 - n_{\text{eff}}/n_f$ of the starters to finish, less departures without replacement. Read on the field at the final whistle, as the main text and its Figure 5 read them, football's starters sit 0.4 percentage points above that least under three substitutions and 5.3 under five, and rugby union's 3.3 to 5.6, while the NRL's starters sit 25 points above the least an interchange cap allows (Table EC14). Scaled by the room above the floor, as indices bounded by a league's structure are scaled (Manasis et al., 2022), every one-way league here sits within 0.12 of its floor on a scale from 0 at the floor to 1 where every starter finishes.

Table EC14: How far above its floor each league sits. The floor is 1− the limit per player, the least share of starters who finish where a replaced player cannot return; in rugby league, where players return, it bounds the starters an interchange can take off. The share is the one Figure 5 of the main text draws. The distance is in percentage points, and the last column scales it by the room above the floor, (share − floor)/(1 − floor), from 0 at the floor to 1 when every starter finishes, as indices bounded by a league's structure are scaled (Manasis et al., 2022).

Sport and limit	Limit per player	Floor	Share	Above, points	Above, scaled
football, three	0.273	72.7	73.1	0.4	0.01
football, five	0.455	54.5	59.9	5.3	0.12
rugby union, old lineups	0.533	46.7	52.3	5.6	0.11
rugby union, Top 14 newest	0.533	46.7	49.9	3.3	0.06
rugby league, eight	0.615	38.5	63.4	25.0	0.41
rugby league, ten	0.769	23.1	64.1	41.0	0.53

Rugby league breaks the ordering, because 0.634 of its starters play the whole match under eight interchanges among 13 players, as many as football's. Australian rules leaves 0.010 of its appearances under a cap of 90 and 0.027 under 75. A reading made after the failure puts the boundary at the size of the limit against the team: under each cap of ten or fewer measured here most starters finish, and under a cap several times the team almost no one does.

Where players may return, the cap on interchanges limits rotations instead, and most NRL starters still play the whole match under caps of ten and eight. Where the rules set no cap and players may return, no one need play the whole match. American football, for example, lets any number of substitutes enter while the ball is dead (National Football League, 2026), and ice hockey and basketball change players on the fly.

The NRL reading was corrected twice. The first run pooled a season under 10 interchanges with one under eight, and the seasons were re-cut by the rulebook's cap. The second found that the apparent rise in finishing after the cut came from the short records of 2015. Once those sides are set

aside, the 10-interchange seasons leave 0.641 of their starters to finish and the eight-interchange seasons 0.634.

Table EC15: Who plays the whole match, by sport-regime. The limit per player is $n_{\max}/n_f$ in the one-way sports and n_e/n_f in the re-entry ones. The starters' and the appearances' shares are the record's. The last column is the share Figure 5 of the main text draws: football's starters on the field at the final whistle (Table EC13), rugby union's one less the withdrawal rate on the old lineups and on the Top 14's newest paired lineups (Table EC7), and the AFL's appearances.

Sport-regime	Limit per player	Bench per player	Starters finish	Appearances finish	Figure 5 draws
football, three	0.273	–	0.746	0.595	0.731
football, five	0.455	–	0.623	0.453	0.599
rugby union	0.533	–	0.513	0.341	0.523, 0.499
rugby league, eight	0.615	0.308	0.634	0.487	0.634
rugby league, ten	0.769	0.308	0.641	0.502	0.641
AFL, cap 75	4.167	0.278	–	0.027	0.027
AFL, cap 90	5.000	0.222	–	0.010	0.010

Table EC16: The NRL by season. Sides set aside are those on which no starter is recorded at the full match or whose minutes are incomplete; the last column is the starters' share as recorded, before any side is set aside.

Season	Interchanges	Sides read	Set aside	Starters finish	Appearances finish	As recorded
2013	10	368	34	0.688	0.557	0.630
2014	10	370	32	0.610	0.468	0.562
2015	10	314	88	0.625	0.482	0.488
2016	8	402	0	0.640	0.491	0.640
2017	8	400	2	0.647	0.498	0.647
2018	8	401	1	0.641	0.493	0.640
2019	8	400	2	0.645	0.494	0.641
2020	8	336	2	0.639	0.490	0.639
2021	8	390	12	0.612	0.468	0.612
2022	8	402	0	0.624	0.478	0.624
2023	8	426	0	0.630	0.483	0.630
2024	8	426	0	0.633	0.485	0.633
2025	8	426	0	0.632	0.484	0.632

EC7 Who plays the whole match, by position

The same question is read inside each sport, by the position the record gives a player before the match starts, on 1,454,066 player-games in six sports. The groups are each sport's coarsest standard units. Rugby league's are the backs, the forwards and the interchange on the NRL's team lists, by the cap the rulebook dates. Football's are the goalkeepers and the outfield starters in the Transfermarkt lineups of the paired records' eight European leagues. Baseball's are the starting batters, by the first position the 2024

box scores list for them, and the starting pitchers. Australian rules' are the three starting lines and the interchange on the AFL's team lists, with time on ground by the game's own clock. Ice hockey's are the NHL's forwards and defencemen, and basketball's the EuroLeague's starters. A batter lasts the game when no substitute takes the batting slot, and a pitcher when the team uses no other pitcher. The AFL's records are its team lists and time on ground in the seasons 2017 to 2025, and the NHL's its game sheets from 2008 to 2024, each season labelled by the year it starts, all read from the leagues' own pages (Australian Football League, 2026; National Hockey League, 2026). Table EC17 gives every group, and Table EC18 the seven predictions, pre-specified before the software read any record by position. Every prediction held. They compared groups whose difference the rules all but guarantee, a single goalkeeper against ten outfield players for example, so they could hardly have failed and carry little evidence. They were registered on the record's minutes. Read on the field at the final whistle, outfield starters play the whole match in 71 per cent of their games under three substitutions and 56 per cent under five, against 72 and 58 per cent recorded at the full match. Under five substitutions 2.5 per cent of outfield starters are recorded at the full match yet were taken off or sent off before the end, and goalkeepers play the whole match in 99 per cent of their games, against 99 per cent recorded. Every prediction holds either way. At EURO 2024 one of 466 substitutions replaced a goalkeeper (Gabrys et al., 2025). Table EC19 sets the spread between positions against the move a change of cap made, in percentage points, in log-odds and, for the AFL, on median time on ground. In football and rugby league the spread is larger than the move on every scale. In the AFL it is not on the share who play the whole match, where both are a few points, and it is on median time on ground. In baseball the 2024 box scores give 28 complete games, a starting pitcher who pitches the whole game, in 4,858 starts, the record low of 28 that the Associated Press reported for that season (The Associated Press, 2026). In Australian rules a starting line spends a median 82 to 86 per cent of the match on the ground under a cap of 75, against 73 per cent for the interchange players. Few players in any line play the whole match, a line being the backs, the midfield or the forwards, and the backs do so most often, in 7.5 per cent of their games under a cap of 75. Starters in the EuroLeague, Europe's top club basketball competition, played every minute of a game without overtime in 0.1 per cent of their games. In the NHL a forward spends a median 25 per cent of a game on the ice and a defenceman 33 per cent, over 730,501 skater-games, and no skater, any player but the goalkeeper, played a whole game.

Table EC17: Who plays the whole match, by position. Whole is the share of a group's games that last the record's full length; median is the median share of the match played, where the record gives the time. Baseball's rows read the batting slot and the pitchers used, so they carry no time.

Sport	Rule	Group	Games	Whole (%)	Median (%)
Rugby league	eight interchanges	backs	28,063	91.0	100
Rugby league	eight interchanges	forwards	24,054	31.3	78
Rugby league	eight interchanges	interchange	15,836	0.0	39
Rugby league	ten interchanges	backs	7,364	92.0	100
Rugby league	ten interchanges	forwards	6,312	31.7	76
Rugby league	ten interchanges	interchange	3,809	0.0	42
Football	five substitutions	goalkeepers	25,972	98.7	100
Football	five substitutions	outfield starters	259,925	58.4	100
Football	three substitutions	goalkeepers	16,198	98.9	100
Football	three substitutions	outfield starters	162,309	72.3	100
Baseball	no cap	catchers	4,858	84.4	–
Baseball	no cap	designated hitters	4,858	85.9	–
Baseball	no cap	infielders	19,432	88.0	–
Baseball	no cap	outfielders	14,574	83.6	–
Baseball	no cap	starting pitchers	4,858	0.6	–
Australian rules	cap of 75	backs	12,611	7.5	86
Australian rules	cap of 75	forwards	12,612	1.4	82
Australian rules	cap of 75	interchange	10,115	0.0	73
Australian rules	cap of 75	midfield	12,611	0.3	82
Australian rules	cap of 90	backs	9,192	2.9	87
Australian rules	cap of 90	forwards	9,192	1.3	84
Australian rules	cap of 90	interchange	6,128	0.0	76
Australian rules	cap of 90	midfield	9,192	0.1	83
Ice hockey	no cap	defencemen	245,983	0.0	33
Ice hockey	no cap	forwards	484,518	0.0	25
Basketball	no cap	starters	43,490	0.1	60

Table EC18: The predictions registered before the software read any record by position, each quoted in full as registered.

Id	Outcome	Prediction
PF01	held	In the NRL, under each interchange cap the rulebook dates for the seasons read (eight and ten interchanges), starting backs (Fullback, Winger, Centre, Five-Eighth and Halfback on the team list) play the whole match, 80 minutes recorded, in a larger share of their appearances than starting forwards (Prop, Hooker, 2nd Row and Lock). Sides whose minutes cannot say who finished are set aside as [stage] sets them aside.
PF02	held	In the eight European leagues of the paired records, under three substitutions and under five (the league-seasons [stage] reads, outside the 2020 amendment), starting goalkeepers play the whole match, 90 minutes recorded, in a larger share of their appearances than outfield starters.
PF03	held	In the 2024 MLB season, each fielding group of starting batters, read from the first position the box score lists for each starter (catchers; infielders at first, second and third base and shortstop; outfielders; designated hitters), stays in the game to its end, with no substitute taking the batting slot, in a larger share of games than the starting pitcher pitches the whole game, with no other pitcher used by the team.

Table EC18, continued

Id	Outcome	Prediction
PF04	held	In the AFL, under each interchange cap the rulebook dates for the panel's seasons (90, then 75), each starting line (backs: BPL, FB, BPR, HBFL, CHB, HBFR; midfield: WL, C, WR, RK, RR, R; forwards: HFFL, CHF, HFFR, FPL, FF, FPR) spends a larger median share of the match on the ground than the interchange (INT and SUB).
PF05	held	In the NHL, fewer than one per cent of skaters' games reach the full length of the game on the ice, for forwards (C, L and R on the game sheet) and for defencemen (D) alike.
PF06	held	In the NHL, defencemen spend a larger median share of the game on the ice than forwards.
PF07	held	In the EuroLeague, starters play the whole of a regulation game, all 40 minutes of a game with no overtime, in fewer than five per cent of their games.

Table EC19: How much position matters, against the cap. For each sport and cap, the share of the upper and the lower group of positions who play the whole match, in per cent, the spread between them in percentage points and in log-odds (none where a share is zero), then the move a change of cap made in each group, in points, and for the AFL the median time on ground. Football is read on the field at the final whistle (Table EC13), the other sports as Table EC17 reads them.

Sport	Cap	Upper group	Lower group	Points	Log-odds
football	three substitutions	goalkeepers 98.8	outfield starters 70.8	28.1	3.53
football	five substitutions	goalkeepers 98.5	outfield starters 55.8	42.7	3.98
rugby league	ten interchanges	backs 92.0	forwards 31.7	60.3	3.21
rugby league	eight interchanges	backs 91.0	forwards 31.3	59.7	3.10
Australian rules	cap of 90	backs 2.9	interchange 0.0	2.9	–
Australian rules	cap of 75	backs 7.5	interchange 0.0	7.5	–
football	three to five	outfield starters −14.9	goalkeepers −0.3	move with the cap	
rugby league	ten to eight	backs −1.1	forwards −0.4	move with the cap	
Australian rules	90 to 75	backs +4.6	interchange +0.0	move with the cap	
Australian rules	cap of 90, median time on ground	backs 87	interchange 76	11	–
Australian rules	cap of 75, median time on ground	backs 86	interchange 73	13	–

EC8 A cap on one player: cricket's bowling quota

Limited-overs cricket caps each bowler rather than the team, and it is not one of the nine sports. It is read here because its quota is the same kind of rule applied to one player. No bowler may bowl more than four overs in a T20 match or ten in a one-day match (International Cricket Council,

2025*a*,*b*, clause 13.9.1). A side that bowls its full overs therefore needs at least five bowlers. An over the umpire miscounts stands (Marylebone Cricket Club, 2026, Law 17.5.1), so a bowler can pass the cap in valid balls without passing it in overs. The records are Cricsheet's ball-by-ball archive (Cricsheet, 2026), read over by over and pooled by format and sex, with no league set against another. A full innings has every scheduled over recorded, a last over of at least six valid balls and fewer than ten wickets down. A miscounted over is any over but an innings' last with other than six valid balls. Sides use 6.1 bowlers on average in T20 matches, and 58 per cent of their bowlers bowl the whole quota. The records show 0.04 per cent of overs miscounted. A miscounted over took 44 bowlers past the cap in valid balls. The archive was first read on the day it was downloaded, before this paper's predictions were written, so none of these readings is pre-specified. Table EC20 gives every reading.

Table EC20: Cricket's bowling quota, pooled by format and sex. Full innings ran every scheduled over; five is the share of them with five bowlers, the fewest the cap allows; quota is the share of their bowlers who bowled the whole quota; miscounted overs are those, an innings' last apart, recorded with other than six valid balls; past the cap counts bowlers with more valid balls than six times the quota.

Format	Sex	Innings	Five (%)	Bowlers	Quota (%)	Overs	Miscounted	Past cap
50 overs	female	570	4.2	6.45	40.6	78,653	26	1
50 overs	male	2,606	6.1	6.34	45.5	356,391	167	7
T20	female	3,760	10.5	6.24	54.3	117,611	96	23
T20	male	11,791	15.3	6.10	59.3	372,973	82	13
50 overs	both	3,176	5.8	6.36	44.6	435,044	193	8
T20	both	15,551	14.2	6.13	58.1	490,584	178	36

EC9 The other elements of the decision

Rugby union has changed its laws for welfare on measured evidence (Raftery et al., 2021). The formula of the main text adds an exact ceiling to that evidence before a limit is set. Each football competition now chooses three or five substitutions, so a team can play under different caps in different competitions (The International Football Association Board, 2022). The rulebooks here exempt replacements for head injuries from their limits from different dates: rugby union, rugby league and Australian rules from 2014, football from 2020 at each competition's choice, and the Gaelic games from 2021. Each exempt replacement raises that match's effective limit by one. In 2009 a rugby union club faked a blood injury with a capsule in a European Cup quarter-final, and English club rugby then let the other team's doctor inspect such wounds (Kitson, 2009). An exemption is used honestly only

while the chance of being caught, times the penalty, outweighs the free change it gives (Becker, 1968). Players who have won together also win more often (Mukherjee et al., 2019), so every change can cost a team some of its shared experience. Teams already take longer over restarts when they lead late in a match (Siegle & Lames, 2012). Referees have lengthened or shortened added time in close games to favour the home team (Garicano et al., 2005). Other work has measured how substitutes perform (Chen & Chen, 2025). In Germany's top league five substitutions brought more goals while the chance of each result stayed the same (Dewenter et al., 2026).

EC10 The rotational recount

In a re-entry sport the natural count is rotations per named player. Table EC21 gives five competitions in three sports, a 41-fold range. The substitution rate factorizes as $s_u\, n_{\text{eff}}/n_f$, where the utilization s_u is the share of the limit a team uses, which separates the rule's scale from the coach's use of it. In every capped competition the count is utilization times the cap over the players named, by construction. The utilization sits in a band from 0.891 to 0.961 across five eras of capped sports. A pre-specified prediction that a capped re-entry sport uses one strict constant share of its cap failed (C42 in Table EC52). A pre-specified prediction then says that the next such era read, a competition already played but not yet read here, will land its utilization between 0.85 and 0.98. The recount is a census and carries no law.

Table EC21: Rotations per named player.

Competition	Cap	Rotations per named player
Australian rules, men	90	3.77
Australian rules, men	75	3.22
Rugby league, men	8	0.45
Australian rules, women	60	2.54
Rugby league, women	8	0.51
Ice hockey, men	none	18.41

EC11 The count in full

In plain words, the number of changes a team makes is narrow, and it sits near the cap. A Poisson count, the model of random events, spreads teams far too widely. A binomial count on the cap is too narrow in football and rugby union, where many teams use the whole cap and a tail uses few, and about right in basketball. A narrow count cut off at the cap fits much better.

A related family of counts, the Conway–Maxwell family, fits better still in rugby union and basketball, and in football the narrow count cut off at the cap fits as well. The paragraphs below compare every model on the same footing, test whether some teams always go to the cap whatever happens, and read the width league by league.

The model. The censored normal reads the count as a latent number cut off at the limit. Unlike a demand in inventory theory (Conrad, 1976; Nahmias, 1994), this latent number moves with the limit, as work expands to fill the time available in staffing (Hasija et al., 2010). So its centre and width describe the fitted distribution, and not an intention that holds whatever the cap. A model in this journal has written a sport's rules into a count before, for volleyball's set scores (Ntzoufras et al., 2021). The censored normal is an ordered probit with known thresholds at the half-integers. Its tested alternative, with a mass at the limit, is an ordered probit with an inflated top category (Harris & Zhao, 2007; Brooks et al., 2012). The likelihood factorizes. Whenever s_p is interior, the fitted share at the limit equals the observed share exactly, so μ and σ come from the bins below the limit alone. Under three substitutions those bins hold 15 per cent of team-matches. So the mass is identified only by what the normal's tail puts at the limit, a warning familiar from bunching estimators (Blomquist et al., 2021). Pooled over leagues, the mass is estimated at zero in football under three substitutions and in rugby union, at 0.073 in football under five, where it lowers the AIC by 2.9, and at 0.062 in basketball, where it lowers it by 15.0. The count reads only the matches whose events list a substitution for at least one side, the rule by which the held-out panel sets aside unrecorded matches (its clause CM01). So 39 matches under five substitutions, 30 of them Turkey's, and 48 under three are set aside. Kept in, they made zero changes the largest residual of the five-substitution count.

Narrower than Poisson. A limit forces a count below Poisson only when it lies within one change of the mean, because the variance-to-mean ratio is at most $c-\mu$. In football and rugby union it does, and the ceiling is 0.17 to 0.79. In basketball it does not, because the ceiling is 1.31, and the observed ratio is 0.17. Every bounded reading fills 13 to 41 per cent of the variance its limit allows. A binomial count on the cap, whose variance is $\mu(1-\mu/c)$, fills exactly $1/c$ of that ceiling, so it is the natural yardstick for a count with a ceiling (Table EC22). Against it football's counts are wider, with 1.4 and 1.2 times its variance under five and three substitutions, and so is rugby union's, with 1.8, while basketball's has 0.9 times it. The binomial

misplaces 11.6 per cent of football's teams under five substitutions and 17.5 per cent of rugby union's, more than the Poisson count does. In basketball it misplaces 4.5 per cent, against the Poisson count's 33.7, because the Poisson count ignores the seven places on the bench. So beyond what the cap forces, teams differ: many use the whole cap, 56 per cent of football's teams under five substitutions and 85 per cent under three, and a tail uses few. So basketball tests underdispersion, and so does baseball before the three-batter minimum of 2020 (Major League Baseball, 2020). Baseball's relief pitchers are bounded only by a roster, and their variance-to-mean ratio is 0.48 to 0.58 in every season from 2008 to 2019. Counts of regularly spaced events are narrower than Poisson (Winkelmann, 1995; Baker & Kharrat, 2018), which suits baseball's relief pitchers, who come one after another. So narrowness alone does not show that teams plan their changes. A pre-specified prediction also asked that the median ratio across seasons fall below one half in both sports where no limit binds, and it failed in baseball (C188 in Table EC52). In basketball the dressing rule of 12 players bounds the bench at seven, and 23 per cent of team-games reach it.

Table EC22: The count against its yardsticks. The binomial count on the cap c has the observed mean μ and a variance-to-mean ratio of $1-\mu/c$; the next column is the observed variance over the binomial's. The last three columns are the teams each model puts in the wrong count, in per cent. Baseball's relief pitchers have no cap, so no binomial and no share at the cap.

Reading	SD	Var/ mean	Binomial var/mean	Var over binomial	At cap (%)	Wrong: Poisson	Wrong: binomial	Wrong: normal
football, five substitutions	0.89	0.18	0.13	1.37	56	11.1	11.6	0.6
football, three substitutions	0.44	0.07	0.06	1.22	85	3.1	3.0	0.1
rugby union, eight-bench sides	1.14	0.18	0.10	1.83	56	11.7	17.5	4.2
basketball, bench players used	0.97	0.17	0.19	0.89	23	33.7	4.5	1.6
baseball, relief pitchers	1.25	0.52	–	–	–	17.7	–	2.8

The censored normal. Write Φ for the standard normal distribution function and c for the limit. The probability of k changes is

$$P(k) = (1 - s_p)\left[\Phi\left(\tfrac{k+1/2-\mu}{\sigma}\right) - \Phi\left(\tfrac{k-1/2-\mu}{\sigma}\right)\right] \quad (0 < k < c),$$

with the bin for zero taking the whole lower tail, $P(0) = (1 - s_p)\,\Phi((1/2 - \mu)/\sigma)$, and the limit taking the mass,

$$P(c) = s_p + (1 - s_p)\left[1 - \Phi\left(\tfrac{c-1/2-\mu}{\sigma}\right)\right].$$

The log-likelihood is $\sum_k N_k \log P(k)$, where N_k is the histogram's count at k. With $s_p = 0$ this is the censored normal of the main text, and with s_p free its tested alternative. Rounding a latent continuous variable to counts in this way is the idea of STAR models (Kowal & Canale, 2020). A rule can also set a count's law: in handball the restart rule decides whether scores are Poisson, binomial or Markov-binomial (Singh et al., 2023).

Every family, like for like. Table EC23 gives every fit behind Figure 3 of the main text, and Figure EC3 draws the censored normal and its best rival for each reading. Each family is censored at the limit, with its tail folded into the top bin, and is fitted again with a mass at the limit. The Conway–Maxwell–Poisson is also fitted with masses at zero and at the limit (Arora et al., 2021), and truncated to the support, the one fit renormalized there rather than censored. The binomial and the Conway–Maxwell–binomial live on the support already and are fitted plain and with the mass (Kadane, 2016). Baseball's relief pitchers have no limit in these data, so baseball has neither a mass nor a binomial. Pearson's statistic pools bins until each expects five team-matches. Set beside the censored CMP, the truncated one shows how much of a ranking the boundary's treatment carries. A family and its version with a mass are nested, the mass sitting on the boundary of its space, so the AIC difference between them is not a test. Wilson (2015) makes the same point for the Vuong test. The mass is tested by likelihood ratio instead, below. With like-for-like boundaries, the censored normal has a lower AIC than every Conway–Maxwell family in football, under both rules, and in baseball, where no limit binds; under five substitutions only its own version with a mass at the limit does better. In rugby union and basketball a Conway–Maxwell family fits better, by 21.7 in basketball and 415.9 in rugby union. A Poisson count misplaces 3 to 34 per cent of teams across the five readings. The best-fitting model of each reading misplaces at most 2.8 per cent of teams, and the censored normal at most 4.2. The Poisson count misplaces most in basketball, whose count of bench players used cannot exceed seven, so there it is a weak benchmark, and the binomial fits worse than the censored normal as well. The censored CMP's dispersion runs from 2.0 in baseball to 5.3 in basketball, always narrower than Poisson. The censored CMP gives the same range for the latent count, 1.08 to 1.97 changes, so the width does not rest on the normal. Under three substitutions it puts the latent count's mean at 3.93, above the limit of three. The football histogram of five substitutions holds 59,982 team-matches.

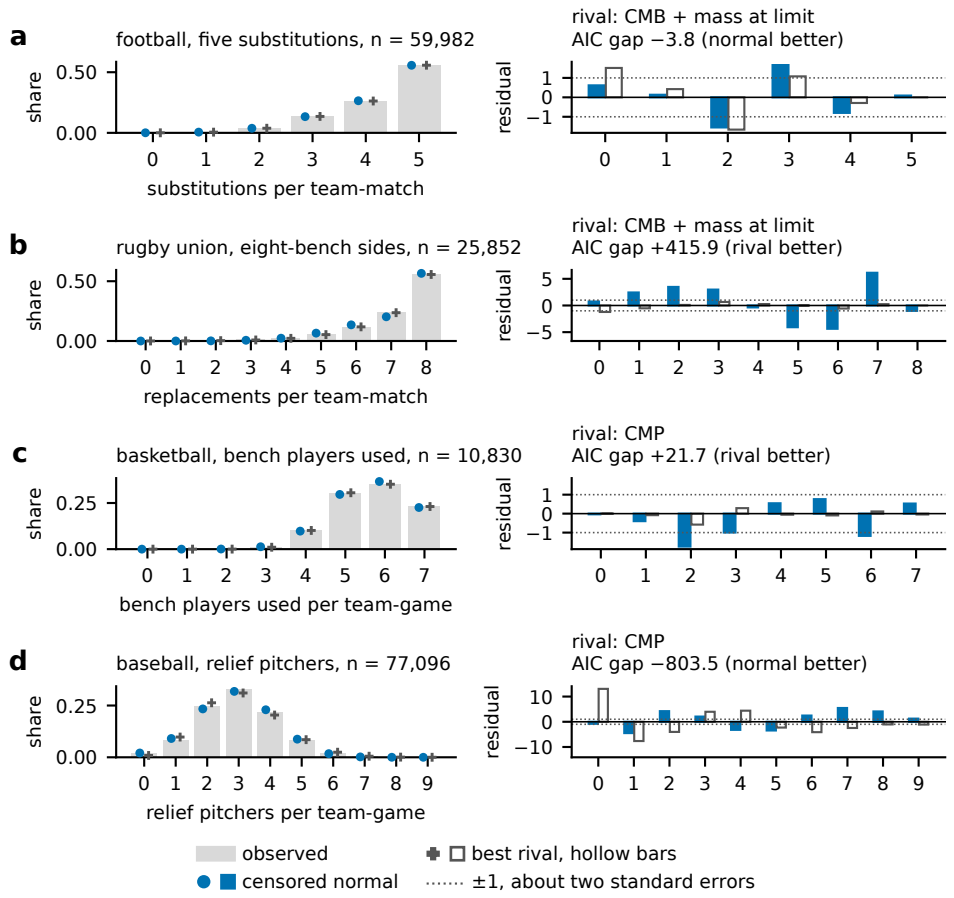


Figure EC3: In rugby union and basketball a Conway–Maxwell rival fits better than the censored normal; in football and in baseball the normal fits as well or better, and in baseball no limit binds. (a) Football. (b) Rugby union. (c) Basketball. (d) Baseball, shown to nine relief pitchers. Left: the observed shares against the censored normal and its best rival among the Conway–Maxwell and binomial families, each fitted with the limit treated the same way; CMP is the Conway–Maxwell–Poisson count, CMB the Conway–Maxwell–binomial, and a mass is extra probability at zero changes or at the limit. Right: the same fits as a suspended rootogram (Kleiber & Zeileis, 2016), the square root of the observed count less that of the expected one. Each names the rival and its AIC gap, the normal's AIC less the rival's, so a positive gap favours the rival. A bar beyond the dotted lines marks a count the model misses; the residual scales differ by row.

Alt text: Four rows, each a histogram on the left and residual bars on the right, the rival's hollow. In football both models miss a little at two and three changes, the rival also at zero. In rugby union the censored normal's residuals run far past the dotted lines and the rival's stay close to them. In basketball the normal passes the lines at two counts. In baseball both miss, the rival most at zero.

Table EC23: Every family, like for like. Every model is censored at the limit except the truncated CMP and the binomial and Conway–Maxwell–binomial (CMB) ones, which live on the support; a mass is at the limit, and the two masses are at zero and at the limit. k is the parameters, and ΔAIC each fit's AIC less its reading's best; Pearson's statistic has its degrees of freedom in brackets. The dispersion is σ/μ for the normal and the exponent ν for the Conway–Maxwell families, where $\nu > 1$ is narrower than Poisson or binomial.

Reading	Model	k	ΔAIC	Pearson	Dispersion	Mass
football, five substitutions	normal	2	2.9	24.8 (3)	0.27	–
football, five substitutions	normal + mass	3	0.0	20.4 (2)	0.27	0.073
football, five substitutions	Poisson	1	6265.7	5,989.3 (4)	–	–
football, five substitutions	Poisson + mass	2	6267.7	5,989.3 (3)	–	0.000
football, five substitutions	CMP, truncated	2	327.8	352.7 (3)	1.02	–
football, five substitutions	CMP	2	172.6	343.2 (3)	2.39	–
football, five substitutions	CMP + mass	3	174.6	343.2 (2)	2.39	0.000
football, five substitutions	CMP + two masses	4	73.4	101.0 (1)	2.45	0.000
football, five substitutions	binomial	1	4622.4	5,144.0 (3)	–	–
football, five substitutions	binomial + mass	2	24.7	58.9 (3)	–	0.364
football, five substitutions	CMB	2	569.3	587.3 (3)	0.33	–
football, five substitutions	CMB + mass	3	6.7	28.6 (2)	0.88	0.330
football, three substitutions	normal	2	0.0	1.7 (1)	0.30	–
football, three substitutions	normal + mass	3	2.0	not testable	0.30	0.000
football, three substitutions	Poisson	1	1413.0	1,284.2 (2)	–	–
football, three substitutions	Poisson + mass	2	1415.0	1,284.2 (1)	–	0.000
football, three substitutions	CMP, truncated	2	40.4	41.9 (1)	0.01	–
football, three substitutions	CMP	2	53.7	63.2 (1)	2.16	–
football, three substitutions	CMP + mass	3	55.7	not testable	2.16	0.000
football, three substitutions	CMP + two masses	4	2.3	not testable	2.39	0.000
football, three substitutions	binomial	1	1381.7	2,268.9 (2)	–	–
football, three substitutions	binomial + mass	2	14.5	17.8 (1)	–	0.632
football, three substitutions	CMB	2	40.5	41.7 (1)	0.00	–

Table EC23, continued

Reading	Model	k	ΔAIC	Pearson	Dispersion	Mass
football, three substitutions	CMB + mass	3	0.3	not testable	0.58	0.481
rugby union, eight-bench sides	normal	2	415.9	475.6 (4)	0.23	–
rugby union, eight-bench sides	normal + mass	3	417.9	475.6 (3)	0.23	0.000
rugby union, eight-bench sides	Poisson	1	2865.5	3,188.7 (7)	–	–
rugby union, eight-bench sides	Poisson + mass	2	2867.5	3,188.7 (6)	–	0.000
rugby union, eight-bench sides	CMP, truncated	2	39.2	47.2 (6)	0.19	–
rugby union, eight-bench sides	CMP	2	779.4	979.4 (4)	2.11	–
rugby union, eight-bench sides	CMP + mass	3	781.4	979.4 (3)	2.11	0.000
rugby union, eight-bench sides	CMP + two masses	4	766.1	930.7 (2)	2.13	0.000
rugby union, eight-bench sides	binomial	1	5847.6	15,100.5 (4)	–	–
rugby union, eight-bench sides	binomial + mass	2	1043.2	1,827.6 (4)	–	0.436
rugby union, eight-bench sides	CMB	2	59.1	62.4 (6)	0.04	–
rugby union, eight-bench sides	CMB + mass	3	0.0	8.2 (5)	0.18	0.147
basketball, bench players used	normal	2	21.7	19.5 (3)	0.18	–
basketball, bench players used	normal + mass	3	6.7	1.9 (1)	0.17	0.062
basketball, bench players used	Poisson	1	6437.1	6,146.6 (6)	–	–
basketball, bench players used	Poisson + mass	2	6439.1	6,146.6 (5)	–	0.000
basketball, bench players used	CMP, truncated	2	15.7	14.5 (2)	4.67	–
basketball, bench players used	CMP	2	0.0	0.3 (2)	5.28	–
basketball, bench players used	CMP + mass	3	2.0	0.3 (1)	5.28	0.000
basketball, bench players used	CMP + two masses	4	4.0	not testable	5.28	0.000
basketball, bench players used	binomial	1	241.7	202.2 (4)	–	–
basketball, bench players used	binomial + mass	2	243.7	202.2 (3)	–	0.000
basketball, bench players used	CMB	2	178.9	160.5 (3)	1.18	–
basketball, bench players used	CMB + mass	3	12.7	9.8 (2)	1.80	0.144
baseball, relief pitchers	normal	2	0.0	693.9 (6)	0.41	–
baseball, relief pitchers	normal + mass	2	0.0	693.9 (6)	0.41	0.000
baseball, relief pitchers	Poisson	1	13884.3	12,458.1 (11)	–	–
baseball, relief pitchers	CMP, truncated	2	803.5	1,595.2 (7)	2.04	–
baseball, relief pitchers	CMP	2	803.5	1,595.2 (7)	2.04	–

The mass at the limit, league by league. Table EC24 gives each football league's fit under five substitutions. Fitted league by league across 29 football leagues under five substitutions, s_p is zero in 10. Brazil's estimate is the largest, 0.52, and its test gives $p = 0.23$. A percentile bootstrap is inconsistent when a parameter sits on its boundary (Andrews, 2000), so the mass is tested by likelihood ratio against the 50:50 mixture of a point mass at zero and a chi-square on one degree of freedom (Self & Liang, 1987). Of the 29 leagues, four reject $s_p = 0$ at the 5 per cent level. If no league had a mass, at least as many rejections would come by chance with probability 0.055. Yet two of them, Belgium's and Scotland's, survive a Bonferroni correction across the 29 leagues. Both had seasons refused at the change of rule (Table EC3), so their panels may mix two rules, and the evidence for a mass is mixed. The relative dispersion σ/μ keeps its bootstrap interval, registered as C220, because it is interior. It runs from 0.21 to 0.31 between leagues, with $I^2 = 0.90$, so it is not one constant, and the pre-specified C152, that it was, failed. A pooled calibration across eight pairings of a sport and a limit does not identify the dispersion. It would put the mass at 0.56, above every league's estimate, and it is not used. Three pre-specified calibrations of a pooled share at the limit failed (C218, C219 and C229). Earlier measurements of how often teams use the whole limit exist: under-20 football's six (Ribeiro et al., 2020), the three of Germany's top league, the Bundesliga, in 88 per cent of matches (Meyer & Klatt, 2021), and rugby union's eight in 54 per cent of team-matches (Roberts et al., 2024). At the 2019 Rugby World Cup the tier-one teams used 97 per cent of their available replacements, and all eight in 81 per cent of cases (World Rugby, 2020). The Premier League's share at the limit in Table EC24 pools its five-substitution seasons, and season by season it rose from 0.37 in 2022–23 and 0.41 in 2023–24 to 0.51 in 2024–25, with 0.47 in 2025–26. An analyst writing in February 2024 put it at about one team in three, and the share of matches in which a team made fewer than three substitutions at one in ten (Caley, 2024), against 0.09 and 0.10 here in the two seasons before.

Table EC24: The censored normal by football league, five substitutions. The likelihood ratio tests $s_p = 0$ against the 50:50 mixture of χ^2_0 and χ^2_1; σ/μ carries its percentile bootstrap interval over club-seasons (C220). The last two columns are the observed and fitted shares of team-matches at the limit. The leagues carry the source's codes: A1 Austria; ARG1 Argentina; AUS1 Australia; BE1 Belgium; BRA1 Brazil; C1 Switzerland; DK1 Denmark; ES1 Spain; FR1 France; GB1 England; GR1 Greece; IT1 Italy; JAP1 Japan; KR1 Croatia; L1 Germany; MLS1 Major League Soccer; NL1 Netherlands; NO1 Norway; PL1 Poland; PO1 Portugal; RO1 Romania; RU1 Russia; SA1 Saudi Arabia; SC1 Scotland; SE1 Sweden; SER1 Serbia; TR1 Turkey; TS1 Czechia; UKR1 Ukraine.

League	Team-matches	s_p	LR	p	μ	σ/μ	95% bootstrap	Observed	Fitted
A1	712	0.239	1.74	0.093	4.11	0.231	0.198–0.255	0.4972	0.4972
ARG1	836	0.000	0.00	1.000	5.20	0.237	0.180–0.257	0.7153	0.7157
AUS1	650	0.000	0.00	1.000	4.62	0.229	0.185–0.256	0.5400	0.5435
BE1	3,274	0.174	19.79	0.000	3.70	0.263	0.249–0.279	0.3448	0.3448
BRA1	1,114	0.517	0.54	0.232	4.85	0.209	0.145–0.231	0.8232	0.8232
C1	860	0.000	0.00	1.000	5.16	0.242	0.160–0.289	0.7000	0.7019
DK1	1,584	0.035	0.02	0.446	4.66	0.233	0.207–0.244	0.5751	0.5751
ES1	4,560	0.000	0.00	1.000	5.10	0.249	0.236–0.261	0.6818	0.6828
FR1	4,114	0.000	0.00	1.000	4.54	0.276	0.263–0.284	0.5134	0.5134
GB1	3,040	0.000	0.00	1.000	4.29	0.308	0.285–0.326	0.4362	0.4365
GR1	2,182	0.243	3.14	0.038	4.26	0.226	0.192–0.247	0.5481	0.5481
IT1	4,560	0.190	0.81	0.184	4.84	0.244	0.230–0.255	0.6875	0.6875
JAP1	592	0.255	0.09	0.384	5.08	0.239	0.190–0.262	0.7635	0.7635
KR1	696	0.086	0.05	0.416	4.72	0.253	0.212–0.274	0.6106	0.6106
L1	3,672	0.000	0.00	1.000	4.83	0.244	0.231–0.260	0.6092	0.6110
MLS1	1,454	0.000	0.00	1.000	4.71	0.261	0.227–0.281	0.5667	0.5667
NL1	3,672	0.110	1.60	0.103	4.26	0.278	0.263–0.292	0.4820	0.4820
NO1	658	0.000	0.00	1.000	4.74	0.253	0.211–0.280	0.5760	0.5779
PL1	1,224	0.121	0.13	0.359	4.76	0.250	0.193–0.282	0.6364	0.6364
PO1	3,672	0.000	0.00	1.000	4.99	0.228	0.213–0.241	0.6661	0.6666
RO1	960	0.393	3.71	0.027	4.16	0.240	0.209–0.264	0.6156	0.6156
RU1	2,876	0.011	0.01	0.462	4.41	0.290	0.267–0.303	0.4781	0.4781
SA1	1,224	0.141	2.41	0.060	3.92	0.269	0.244–0.289	0.3913	0.3913
SC1	2,376	0.178	17.16	0.000	3.60	0.298	0.275–0.316	0.3430	0.3430
SE1	658	0.159	0.48	0.244	4.28	0.250	0.224–0.272	0.5106	0.5106
SER1	960	0.212	0.67	0.206	4.43	0.240	0.211–0.258	0.5854	0.5854
TR1	4,280	0.054	0.20	0.329	4.54	0.256	0.234–0.272	0.5409	0.5409
TS1	960	0.308	0.63	0.214	4.56	0.212	0.173–0.235	0.6708	0.6708
UKR1	2,562	0.032	0.06	0.401	4.46	0.298	0.273–0.316	0.5031	0.5031
pooled	59,982	0.073	4.86	0.014	4.57	0.271	–	0.5578	0.5578

Variance to mean. Table EC25 gives the variance-to-mean ratio of relief pitchers in every baseball season and of bench players used in every EuroLeague and EuroCup season. Every one is below one, in baseball after the three-batter minimum of 2020 as before it, a rule expected to change little (Clemens, 2019).

Table EC25: Variance to mean of relief pitchers per nine-inning team-game in baseball, and of bench players used in basketball's EuroLeague and EuroCup.

Sport	Season	Team-games	Mean	Variance/mean
baseball	2008	4,440	2.75	0.534
baseball	2009	4,470	2.80	0.542
baseball	2010	4,420	2.72	0.569
baseball	2011	4,384	2.68	0.574
baseball	2012	4,476	2.84	0.559
baseball	2013	4,376	2.76	0.542
baseball	2014	4,396	2.78	0.538
baseball	2015	4,440	2.94	0.581
baseball	2016	4,486	3.00	0.517
baseball	2017	4,496	3.10	0.484
baseball	2018	4,430	3.20	0.502
baseball	2019	4,442	3.23	0.505
baseball	2020	1,660	3.31	0.539
baseball	2021	4,426	3.29	0.475
baseball	2022	4,428	3.17	0.457
baseball	2023	4,458	3.12	0.433
baseball	2024	4,426	3.12	0.403
baseball	2025	4,442	3.16	0.421
basketball	EuroCup 2013	424	5.38	0.188
basketball	EuroCup 2014	331	5.44	0.166
basketball	EuroCup 2015	328	5.54	0.204
basketball	EuroCup 2017	237	5.71	0.150
basketball	EuroCup 2018	223	5.99	0.138
basketball	EuroCup 2019	241	5.58	0.172
basketball	EuroCup 2020	225	5.71	0.165
basketball	EuroCup 2021	219	5.49	0.218
basketball	EuroCup 2022	224	5.46	0.193
basketball	EuroCup 2023	237	5.70	0.174
basketball	EuroCup 2024	268	5.67	0.164
basketball	EuroLeague 2007	305	5.33	0.163
basketball	EuroLeague 2008	265	5.37	0.174
basketball	EuroLeague 2009	260	5.56	0.187
basketball	EuroLeague 2010	282	5.46	0.178
basketball	EuroLeague 2011	314	5.68	0.168
basketball	EuroLeague 2012	405	5.56	0.171
basketball	EuroLeague 2013	400	5.62	0.173
basketball	EuroLeague 2014	421	5.65	0.148
basketball	EuroLeague 2015	422	5.64	0.185
basketball	EuroLeague 2016	462	5.61	0.169
basketball	EuroLeague 2017	485	5.82	0.142
basketball	EuroLeague 2018	482	5.90	0.149
basketball	EuroLeague 2019	465	6.00	0.126
basketball	EuroLeague 2020	583	5.90	0.140
basketball	EuroLeague 2021	515	5.80	0.145
basketball	EuroLeague 2022	586	5.84	0.138
basketball	EuroLeague 2023	613	5.89	0.133
basketball	EuroLeague 2024	608	5.82	0.151

The widths. The censored normal's width is 1.24 changes in football under five substitutions and 1.14 under three, 1.76 in rugby union, 0.95 in basketball and 1.22 in baseball. It is the standard deviation of the nor-

mal before the cap cuts it, so it exceeds the observed spread wherever the cap binds. The observed standard deviations are 0.89 and 0.44 changes in football, 1.14 in rugby union, 0.97 in basketball and 1.25 in baseball, about one change or less. Table EC26 gives the censored normal's width in each reading, beside the censored CMP's latent count and the ceiling a limit puts on the variance. The pre-specified band was a factor of 2.5 around the median, which admits standard deviations from 0.49 to 3.05 changes, so it could hardly fail. Every width lies within a factor of 1.44 of their median, 1.22. Its uncertainty was tested too. Resampling team-matches, the five intervals for the width share no common value, so the pre-specified C246 failed and the band rests on five point estimates. Where no limit binds, the width can be read off the histogram directly, and baseball's observed standard deviation less Sheppard's correction, 1.21, is close to the censored normal's 1.22. The same basketball team-games, counted from the bench and from the whole roster, give relative dispersions in the ratio of their centres, median 1.90. The standard deviation is the same both ways, which is why the width is read in changes. Any count on $0, \ldots, c$ with mean μ has a variance of at most $(c - \mu)\mu$ (Bhatia & Davis, 2000), the ceiling the table gives as $c - \mu$ for the variance-to-mean ratio.

Table EC26: The width by reading. The censored normal's centre μ and width $\sigma = (\sigma/\mu)\,\mu$, in changes; the censored Conway–Maxwell–Poisson's latent count, its mean and standard deviation before the limit folds its tail; the observed variance to mean; and the ceiling $c - \mu$ that any count on $0, \ldots, c$ with mean μ puts on that ratio, with the per cent of the largest variance, $(c - \mu)\mu$, that the reading fills.

Reading	n	μ	σ	CMP mean	CMP s.d.	Var./mean	Ceiling	Filled (%)
football, five substitutions	59,982	4.57	1.24	4.79	1.46	0.18	0.67	27
football, three substitutions	58,328	3.69	1.14	3.93	1.40	0.07	0.17	41
rugby union, eight-bench sides	25,852	7.79	1.76	7.92	1.97	0.18	0.79	23
basketball, bench players used	10,830	5.63	0.95	5.75	1.08	0.17	1.31	13
baseball, relief pitchers	77,096	2.99	1.22	2.99	1.27	0.52	–	–

Australian rules' centre. Table EC27 gives the censored normal fitted to the AFL feed's sampled team-games under each cap. The fitted centres fell from 109.6 interchanges under a cap of 120 to 85.4 under 90 and 70.2 under 75. So the fitted centre fell with each cut of the cap, and each centre is 0.91 to 0.95 of its cap. Teams made on average 109.2, 85.1 and 70.0 interchanges a team-game under the three caps, and each fitted centre lies

just above the observed mean, as a count cut off at the cap requires. The recount of Table EC21 reads another sample of the same competition, so its rotations per named player do not reproduce these means.

Table EC27: Australian rules: the censored normal fitted to the interchanges of the AFL feed's sampled team-games under each cap. The relative width is the standard deviation over the centre.

Cap	Team-games	Mean	Centre	Centre over cap	Relative width	Variance to mean
120	826	109.2	109.6	0.91	0.075	0.85
90	828	85.1	85.4	0.95	0.063	0.29
75	864	70.0	70.2	0.94	0.058	0.28

Rugby union's centre by league. Table EC28 gives the rise in each league, from 2.10 to 2.41. The five agree, so the pooled error of 0.08 describes them all. The bench of eight came after the bench of seven, so the rise is confounded with time. A step at the change in the first two eight-bench seasons was pre-specified and held (C215). A drift within the eight-bench seasons smaller than that step was pre-specified too, and it failed (C216).

Table EC28: Rugby union's centre under a bench of seven and of eight, league by league, and the rise with its standard error. Cochran's $Q = 1.3$ on four degrees of freedom ($p = 0.86$).

League	Team-matches	Seven	Eight	Rise	s.e.
Premiership	5,595	5.14	7.27	2.13	0.19
Pro D2	9,693	5.73	7.88	2.15	0.17
Super Rugby	4,278	5.90	8.12	2.22	0.13
Top 14	8,169	5.67	7.77	2.10	0.27
URC	5,256	5.25	7.66	2.41	0.22

The women's game, refused. The women's appearance records hold 998 matches and 29,799 appearances in seven leagues. They name the player and the match and carry no club, so a team-match cannot be formed from them, and an unused substitute is not recorded at all. So the count's width and the window are refused for the women's game rather than forced.

EC12 The window in full

In plain words, suppose a team that makes n changes places them like n random moments in one stretch of the match. Then the average minute of each change falls on a straight line, and the spread of each change is fixed as well. The subsections below check the line's form, fit it with the right allowance for the teams that appear in several cells, and ask why the first

and the last changes leave it. Injuries move the first changes early, and the source's listing of added time at minute 90 piles the last ones up at that minute.

EC12.1 The form, and what it rules out

Uniform order statistics given the count are the order-statistic property of a point process. The processes that have it are mixed Poisson processes, up to a change of time, and mixed sample processes, in which a number is drawn first and the times are then placed uniformly (Feigin, 1979; Puri, 1982; Deffner & Haeusler, 1985). A mixed Poisson count is at least as wide as Poisson. But a count cut off at a cap can be narrower than Poisson whatever process made it, and the tests below reject the order-statistic property itself. So this argument does not single out a process, and the model of Section EC13 draws stoppages instead. A renewal process whose events are uniform order statistics is Poisson (Liberman, 1985), so it cannot also give the timing. Lorentzen (2017) wrote the positions $k/(n+1)$ down as a prescription for dividing playing time equally among children.

A uniform window fixes the spread of each change as well as its mean. Write the minute of the k-th of n changes as $X_{k:n} = a + b\,U_{k:n}$, with the window's opening a and width b varying between matches and $U_{k:n}$ the k-th of n uniform draws, independent of (a, b). Then

$$\begin{aligned} \operatorname{Var}(X_{k:n}) &= \operatorname{E}[b^2]\, v_{k:n} + \operatorname{Var}(a + b\,\mu_{k:n}) \;\geq\; (\operatorname{E}[b])^2\, v_{k:n}, \\ v_{k:n} &= \frac{k\,(n-k+1)}{(n+1)^2\,(n+2)}, \end{aligned} \tag{EC1}$$

where $\mu_{k:n} = k/(n+1)$. The window's width estimates $\operatorname{E}[b]$, so the ratio of a cell's observed spread to $b\sqrt{v_{k:n}}$ cannot fall below one when the draws are uniform and independent. Testing whether events fall at random by the spacing of order statistics is an old idea (Greenwood, 1946). Here the width is identified twice, by the slope of the means and by the spread inside each cell (Sargan, 1958). Service operations test arrivals the same way, and such tests reject when times are rounded, rates vary within an interval or periods are pooled (Kim & Whitt, 2014). Here those are the recorded minute, bundling and a window that varies between matches. Three things can lower the ratio: a density peaked inside the window (Mosteller, 1946), negative dependence between a team's own changes, and a mass at a boundary.

EC12.2 The cells

Table EC29 gives every cell. The required standard deviation is $b\sqrt{v_{k:n}}$ at the interior window's width, and the implied width is the cell's standard deviation over $\sqrt{v_{k:n}}$. Integer minutes are negligible, because Sheppard's correction changes no cell's standard deviation by more than 0.1 per cent. The cells' mean minutes run from 54.0 to 84.0 in football, over 15 cells and 257,536 changes in the five-substitution era. They are read on every five-substitution team-match of the 29 leagues with at least 500 of them, while predictions C201, C236 and C238 read the leagues with enough team-matches under each rule, as they name them. Spreading the changes evenly over the whole match would miss the cells by 25.4 minutes in football and 17.0 in rugby union, each cell weighted by its changes. A single change strays further from its cell's mean than the cells do from the line. The line misses a single change by 7.8 minutes on average in football, against 11.4 for the average change's minute and 7.8 for each cell's own mean, the least any placement by cell can miss. Teams that start changing early also have more time left for further changes, so the opening that falls with the count may run from timing to count as well as from count to timing. Losing teams change earlier (Gómez et al., 2016), and the registered model does not read the score (Section EC13.4 reads it). In rugby union they run from 38.5 to 70.8 minutes of 80, over 33 cells and 186,145 changes. One line through every cell (Figure EC4a) leaves residuals of 1.7 and 2.1 minutes in root mean square, from −2.6 to 2.7 in football and −6.3 to 1.9 in rugby union.

Table EC29: The cells. Mean, standard error and standard deviation are in minutes; the required standard deviation is $b\sqrt{v_{k:n}}$ at the interior window's width, the ratio the standard deviation over it, the bundled ratio what bundling alone predicts, and the implied width $\mathrm{sd}/\sqrt{v_{k:n}}$ in minutes. The last two columns are the shares of a cell booked at full time and at half-time.

Sport	k	n	Changes	Mean	s.e.	s.d.	Required	Ratio	Bundled	Width	Full (%)	Half (%)
football	1	1	365	71.8	0.73	14.0	12.6	1.11	1.00	48.5	7.1	6.3
football	1	2	2,125	66.0	0.29	13.6	10.3	1.32	1.08	57.5	1.0	9.6
football	2	2	2,125	79.2	0.20	9.1	10.3	0.88	1.08	38.6	13.3	1.4
football	1	3	8,316	60.8	0.16	14.1	8.5	1.67	1.13	73.0	0.2	13.3
football	2	3	8,316	73.2	0.11	10.0	9.8	1.02	1.13	44.7	2.4	3.2
football	3	3	8,316	82.6	0.08	7.4	8.5	0.87	1.13	38.0	21.2	0.3
football	1	4	15,677	57.5	0.11	14.2	7.1	2.00	1.23	87.2	0.0	18.3
football	2	4	15,677	68.4	0.08	10.2	8.7	1.17	1.21	51.1	0.5	6.5
football	3	4	15,677	77.0	0.07	8.8	8.7	1.01	1.22	44.2	6.1	1.1
football	4	4	15,677	84.0	0.05	6.1	7.1	0.86	1.23	37.5	23.9	0.1
football	1	5	33,053	54.0	0.08	13.9	6.2	2.25	1.33	98.5	0.0	30.5
football	2	5	33,053	63.0	0.06	10.5	7.8	1.35	1.29	58.9	0.1	16.3
football	3	5	33,053	70.2	0.05	9.7	8.3	1.17	1.26	51.3	1.0	4.7
football	4	5	33,053	77.6	0.05	8.2	7.8	1.06	1.30	46.3	6.0	0.4
football	5	5	33,053	83.4	0.03	5.5	6.2	0.90	1.33	39.4	16.3	0.0

Table EC29, continued

Sport	k	n	Changes	Mean	s.e.	s.d.	Required	Ratio	Bundled	Width	Full (%)	Half (%)
union	1	3	241	47.0	0.98	15.2	6.0	2.53	1.11	78.3	0.0	–
union	2	3	241	59.2	0.62	9.6	6.9	1.39	1.10	43.1	0.0	–
union	3	3	241	68.0	0.46	7.1	6.0	1.19	1.11	36.8	0.0	–
union	1	4	588	44.8	0.59	14.4	5.1	2.84	1.17	88.0	0.0	–
union	2	4	588	55.7	0.36	8.8	6.2	1.43	1.14	44.2	0.0	–
union	3	4	588	62.4	0.31	7.6	6.2	1.23	1.16	38.2	0.0	–
union	4	4	588	69.0	0.26	6.4	5.1	1.26	1.17	39.2	0.0	–
union	1	5	1,382	43.6	0.37	13.7	4.4	3.13	1.23	97.1	0.0	–
union	2	5	1,382	53.9	0.23	8.4	5.5	1.53	1.19	47.3	0.0	–
union	3	5	1,382	59.7	0.19	7.1	5.9	1.22	1.19	37.8	0.0	–
union	4	5	1,382	64.9	0.18	6.6	5.5	1.20	1.20	37.1	0.0	–
union	5	5	1,382	70.2	0.16	5.9	4.4	1.35	1.23	41.9	0.1	–
union	1	6	2,995	41.5	0.26	14.0	3.8	3.66	1.26	113.3	0.0	–
union	2	6	2,995	51.4	0.15	8.3	4.9	1.68	1.22	52.0	0.0	–
union	3	6	2,995	56.6	0.13	7.0	5.4	1.29	1.22	39.9	0.0	–
union	4	6	2,995	61.0	0.12	6.6	5.4	1.21	1.22	37.4	0.0	–
union	5	6	2,995	65.6	0.11	6.2	4.9	1.26	1.24	39.0	0.0	–
union	6	6	2,995	70.8	0.10	5.5	3.8	1.44	1.26	44.7	0.2	–
union	1	7	6,162	39.6	0.18	13.9	3.4	4.07	1.33	126.1	0.0	–
union	2	7	6,162	49.5	0.11	8.3	4.5	1.85	1.28	57.2	0.0	–
union	3	7	6,162	54.3	0.09	6.7	5.0	1.34	1.26	41.5	0.0	–
union	4	7	6,162	58.0	0.08	6.5	5.2	1.26	1.26	39.1	0.0	–
union	5	7	6,162	61.8	0.08	6.5	5.0	1.30	1.27	40.2	0.0	–
union	6	7	6,162	65.9	0.08	6.2	4.5	1.39	1.29	43.1	0.0	–
union	7	7	6,162	70.6	0.07	5.7	3.4	1.66	1.33	51.6	0.4	–
union	1	8	14,382	38.5	0.12	14.0	3.1	4.53	1.42	140.5	0.0	–
union	2	8	14,382	48.2	0.07	8.2	4.1	2.02	1.35	62.7	0.0	–
union	3	8	14,382	52.5	0.05	6.5	4.6	1.41	1.32	43.8	0.0	–
union	4	8	14,382	55.7	0.05	6.4	4.9	1.31	1.30	40.6	0.0	–
union	5	8	14,382	58.9	0.05	6.4	4.9	1.32	1.31	40.8	0.0	–
union	6	8	14,382	62.1	0.05	6.4	4.6	1.39	1.33	43.1	0.0	–
union	7	8	14,382	65.7	0.05	6.4	4.1	1.56	1.36	48.4	0.0	–
union	8	8	14,382	70.1	0.05	6.1	3.1	1.97	1.42	61.0	0.4	–

EC12.3 The lines, with the covariance

Within one count the cells $(1, n), \dots, (n, n)$ average the same teams. So the covariance of two cells' means is the covariance of the teams' k-th and k'-th minutes over the teams, divided by their number. Cells of different counts share no team. Every line is fitted by generalized least squares with that block covariance, and its χ^2 is $r^\top V^{-1} r$ for residuals r and covariance V. Neighbouring cells' errors are correlated by 0.48 to 0.66 in football and 0.49 to 0.83 in rugby union. Against the covariance the one line leaves $\chi^2 = 5{,}519$ on 13 degrees of freedom in football and 8,256 on 31 in rugby union, so it is a summary, not a law. Table EC30 gives each reading. For uniform order statistics the weights of the best linear estimate of location and scale fall on the first and last order statistics alone (Lloyd, 1952). The interior cells then only check the window. The end cells are contaminated,

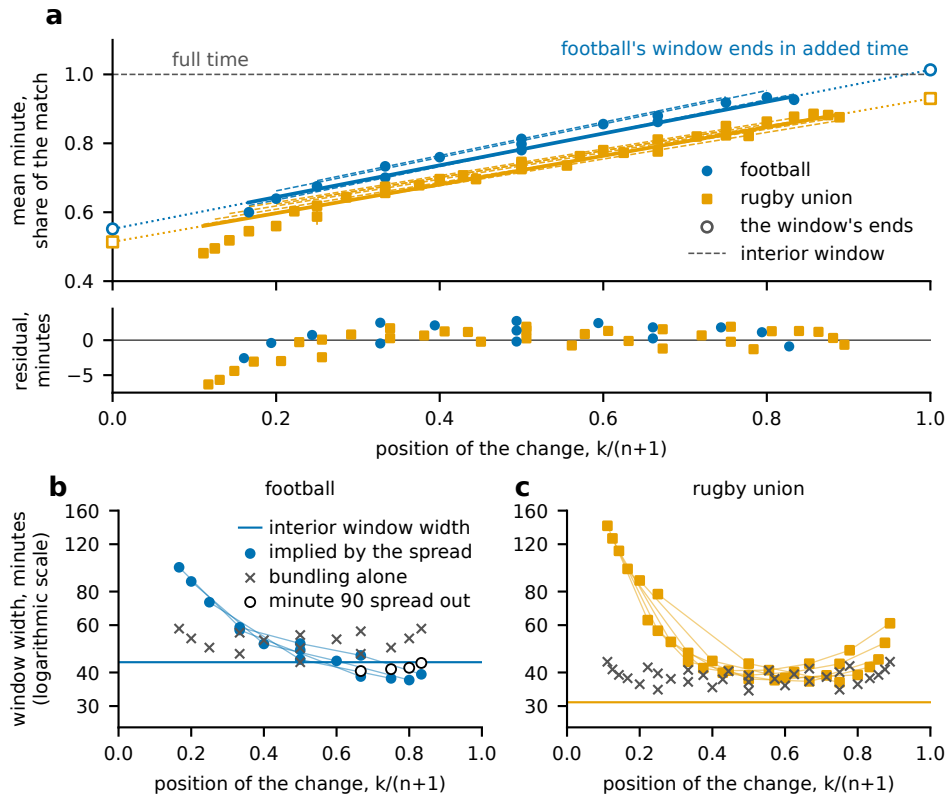


Figure EC4: The line summarizes the cells' averages, and first changes come earlier than it says. (a) The mean minute of the k-th of n changes, as a share of the match, against $k/(n+1)$; the 95 per cent intervals are drawn, and all but a few of the early cells' are smaller than the markers. The thick line is the one line fitted to every cell, dotted where it is carried to the window's ends at 0 and 1 (open markers), and football's window ends in added time, past full time. The dashed lines are the interior window, an opening for each n with one width, and the strip below gives each cell's residual from the thick line in minutes. (b, c) The width each cell's spread implies, on a logarithmic axis, against the interior window's width, the horizontal line, whose 95 per cent band is no wider than the line; thin lines join the cells of one n, and crosses are what bundling alone predicts with a fixed window. In football, open markers are the last-change cells with the changes listed at minute 90 spread over five minutes.

Alt text: Panel a: two rising lines of points, football above rugby union, with dashed lines close to the points of each count; the residuals are within a few minutes except rugby union's earliest cells, several minutes early. Panels b and c: implied widths on a logarithmic scale, falling steeply from the first change and then flattening, near the interior window's width in football and above it in rugby union, where they rise again at the last changes.

so the window is estimated on the interior with an opening for each count and one width (Table EC31), and the ends are read against it (Table EC32). Football's interior is rejected at the 5 per cent level, $\chi^2 = 7.6$ on two degrees of freedom ($p = 0.022$). Rugby union's interior fits less well, $\chi^2 = 661$ on 14, with openings falling from 43.7 to 41.8 minutes and a width of 31.0. Against the interior window, football's first and last changes come 1.4 to 2.0 minutes early, and rugby union's first changes 4.4 to 6.8 minutes early. One club's first, second and third changes have been timed at 59, 71 and 77 minutes under the three-substitution rule (Hills et al., 2019), against 56.7, 70.1 and 81.3 across the three-substitution seasons here. Each end residual's variance comes from the same covariance, because an end cell shares its teams with the interior cells of its count. Jointly the ends reject the interior window, $\chi^2 = 3{,}348$ on six degrees of freedom in football and 7,677 on 12 in rugby union. Cells with different (k, n) can share one position. At such positions the cells lie 0.018 to 0.034 of the match apart against a worst single-cell error of 0.012, so the count n carries information of its own. Fitted on Blom's expected normal order statistics instead, the cells give $\chi^2 = 10{,}483$ in football against 5,519 for the uniform line, and 6,218 in rugby union against 8,256. So the means alone cannot choose between a uniform window and a bell-shaped spread of changes over the match (Figure EC5). A mixture of two normal curves has been fitted to the minutes of second-half substitutions pooled over the count (Sun et al., 2024), and the cells here keep each count apart.

Table EC30: The window's lines, each fitted by generalized least squares on the cells' mean minutes with the covariance the teams give: within one count the cells average the same teams, and cells of different counts share none. The interior is the cells from the second change to the one before last. The end rows test the first and last cells against the interior window, with the variance the shared teams give their residuals.

Sport	Reading	Parameters	Cells	χ^2	d.f.
football	one line, every cell	2	15	5,519	13
football	an opening for each n, one width	6	15	3,365	9
football	a line for each n	8	14	3,339	6
football	one line, interior	2	6	1,239	4
football	an opening for each n, one width, interior	4	6	7.6	2
football	the first changes against the interior	0	3	864	3
football	the last changes against the interior	0	3	1,294	3
football	both ends against the interior	0	6	3,348	6
football	one line on Blom's normal positions	2	15	10,483	13
union	one line, every cell	2	33	8,256	31
union	an opening for each n, one width	7	33	7,866	26
union	a line for each n	12	33	7,716	21
union	one line, interior	2	21	1,037	19
union	an opening for each n, one width, interior	7	21	661	14

Table EC30, continued

Sport	Reading	Parameters	Cells	χ^2	d.f.
union	the first changes against the interior	0	6	7,294	6
union	the last changes against the interior	0	6	639	6
union	both ends against the interior	0	12	7,677	12
union	one line on Blom's normal positions	2	33	6,218	31

Table EC31: The interior window by count: where it opens and closes, in minutes, with one width for each sport, each with its standard error.

Sport	n	Opens	s.e.	Closes	Width	s.e.
football	3	51.3	0.13	95.0	43.7	0.14
football	4	50.8	0.10	94.5	43.7	0.14
football	5	48.5	0.09	92.1	43.7	0.14
union	3	43.7	0.62	74.7	31.0	0.09
union	4	43.7	0.31	74.7	31.0	0.09
union	5	44.1	0.17	75.1	31.0	0.09
union	6	43.3	0.12	74.2	31.0	0.09
union	7	42.5	0.09	73.5	31.0	0.09
union	8	41.8	0.07	72.7	31.0	0.09

Table EC32: The end cells against the interior window: the observed mean minute, the minute the interior window puts it at, the residual in minutes and its z.

Sport	k	n	Observed	Interior	Residual	z
football	1	3	60.8	62.2	−1.5	−10.5
football	3	3	82.6	84.1	−1.4	−14.8
football	1	4	57.5	59.6	−2.0	−19.1
football	4	4	84.0	85.8	−1.8	−26.7
football	1	5	54.0	55.7	−1.7	−23.7
football	5	5	83.4	84.9	−1.5	−31.4
union	1	3	47.0	51.4	−4.4	−5.3
union	3	3	68.0	66.9	1.1	1.9
union	1	4	44.8	49.9	−5.1	−9.4
union	4	4	69.0	68.5	0.6	2.4
union	1	5	43.6	49.3	−5.7	−16.4
union	5	5	70.2	69.9	0.3	1.9
union	1	6	41.5	47.7	−6.2	−26.3
union	6	6	70.8	69.8	1.0	10.9
union	1	7	39.6	46.4	−6.8	−41.7
union	7	7	70.6	69.6	0.9	15.0
union	1	8	38.5	45.2	−6.7	−64.1
union	8	8	70.1	69.3	0.8	18.6

EC12.4 The spread

Figure EC4b and c give the width each cell's spread implies, $\mathrm{sd}/\sqrt{v_{k:n}}$, against the interior window's width. In football's five-change sequence the implied width falls from 98.5 to 39.4 minutes against 43.7, so football's ratio

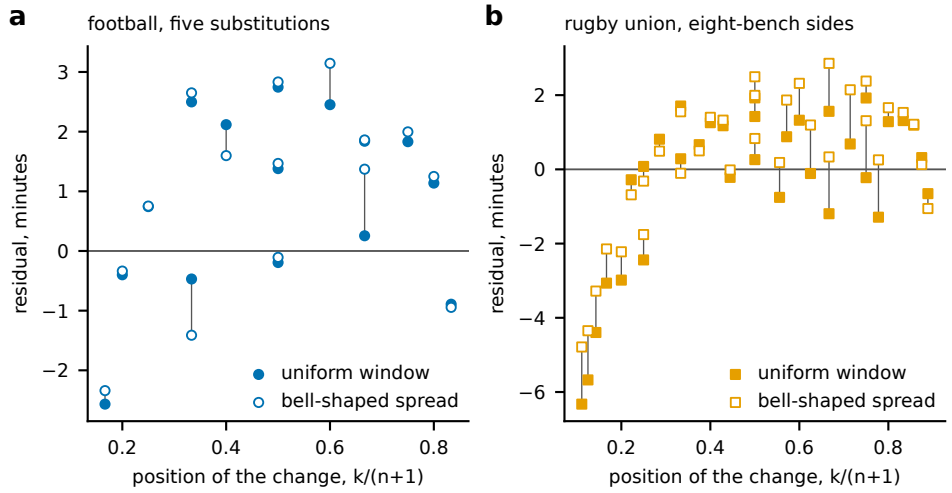


Figure EC5: The cells' averages alone cannot choose between a uniform window and a bell-shaped spread of changes over the match. Each cell's residual in minutes under the uniform window (circles) and under the bell-shaped spread, fitted on Blom's expected normal order statistics (squares), both by generalized least squares with the covariance the teams give. (a) Football, five substitutions. (b) Rugby union, eight-bench sides.

Alt text: Panels a and b: scatters of residuals in minutes against the position of the change. Under both shapes the residuals track each other: a few minutes either side of zero in football, and most negative at rugby union's earliest positions.

falls with the position of the change. In rugby union's eight-change sequence it falls from 140.5 to 40.6 and rises to 61.0 against 31.0, a U. The interior widths carry errors of 0.3 and 0.3 per cent. Over the 27 interior cells, 21 of them rugby union's, the ratio is 1.01 to 2.02, with a median of 1.31. That is above the floor, as a uniform window with some variation between matches requires. In football five of the six interior cells are no wider than bundling alone predicts, so bundling leaves no room for a window that varies between matches. In rugby union the cells from the third change on lie within 0.20 of bundling's prediction, and the second changes are wider, at 1.39 to 2.02. So the spread cannot separate a varying window from a fixed one whose draws are bundled, by the rule in football and by choice in rugby union.

EC12.5 Bundling

From 2020 a football team making five substitutions may use at most three stoppages during play, to reduce disruption to the match (The International Football Association Board, 2020). So a team that makes more changes than stoppages must make some together. Given each team's own use of half-time, the rule forces at least 35.0 per cent of consecutive changes into one stoppage, which is Proposition 2 of the main text summed over the teams. Holding each team's count and half-time changes fixed, a limit of one stoppage in play would force 92 per cent, two 62, four 14 and five 3.4. Table EC33 gives the share of consecutive changes the rule forces into one stoppage, the share recorded in one minute, the share at half-time and the share of used stoppages with two or more changes. Table EC34 reads each team's distinct in-play minutes as its stoppages. The record shows 37.4 per cent of consecutive changes in the same minute, 32 to 43 per cent by league, against 8 per cent in the three-substitution seasons, when no such rule applied. At the World Cups, the share of used substitution opportunities with two or more players rose from 3.6–5.2 per cent to 38.2 per cent when five substitutions came (Xiao & Zhang, 2024). On the same base the leagues here show 5.1 per cent under three substitutions and 35.5 per cent under five. A team whose n changes fall in $m < n$ stoppages draws m times, so its k-th change is the j-th of m uniform draws. Read on the teams' own groupings of changes into minutes, bundling alone predicts spread ratios of 1.00 to 1.33 in football and 1.10 to 1.42 in rugby union, with no variation in the window at all. Against the interior window's width, the j-th of three stoppages has a spread ratio of 1.69, 0.79 and 0.61, so once bundling is taken out the later stoppages fall below the floor. In rugby union, where no stoppage rule applies, 32 per cent of consecutive replacements share a recorded minute. At the 2019 Rugby World Cup 94 per cent of tier-one

replacements came in the second half (World Rugby, 2020).

Table EC33: Bundling. Shares of consecutive pairs of changes that the rule forces into one stoppage, that the record books in one minute and that the record books both at half-time, the share of teams that change at half-time, and the share of a team's used stoppages with two or more changes (Two+), all in per cent. The rule applies to football's five-substitution seasons only.

Sport	Rule	Teams	Pairs	Forced	Same	Half pair	Half use	Two+
football	five	59,171	198,000	35.0	37.4	3.4	27.3	35.5
football	three	56,740	106,311	–	7.6	1.6	19.0	5.1
union	eight	25,830	160,475	–	32.4	0.7	8.0	27.8

Table EC34: Football's stoppages. Each team's distinct in-play substitution minutes are read as its stoppages, and the j-th of m is set against $b\sqrt{v_{j:m}}$ at the interior window's width.

m	j	Teams	Mean	s.d.	Required	Ratio
1	1	2,015	73.3	10.6	12.6	0.84
2	1	13,886	66.1	10.8	10.3	1.05
2	2	13,886	81.1	6.8	10.3	0.66
3	1	37,020	59.3	14.3	8.5	1.69
3	2	37,020	73.9	7.7	9.8	0.79
3	3	37,020	84.4	5.2	8.5	0.61
4	1	6,263	58.5	14.3	7.1	2.00
4	2	6,263	69.5	8.2	8.7	0.94
4	3	6,263	77.3	7.0	8.7	0.80
4	4	6,263	84.5	4.8	7.1	0.68

The stoppage under Law 3.2. Read as the changes a team books in one minute, 15.9 per cent of the five-change teams and 8.2 per cent of the four-change teams show more stoppages in play than the three the Laws allow (Table EC35). The source marks no concussion substitution, the one change the Laws leave outside the opportunities, so most of the excess is the reading's: a stoppage that straddles a minute boundary books its changes in two minutes. Read with a stoppage that opens at minute t taking the changes booked at t or $t+1$, only 0.6 per cent of the five-change teams remain above three, and 0.8 per cent on the held-out matches, against 20.7 per cent minute by minute. Under that reading 42 per cent of consecutive changes share a stoppage, against the 35.0 per cent the rule forces, and joining the few teams still above three changes it little, to 42 per cent. So the minute-based share of 37.4 per cent understates how often teams combine changes. Table EC36 gives the stoppages' spreads under each reading.

Table EC35: The stoppage under Law 3.2. A stoppage is the changes a team books in one minute (the registered reading), or merged, a stoppage that opens at minute t taking the changes booked at t or $t+1$, or merged and capped, a team still above three having its two closest stoppages joined until three remain. The first rows give the teams that made four or five changes and the share of them, in per cent, with more in-play stoppages than the three the Laws allow. The last rows give, over consecutive pairs of changes, the share Proposition 2 of the main text forces into one stoppage and the shares in one stoppage and in one minute.

Reading	n	Teams	One minute	Merged	Capped
held out	4	338	13.3	0.0	0.0
held out	5	644	20.7	0.8	0.0
in sample	4	15,677	8.2	0.4	0.0
in sample	5	33,053	15.9	0.6	0.0
in sample	pairs	198,000	one minute: forced 35.0, one stoppage 37.4, one minute 37.4		
in sample	pairs	198,000	merged: forced 35.0, one stoppage 42.2, one minute 37.4		
in sample	pairs	198,000	merged and capped: forced 35.0, one stoppage 42.3, one minute 37.4		
held out	pairs	4,058	one minute: forced 33.3, one stoppage 33.3, one minute 33.3		
held out	pairs	4,058	merged: forced 33.3, one stoppage 39.4, one minute 33.3		
held out	pairs	4,058	merged and capped: forced 33.3, one stoppage 39.6, one minute 33.3		

Table EC36: Football's stoppages under each reading, in sample: the j-th of m stoppages' opening minute, its spread and the ratio of that spread to $b\sqrt{v_{j:m}}$ at the interior window's width.

Reading	m	j	Teams	Mean	SD	Ratio
one minute	1	1	2,016	73.3	10.6	0.84
one minute	2	1	13,911	66.1	10.8	1.05
one minute	2	2	13,911	81.1	6.8	0.66
one minute	3	1	37,134	59.2	14.3	1.69
one minute	3	2	37,134	73.9	7.7	0.79
one minute	3	3	37,134	84.4	5.2	0.61
one minute	4	1	6,489	58.2	14.6	2.05
one minute	4	2	6,489	69.4	8.3	0.95
one minute	4	3	6,489	77.3	7.0	0.80
one minute	4	4	6,489	84.6	4.8	0.68
merged	1	1	2,248	73.3	10.4	0.83
merged	2	1	16,378	66.1	10.5	1.02
merged	2	2	16,378	81.2	6.6	0.64
merged	3	1	40,790	58.7	14.4	1.70
merged	3	2	40,790	73.8	7.5	0.77
merged	3	3	40,790	84.5	5.0	0.59
merged	4	1	460	48.1	18.5	2.60
merged	4	2	460	67.0	9.9	1.13
merged	4	3	460	77.6	7.4	0.85
merged	4	4	460	86.1	4.9	0.68
merged and capped	1	1	2,248	73.3	10.4	0.83
merged and capped	2	1	16,378	66.1	10.5	1.02
merged and capped	2	2	16,378	81.2	6.6	0.64
merged and capped	3	1	41,250	58.6	14.5	1.71
merged and capped	3	2	41,250	73.8	7.5	0.77
merged and capped	3	3	41,250	84.5	5.0	0.60

EC12.6 The first change: injury and position

Table EC37 reads football's first changes with and without the changes the source flags as injuries. Both ends of the sequence depart, and these readings were made after the fact. The first change is too wide in both sports, up to 2.25 times the requirement in football and 4.53 in rugby union. In football 30 per cent of the first changes of five-change teams are booked at half-time. Without them the first change's spread is wider still, 2.50 times the requirement, so half-time does not explain the excess. The source flags 16 per cent of those first changes as injuries, and without them the ratio falls from 2.25 to 1.75. No interior cell moves by more than 0.03. Most changes carry no reason at all, so the flag marks a floor on the changes injury forces, and a first change it misses still widens the cell. Table EC38 reads rugby union's replacements by the position of the starter each one replaced. A replacement takes the position of the starters replaced at its entry minute, forward for shirts 1 to 8 and back for 9 to 15, because the lineups list the starting fifteen in shirt order. A side is kept when every replacement matches starters of one kind, 12,029 sides. International replacements have been timed by position before, backs latest and forwards in two earlier peaks (Lacome et al., 2016). Read on its own window, each position's first change is still 2.06 to 3.37 times its requirement among forwards and 2.00 to 2.90 among backs, against 2.40 to 4.30 for all replacements on the same sides.

Table EC37: The first change and the changes the source flags as injuries, football under five substitutions: the per cent flagged, and the spread ratio with every change, without the flagged changes, and on the teams that made none.

n	Changes	Flagged (%)	Ratio	Without flagged	Teams without	Ratio
1	365	9.9	1.11	0.95	329	0.95
2	2,125	9.2	1.32	1.11	1,811	1.11
3	8,316	14.0	1.67	1.30	6,247	1.31
4	15,677	15.9	2.00	1.51	11,148	1.51
5	33,053	16.1	2.25	1.75	22,833	1.75

Table EC38: Rugby union's replacements by the position of the starter replaced, forwards (shirts 1 to 8) and backs (9 to 15), on the sides whose every replacement has one. Each position's window is its own interior fit, and the ratios are the first change's, the interior's median and the last change's.

Position	n	Sides	Width	First	Interior	Last
all	3	219	33.0	2.40	1.32	1.12
all	4	476	33.0	2.69	1.22	1.18
all	5	991	33.0	2.95	1.11	1.20
all	6	1,859	33.0	3.46	1.17	1.31
all	7	3,058	33.0	3.86	1.23	1.49

Table EC38, continued

Position	n	Sides	Width	First	Interior	Last
all	8	5,426	33.0	4.30	1.28	1.73
back	2	4,445	34.6	2.00	–	1.13
back	3	4,879	34.6	2.36	1.23	1.02
back	4	103	34.6	2.90	1.55	1.09
forward	2	283	27.6	2.06	–	1.52
forward	3	814	27.6	2.47	1.42	1.44
forward	4	2,279	27.6	2.91	1.42	1.58
forward	5	6,612	27.6	3.29	1.47	1.82
forward	6	2,001	27.6	3.37	1.45	2.12

EC12.7 The 90th minute

The source books every change in added time at the 90th minute. Football's last-change cells sit below the floor, at 0.86 to 0.90, and up to 24 per cent of a last-change cell is booked at the 90th minute. Booking an unknown minute at the end of the half is coarsened data in the sense of Heitjan & Rubin (1991). Spreading the booked changes over a stoppage imputes their minutes, and the imputation assumes the data are coarsened at random. In Major League Soccer (MLS), North America's top football league, capping each half at 45 minutes cuts the minutes played by about 8 per cent in 2013 to 2017 and by over 13 per cent in 2024 (McKinley, 2024). Let a cell carry a share q of its changes at the full match T, and spread them uniformly over a stoppage of S minutes. The spread part has mean $T + S/2$ and variance $S^2/12$, and the rest has mean $\bar{t}_0$ and variance σ_0^2. The observed mean is $\bar{t} = (1 - q)\bar{t}_0 + qT$ and the new mean $\bar{t} + qS/2$. The law of total variance then gives the new variance, $\mathrm{Var} + qS(T - \bar{t}) + qS^2/3 - (qS)^2/4$, where Var is the observed variance. In football's four last-change cells the booked mass pushes the spread ratio below its floor. Table EC39 reads each against the interior window's width, which no end cell moves. Over five minutes the four reach 0.93 to 0.99 of the requirement, and four stay below it. Over eight minutes two stay below, at 0.96 to 1.06. The stoppage each cell needs to reach the floor runs from 5.3 to 12.0 minutes. The stoppage is carried over from measurements rather than read here. In six European leagues the second half adds 4.6 minutes on average (Kocsoy, 2025, Table 2). Added time has also been timed from broadcast footage at the 2022 World Cup and Euro 2024 (Butler et al., 2026). The stricter added time of 2022 and 2023 lengthened matches in one league by 2.9 minutes (Turan et al., 2026). Referees award about 2.2 minutes less added time than a match loses (Li et al., 2026).

Table EC39: Football's last-change cells with the changes booked at minute 90 spread over a stoppage, in their spread and their mean, read against the interior window's width, and the stoppage each cell needs to reach the floor.

n	At minute 90 (%)	5 minutes	8 minutes	Needed (minutes)
2	13.3	0.93	0.96	12.0
3	21.2	0.94	0.99	8.4
4	23.9	0.95	1.02	7.1
5	16.3	0.99	1.06	5.3

EC13 The model of stoppages

The order-statistic line describes the average minute of each change, but drawn one by one its uniform moments put almost no two changes in one minute. The model of the main text draws stoppages instead, in three steps read from the five-substitution team-matches of the 29 leagues with at least 500 of them. First, a team's number of changes comes from the censored normal of Section EC11. Second, a team with n changes draws its grouping, the number made at half-time and the sizes of its stoppages in play in time order, from the groupings of the panel's teams with n changes (Table EC40). Third, a team with m stoppages in play draws m minutes independently from the minutes at which the panel's teams with m stoppages stopped play, redraws any set with a repeated minute, and sorts them (Table EC41). Teams with more than four stoppages in play draw from the pool for four. Each change takes its stoppage's minute, and the k-th change is the k-th smallest of the team's change minutes. Beyond these frequency tables, the model's only fitted part is the censored normal of its count. The provider books a change at half-time at minute 46, and half-time is not one of the three stoppages the Laws allow. Drawn with the counts of the panel's own teams, the model misses the cells' mean minutes by 0.6 minutes, each cell weighted by its changes, against 1.1 for the line, and by 0.7 against 1.4 unweighted. It puts 37 per cent of consecutive changes in one minute, against 37 per cent in the records and 6 per cent for independent uniform moments on the window. On these seasons it does so by construction, because it draws each team's grouping from them. In the records 35 per cent of the stoppages a team uses hold two or more changes. Its predicted spreads run a little wide, because a team's real stoppages are spaced a little more evenly than independent draws (Table EC42). It misses a single change by 7.8 minutes on average, about as much as the line, because a single change's own spread dominates both.

The held-out matches are the 2019–20 domestic-league matches played under the temporary amendment of 8 May to 31 July 2020, which allowed

five substitutions in at most three opportunities during play. No stage of this paper had read them, because its three-substitution panels leave out the amendment window and its count and timing panels leave out the 2019–20 season. One stage of an earlier analysis of starters' exit minutes had pooled their exits with those of later seasons, and it read neither their counts by team-match, nor their cells, nor their stoppages. The model was frozen, and six predictions with their pass marks were committed to the register, before any held-out record was read (Table EC43). The predictions were committed on 15 October 2026, about half an hour before the matches were read, and the software refuses to read them unless the register and the frozen tables are committed and unchanged. The register prints the software's and the result files' names as [stage] and [deposit], as the main register does. The held-out panel holds 635 matches in eight leagues, 1,270 team-matches, after four matches with no substitution recorded for either side were set aside, and every league adopted the amendment. All six predictions held. The marks were set after the model had been read on the fitted seasons, as round numbers looser than its errors there, 2.5 minutes against 0.6 for the cells and 5 per cent against 0.6 for the count. They asked whether the model carried to new matches with a tolerable loss, and only CM03 set it against a rival, the line. The censored normal misplaced 4.9 per cent of the held-out teams, against a mark of 5, and the Poisson count 13 per cent. On the 14 cells with at least 50 changes the model missed the mean minutes by 0.9 minutes, against a mark of 2.5 and 1.5 for the line, each cell weighted by its changes. The median ratio of observed to predicted spread was 0.88, and 93 per cent of the cells lay in the wide band. The model put 37 per cent of consecutive changes in one minute against 33 per cent observed, and 34 per cent of stoppages above one change against 31, both within the mark of 5 points. It put 8.7 per cent of the changes at half-time against 9.5 per cent observed, within 3 points. Taking each change on its own, it missed by 7.7 minutes, against 7.8 for the line.

Table EC40: How the fitting panel's teams group their changes, by the number of changes n: the share of teams that make a change at half-time, and the share whose changes in play fall in m distinct minutes. The model draws a team's grouping from these frequencies, kept pattern by pattern with the sizes of the stoppages in time order.

n	Teams	Half-time (%)	$m = 0$	$m = 1$	$m = 2$	$m = 3$	$m \geq 4$
1	365	6.3	6.3	93.7	0.0	0.0	0.0
2	2,125	9.9	1.1	24.8	74.1	0.0	0.0
3	8,316	14.1	0.2	6.3	44.4	49.1	0.0
4	15,677	20.4	0.0	2.3	22.8	66.7	8.2
5	33,053	35.1	0.0	0.8	15.3	68.0	15.9

Table EC41: The minutes at which the fitting panel's teams stop play to make changes, by the number m of their stoppages in play, half-time left out. The model draws a team's m stoppage minutes from the pool for m.

m	Stoppages	Mean minute	Lower quartile	Median	Upper quartile	At minute 90 (%)
1	2,016	73.3	66	74	81	6.5
2	27,822	73.6	66	75	82	6.9
3	111,402	72.5	65	75	83	7.2
4 or more	27,587	72.4	65	74	82	4.9

Table EC42: The model of stoppages cell by cell, the k-th of n changes, in minutes: the observed and predicted mean and standard deviation on the fitting panel, and on the held-out matches of 2020 with their number of changes.

k	n	Mean	Model	SD	Model	Held out	Mean	Model	SD	Model
1	1	71.8	71.6	14.0	12.2	15	76.7	71.7	9.3	12.1
1	2	66.0	65.9	13.6	12.9	72	62.7	65.9	13.2	12.8
2	2	79.2	78.0	9.1	9.6	72	77.7	78.0	9.3	9.6
1	3	60.8	60.6	14.1	14.6	198	59.5	60.6	14.1	14.6
2	3	73.2	72.2	10.0	10.8	198	71.8	72.1	10.6	10.9
3	3	82.6	81.0	7.4	8.0	198	82.2	81.0	7.7	7.9
1	4	57.5	57.5	14.2	15.0	338	57.3	57.5	12.8	15.1
2	4	68.4	67.5	10.2	12.8	338	68.2	67.5	10.3	12.8
3	4	77.0	75.8	8.8	9.8	338	77.6	75.8	8.5	9.8
4	4	84.0	82.2	6.1	7.1	338	84.7	82.2	5.5	7.2
1	5	54.0	54.2	13.9	14.8	644	53.3	54.2	13.5	14.8
2	5	63.0	62.5	10.5	14.2	644	61.9	62.6	10.3	14.1
3	5	70.2	70.4	9.7	11.4	644	69.2	70.4	9.8	11.4
4	5	77.6	77.1	8.2	9.3	644	77.4	77.2	8.3	9.2
5	5	83.4	82.7	5.5	6.8	644	83.0	82.7	5.6	6.8

Table EC43: The model's predictions for the held-out matches, each quoted in full as registered, with its outcome and the reading it was judged on.

Id	Outcome	Reading	Prediction
CM01	held	normal 0.0492; Poisson 0.1317	On the held-out matches, the 2019-20 domestic-league matches of the provider's snapshot of 3 August 2026 played from 8 May to 31 July 2020 under IFAB's temporary amendment of 8 May 2020 (five substitutions, in at most three opportunities during play), in the leagues whose restart shows at least one side making more than three changes, with any match whose events list no substitution for either side set aside as unrecorded and counted, the censored normal that [stage] fitted to the five-substitution team-matches of the leagues with at least 500 of them, frozen in [deposit], places at most 5 per cent of the held-out team-matches in a count other than the observed one (half the summed absolute difference between the observed shares of 0 to 5 changes, more than five counted at five, and the frozen shares), and fewer than the Poisson count frozen beside it.
CM02	held	0.88 minutes on 14 cells	On the same held-out matches, for every cell (the k-th of n changes, averaged over the sides that made exactly n, for n from 1 to 5) holding at least 50 changes, the change model frozen in [deposit] predicts the cell's mean minute with a mean miss of at most 2.5 minutes, each cell weighted by its changes. The model draws a side's grouping of its n changes (how many at half-time, which the provider books at minute 46, and the sizes of its in-play stoppages in time order) from the fitting panel's frequencies for n [deposit]. It draws the side's m in-play stoppage minutes independently and without repeats from the minutes at which the fitting panel's sides with m in-play stoppages stopped play, the pool for four standing for four or more [deposit], and sorts them. The k-th change is the k-th smallest change minute. Each count is simulated 200,000 times with seed 333 [stage].
CM03	held	model 0.88; line 1.51	On the same cells, the change model's weighted mean miss is no larger than the miss of the order-statistic line that [stage] fitted to the fitting panel's cells (one line on uniform positions, by generalised least squares), frozen in [deposit]
CM04	held	median 0.884; within the wide band 0.929	On the same cells, the ratio of each cell's observed standard deviation to the change model's predicted standard deviation has a median between 0.85 and 1.15, and at least 80 per cent of the cells have a ratio between 0.75 and 4/3.
CM05	held	same minute 0.3332 against 0.3685; shared stoppages 0.3058 against 0.3445	On the held-out sides that made 1 to 5 changes, the share of consecutive pairs of a side's changes booked in one minute is within 5 percentage points of the change model's prediction for sides with the same counts, and so is the share of the sides' stoppages (each side's distinct change minutes, half-time included) that hold two or more changes.

Table EC43, continued

Id	Outcome	Reading	Prediction
CM06	held	0.0946 against 0.0873	On the same sides, the share of changes booked at half-time, minute 46, is within 3 percentage points of the change model's prediction for sides with the same counts.

EC13.1 Against the fitted seasons' own tables

Table EC44 sets the model against the simplest rival a simulator has, the fitted seasons' own tables applied to the held-out matches. Most of the fitted seasons were played after the held-out matches, so the tables are carried back as much as forward, and the predictions for 2026–27 will be the first test forward in time. Their cell means missed the held-out cells by 0.7 minutes, each cell weighted by its changes, against the model's 0.9, and by 0.9 against 1.0 unweighted. Every change placed at the average minute missed by 9.1. Their histogram misplaced 4.96 per cent of the held-out teams, against the censored normal's 4.92. Their share of consecutive changes in one minute, 37.4 per cent, is the model's by construction. A paired bootstrap over the held-out matches, both sides of a match kept together, gives each comparison its 95 per cent interval (Table EC45). The model's miss less the line's is −0.6 minutes (−0.8 to −0.4), and less the fitted seasons' own means' it is 0.19 (−0.01 to 0.39), and the two counts misplace as many teams. So on the held-out matches the model predicts the cells clearly better than the line and about as well as the tables it is built from, and the test shows that the restart resembled the seasons around it. What the model adds is a generator, a grouping table and four pools of minutes from which a simulator draws whole sequences of changes. It keeps the rule's part, how changes group into at most three stoppages during play, apart from the timing. Resampling each team's whole vector of stoppage minutes from the fitted teams with the same count and the same number of stoppages, with no parameter at all, missed the held-out cells by 0.5 minutes (Table EC46). So the model's independent draws give up the spacing between a team's own stoppages. Taking each change on its own, the fitted seasons' own cell means miss by 7.7 minutes, against the model's 7.7 and the line's 7.8.

Table EC44: The model of stoppages against its benchmarks, on the fitted seasons and on the held-out matches of 2020. The first four rows give how far each placement misses the cells' mean minutes, in minutes, each cell weighted by its changes (as clause CM02 reads) and unweighted; the held-out cells are those with at least 50 changes. The next rows give the teams each count misplaces and the share of consecutive changes in one minute, in per cent. The fitted seasons, most of them played after the held-out matches, give those matches their own cell means, histogram and share; Table EC45 gives the held-out comparisons' intervals. The count rows read the count registered before the held-out test, fitted on 60,060 team-matches; Figure 3 of the main text sets aside the 78 of them from matches with no recorded change, as clause CM01 does, so its shares differ slightly.

Placement	Fitted, weighted	Fitted, unweighted	Held out, weighted	Held out, unweighted
the model of stoppages	0.59	0.67	0.88	1.05
the order-statistic line	1.14	1.43	1.51	1.38
every change at the average minute	8.71	7.89	9.09	8.74
the fitted seasons' own cell means	–	–	0.69	0.91
count: the censored normal	0.57		4.92	
count: the Poisson count	10.92		13.17	
count: the fitted seasons' histogram	–		4.96	
one minute: the records	37.4		33.3	
one minute: the model	37.4		36.8	
one minute: random moments	6.3		6.3	
one minute: the fitted seasons' share	–		37.4	

Table EC45: The held-out comparisons with their 95 per cent intervals, from a paired bootstrap over the held-out matches (2,000 resamples of 635 matches, both sides of a match kept together). The cells are the 14 the test read, each weighted by its changes, in minutes; the counts give the teams misplaced, in per cent. The predictions are fixed and only the matches are resampled.

Quantity	Value	Low	High
cells: the registered model	0.88	0.69	1.25
cells: the order−statistic line	1.51	1.26	1.92
cells: the fitted seasons' own means	0.69	0.47	1.12
count: the frozen censored normal	4.92	2.62	7.90
count: the fitted seasons' histogram	4.96	2.73	7.87
cells: model less line	−0.62	−0.78	−0.44
cells: model less the fitted seasons' means	0.19	−0.01	0.39
count: normal less the fitted histogram	−0.04	−0.35	0.08

EC13.2 The count of changes and the stoppage minutes

The pools of stoppage minutes depend on m alone, but teams that make more changes open their window earlier (Table EC31). The model's largest errors sit there: it puts the last of three or four changes up to 1.8 minutes

early in sample and up to 2.4 on the held-out matches (Table EC42). Drawing the stoppage minutes from the teams with the same count and the same m cuts that error by 35 per cent in sample. Drawing each team's whole vector removes it in sample, so most of the error comes from drawing a team's stoppages independently, which spreads them more than real teams do. The model was also refitted on stoppages merged and capped under Law 3.2, each change booked at its stoppage's opening or a minute after, as the fitted teams booked theirs. It then misses the cells by 0.8 minutes in sample and 0.7 on the held-out matches, and it draws no team with more than three stoppages in play. Table EC46 gives every variant on both sets of matches.

Table EC46: The model of stoppages with each way of drawing the stoppage minutes, read after the held-out matches: the miss of the cells' mean minutes weighted by their changes and unweighted, the median ratio of observed to predicted spread and the cells inside the band of 0.75 to 4/3 (in per cent), the predicted share of consecutive changes in one minute (per cent) and the miss of each change on its own, in minutes. Every variant draws the count and the grouping as the registered model does.

Stoppage minutes	Reading	Weighted	Unweighted	Ratio	In band	One minute	Single
model of stoppages (the registered model, redrawn)	in sample	0.59	0.68	0.92	93	37.4	7.84
model of stoppages (the registered model, redrawn)	held out	0.87	1.02	0.88	93	36.8	7.74
draws from teams with the same count	in sample	0.63	0.55	0.90	100	37.3	7.84
draws from teams with the same count	held out	0.84	1.07	0.89	93	36.8	7.72
a whole vector from one such team	in sample	0.32	0.21	0.99	100	37.4	7.79
a whole vector from one such team	held out	0.51	0.80	0.96	100	36.9	7.67
the spacing model, fitted α	in sample	0.81	0.58	0.99	100	37.4	7.87
the spacing model, fitted α	held out	0.76	0.97	0.90	93	36.8	7.72
the spacing model, $\alpha = 1$	in sample	0.87	0.63	0.99	93	37.4	7.88
the spacing model, $\alpha = 1$	held out	0.79	0.97	0.89	86	36.8	7.72
independent draws, merged and capped stoppages	in sample	0.84	0.87	0.91	93	37.6	7.89
independent draws, merged and capped stoppages	held out	0.71	0.93	0.85	93	37.1	7.77

EC13.3 How evenly a team spaces its stoppages

Let a team's m stoppages in play fall in a window $[a, a+b]$, and let the $m+1$ spacings between the window's ends and the stoppages, as shares of b, follow a symmetric Dirichlet distribution with concentration α. Then the j-th stoppage has

$$\mathrm{E}[\tau_{j:m} \mid a, b] = a + b\,\frac{j}{m+1}, \qquad \mathrm{Var}(\tau_{j:m} \mid a, b) = \frac{b^2\, j\,(m+1-j)}{(m+1)^2\,\big((m+1)\alpha+1\big)}, \tag{EC2}$$

so the order-statistic line holds for every α (David & Nagaraja, 2003). At $\alpha = 1$ the variance is $b^2 v_{j:m}$ of equation (EC1), the uniform order statistics, and $\alpha > 1$ narrows every spread by the factor $\sqrt{(m+2)/((m+1)\alpha+1)}$. Independent gamma gaps with shape α, divided by their sum, are Dirichlet, so the gaps between a team's first and last stoppage, given their sum, read α as the shape of a gamma renewal process, the mechanism that narrows counts of regularly spaced events (Winkelmann, 1995; Baker & Kharrat, 2018). On a fixed window a renewal process's two end spacings follow other laws, so the line $a + b\,j/(m+1)$ is a property of the symmetric Dirichlet, not of a gamma renewal process. Among renewal processes only $\alpha = 1$ gives uniform order statistics (Liberman, 1985). The $m-1$ gaps between a team's first and last stoppage, as shares of their sum, are symmetric Dirichlet with $m-1$ parts whatever a, b and the two end spacings are. So α is estimated by maximum likelihood on the teams with three or more stoppages in play, with no window (Table EC47). Minutes are booked whole, so each is spread over the minute it names, and a booking at minute 90 over the stoppage of the 90th minute above. Football's 43,949 teams give $\alpha = 1.48$, with a standard error of 0.009. Read with minute 90 as its own minute, α is 1.43, and on the held-out matches it is 1.32 (0.05) on 961 teams. On stoppages merged under Law 3.2 it is 3.42, because merging removes the one-minute gaps a straddled stoppage leaves. Rugby union's 25,262 teams, under no limit on stoppages, give 1.58.

The booking itself lifts α above one. Two changes in one minute are one stoppage by definition, so the shortest gaps never appear, and the likelihood is driven by the shortest gaps. So $\alpha = 1$ is the wrong null, and the null is simulated through the same booking. Each team keeps its observed first and last stoppage, its other stoppages fall at random between them in distinct whole minutes, and each of 200 samples is spread and fitted as the records are. Given its first and last stoppage, a team's other stoppages are uniform between them when they fall at random, so the null keeps each team's own span. On football's fitted seasons the null gives 1.29 (95 per cent range 1.28 to 1.30), so most of the departure of 1.48 from one is the booking, and the

rest is evenness the records add. On the held-out matches it gives 1.29 (1.22 to 1.37), and 1.32 lies inside that range, a share 0.28 of the null lying at or above it. With minute 90 read as its own minute the null gives 1.24, on merged stoppages 1.71, and in rugby union 1.35, at most 1.36, each below its estimate. Figure EC6a draws each estimate beside its null. Read as merged stoppages, the estimate is twice its null, so the stoppages the Laws count are spread well apart. Against the null, the fitted α narrows the spread of three stoppages by a factor of 0.94. So teams space their stoppages a little more evenly than random draws on the fitted seasons and in rugby union, and as evenly on the held-out matches.

The symmetric Dirichlet gives every gap the same mean share, and the gaps are not equal. The first is the longest in every group of Table EC48 but one (Figure EC6b): 53.8 per cent of the span when a football team's three stoppages leave two gaps, and 41.3 and 35.3 per cent in rugby union with three and four gaps. In football's 325 teams with four gaps the second is the longest, 28.4 per cent against 27.1 for the first. So the line holds on average only, as its tests found, and a generator with symmetric spacing places the middle stoppages early. One generator draws a team's first and last stoppage together from the fitted teams with the same count and m, and places the stoppages between them at Dirichlet shares of that span. Its cells' spreads have a median ratio of 0.99 in sample, against 0.92 for independent draws. It misses the cells' means by 0.8 minutes in sample because of that early placement (Table EC46), and Figure EC7a draws every variant. If rugby union's stoppages came as a gamma renewal process with shape 1.58, its number of stoppages, 5.1 on average, would have a variance 0.65 of its mean. It has 0.33, so even spacing explains part of the narrow count and not all of it. Football's number of stoppages is bounded by the Laws, so it cannot test this.

Table EC47: How evenly a team spaces its stoppages. The concentration α of a symmetric Dirichlet fitted by maximum likelihood to the gaps between each team's first and last stoppage, as shares of their sum, on the teams with three or more. Each minute is spread over the minute it names, and a booking at minute 90 over the 5 minutes after it or read as its minute; each estimate is the mean of 20 such spreadings, its error combining the spread within and between them. Whole minutes alone lift α above one, so each estimate stands beside its null: the mean and the 95 per cent range of 200 samples in which each team keeps its first and last stoppage and its others fall at random between them in distinct whole minutes, each sample spread and fitted the same way.

Sport	Reading	Stoppage	Minute 90	Teams	α	s.e.	Null	Null, 95%
football	in sample	one minute	minute 90 spread over the stoppage	43,949	1.485	0.009	1.29	1.28–1.30
football	in sample	one minute	minute 90 read as its minute	43,949	1.426	0.008	1.24	1.23–1.26
football	in sample	merged and capped	minute 90 spread over the stoppage	41,250	3.416	0.023	1.71	1.69–1.72
football	in sample	merged and capped	minute 90 read as its minute	41,250	3.145	0.021	–	–
football	held out	one minute	minute 90 spread over the stoppage	961	1.317	0.052	1.29	1.22–1.37
football	held out	one minute	minute 90 read as its minute	961	1.292	0.049	–	–
rugby union	in sample	one minute	minute 90 read as its minute	25,262	1.576	0.008	1.35	1.35–1.36

Table EC48: The mean share of each gap between a team's first and last stoppage, in time order and in per cent, by the number of gaps, against the equal shares a symmetric Dirichlet gives every gap whatever α is; groups of at least 100 teams.

Sport	Gaps	Teams	Mean shares (%)	Equal share
football	2	37,134	53.8, 46.2	50.0
football	3	6,489	36.1, 32.3, 31.6	33.3
football	4	325	27.1, 28.4, 24.9, 19.6	25.0
rugby union	2	2,103	54.2, 45.8	50.0
rugby union	3	5,301	41.3, 29.0, 29.7	33.3
rugby union	4	7,622	35.3, 22.7, 20.1, 21.9	25.0
rugby union	5	6,678	31.9, 19.0, 16.0, 15.7, 17.4	20.0
rugby union	6	2,969	30.0, 17.2, 13.2, 12.3, 12.4, 14.7	16.7
rugby union	7	589	28.4, 16.1, 11.7, 10.4, 10.1, 10.4, 12.8	14.3

EC13.4 The score

The model is marginal over the score, yet teams that are behind change earlier (Del Corral et al., 2008; Gómez et al., 2016). In a version for in-play use, each team draws its grouping from the fitted teams with its count and its

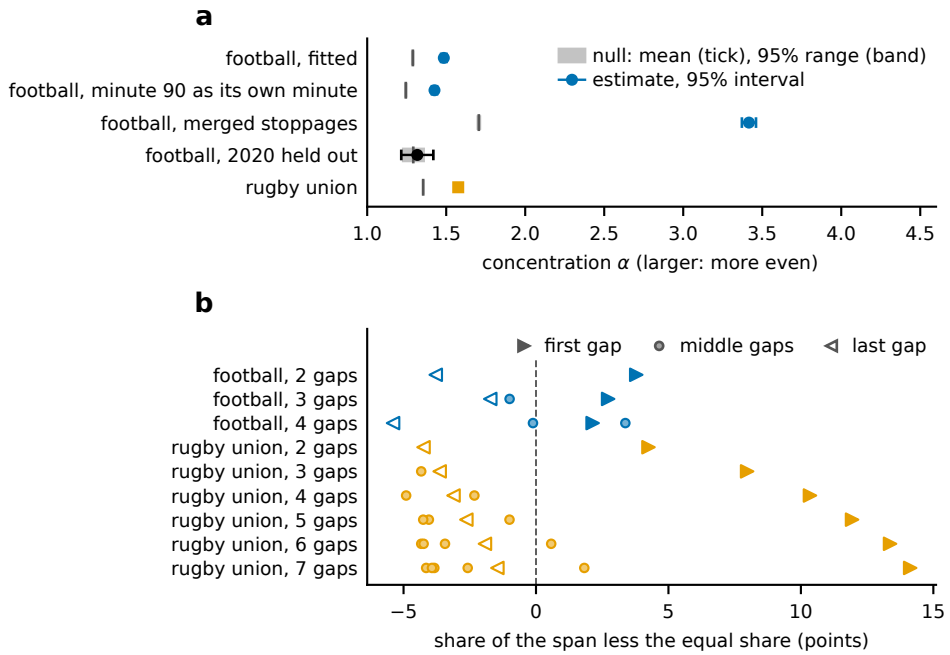


Figure EC6: Read through the same whole minutes, teams space their stoppages a little more evenly than random on the fitted seasons and as evenly on the held-out matches, and their merged stoppages much more evenly. (a) The concentration α with its 95 per cent interval for each reading, beside its null (grey): the mean (tick) and the 95 per cent range of the concentration when each team keeps its first and last stoppage and its other stoppages fall at random between them in distinct whole minutes, read the same way. (b) Each gap's mean share of the span between a team's first and last stoppage, less the equal share of a symmetric Dirichlet, in percentage points, for every group of Table EC48: the first gap, the middle gaps and the last gap; the dashed line is the equal share. Football's groups of three and four gaps are teams with four and five stoppages read minute by minute, more than the Laws allow. The held-out estimate was read after the test.

Alt text: Panel a: points to the right of narrow grey null bars, except the held-out point, which sits on its bar, and farthest right for the merged stoppages. Panel b: rows of markers, the first gap to the right of a dashed line at zero in every row and most of the others to the left of it.

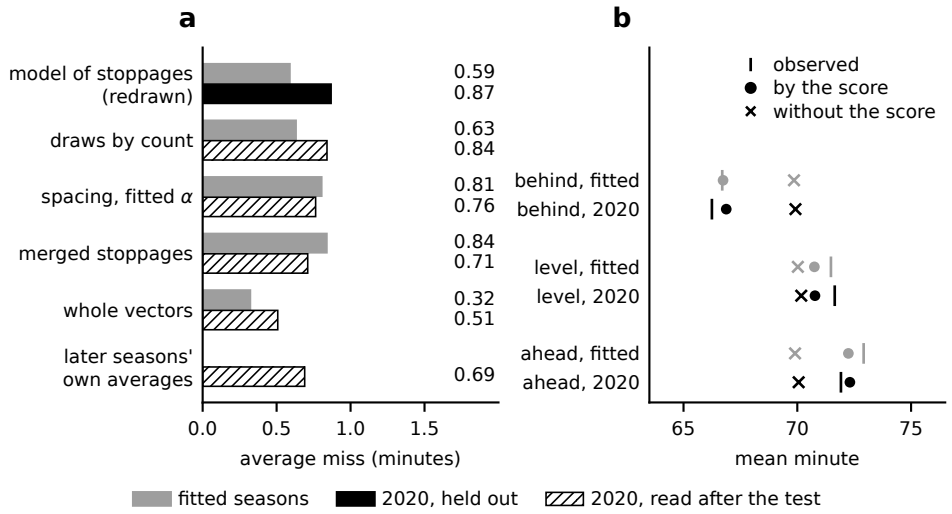


Figure EC7: Among the readings made after the test, drawing whole sequences misses the cells least, and the score moves the changes' minutes. (a) How far each way of drawing the stoppage minutes misses the cells' mean minutes, each cell weighted by its changes, on the fitted seasons (grey) and on the held-out matches of 2020. The top row is the main text's model redrawn, so its held-out miss differs from the registered reading by the draws. Draws by count use the minutes of teams with the same number of changes, spacing uses Dirichlet shares, merged stoppages read Law 3.2, and whole vectors copy a fitted team's minutes; the later fitted seasons' own averages have no bar on the fitted seasons, which are their source. These variants and the fitted seasons' means were read after the held-out matches had been read, so their held-out bars are hatched: they test nothing. (b) The mean minute of the changes of football teams behind, level and ahead at half-time, on the fitted seasons and on the held-out matches of 2020: observed, drawn by the score and drawn without it. Both panels describe; neither is a test.

Alt text: Panel a: pairs of bars, shortest for the whole vectors, with their values in a column at the right. Panel b: ticks for the observed minutes, earliest behind and latest ahead, matched by the dots drawn by the score and not by the crosses.

score at half-time, behind, level or ahead. It draws its stoppage minutes with weights $f_m(t)\, r_m(s(t), t)$, where f_m is the pool of minutes for m stoppages and $s(t)$ the team's own score at minute t, counted from the goals before it. The factor r_m is the rate at which teams with m stoppages stop play in that score against all scores, in each sixth of the match, an intensity in the spirit of this journal's point-process model of goals (Volf, 2009). Teams behind at half-time make their changes on average at minute 66.7, level teams at 71.5 and teams ahead at 72.9 (Table EC49; Figure EC7b). In a reading made after the test, on the cells of each half-time score, the model drawn by the score misses by 0.7 minutes in sample and 1.0 on the held-out matches, against 2.4 and 2.3 for the registered model, each cell weighted by its changes. So the score is the input an in-play simulator should add, and a simulator that runs before the match, without the score, can use the registered model.

Table EC49: The model drawn by the score. By the score at half-time: the changes read, their mean minute observed, under the model that draws stoppages by the score and under the registered model, and how far each misses the cells of that score, each cell weighted by its changes; held-out cells with at least 50 changes.

Reading	Score at half-time	Changes	Observed	By score	Without	Miss, by score	Miss, without
in sample	behind	79,036	66.7	66.7	69.9	0.51	3.15
in sample	level	100,398	71.5	70.8	70.0	0.77	1.45
in sample	ahead	78,102	72.9	72.2	69.9	0.68	3.01
held out	behind	1,646	66.2	66.8	69.9	1.13	3.69
held out	level	1,950	71.6	70.7	70.1	1.04	1.53
held out	ahead	1,570	71.9	72.2	70.0	0.93	1.94

EC13.5 A test on matches not yet read

The readings of this section after the held-out test were made after those matches had been read, so they describe them. For them, four predictions are registered for the 2026–27 seasons of the panel's leagues, whose records no stage of this paper has read (Table EC50). They say that teams space their stoppages more evenly than independent draws, that merged stoppages leave few teams above three, and that whole vectors place the changes at least as well as the registered model, with spreads close to the records'. The software refuses to read them until the register and the frozen tables are committed and unchanged. A dated addendum, committed on 16 October 2026 before any of their records was read, makes them readable and their outcome interpretable (Table EC51). It reads each league on the season its source labels 2026, which is the 2027 season for the five leagues that play within a calendar year. It reads a league once a match of its following season

is in the snapshot and every test match is dated, a rule that holds where a league changes its calendar, as Japan's J1 does in 2026–27, or its size. It names the source and a fallback, because the source's updates were paused when the predictions were registered. It states that 2026–27 is the first season under Law 3.3's ten-second exit, which can move a planned stoppage into a later minute, so a failure of CM08 to CM10 is the model's or the rule's. CM09 and CM10 compare two ways of drawing the stoppage minutes, so they check the fitted seasons' stability more than they test the model. On the one-minute reading CM07 cannot fail, because the booking alone lifts α above one, so the addendum adds CM07a, which reads α against its null. The addendum's reader was run once on the restart matches, as a dry run of its code on matches already read. CM07 to CM10 would have held, and CM07a would not, α being 1.31 against a null whose 95 per cent range ends at 1.37.

Table EC50: The predictions registered for the 2026–27 seasons, each quoted in full as registered. None has been read.

Id	Outcome	Prediction
CM07	to be read	On the 2026-27 domestic-league team-matches of the 29 leagues of [stage]'s panel (frozen in [deposit]), a league read only when the snapshot given to the reader holds its complete 2026-27 season (as many matches as its 2025-26 season, every one dated), with any match whose events list no substitution for either side set aside as unrecorded and any league with no side above three changes left out, the sides with three or more in-play stoppages (a stoppage being the changes a side books in one minute, and a change booked at minute 46 a half-time change and no in-play stoppage) space them more evenly than uniform order statistics: the gaps between each side's first and last stoppage, as shares of their sum, give a symmetric Dirichlet concentration above 1 by maximum likelihood pooled over the numbers of gaps, with each minute spread uniformly over the minute it names and a booking at minute 90 over the five minutes after it, the estimate being the mean over 20 such spreadings with seed 2371 [stage], and the likelihood-ratio test of a concentration of 1 rejects it at the 5 per cent level.
CM08	to be read	On the same team-matches, when a stoppage that opens at minute t is read as taking every change the side books at t or t + 1, at most 2 per cent of the sides that made five changes use more than three in-play stoppages, the three opportunities that Law 3.2 allows.
CM09	to be read	On the same team-matches, for every cell (the k-th of n changes, averaged over the sides that made exactly n, for n from 1 to 5) holding at least 50 changes, the side's in-play stoppage minutes drawn as one whole vector from a fitting side with the same count and the same number of in-play stoppages ([deposit]), with the side's grouping drawn from [stage]'s frozen groupings ([deposit]), miss the cells' mean minutes by no more than [stage]'s frozen model, which draws the stoppage minutes independently and without repeats from its frozen pools ([deposit]), each cell weighted by its changes, both simulated 200,000 times per count with seed 2371.

Table EC50, continued

Id	Outcome	Prediction
CM10	to be read	On the same cells, the whole-vector draws of CM09 give a median ratio of each cell's observed to predicted standard deviation between 0.9 and 1.1, and at least 80 per cent of the cells have a ratio between 0.75 and 4/3.

Table EC51: The dated addendum to the predictions for 2026–27, committed before any of their records was read, each item quoted in full as registered; a stage of the software is printed as [stage].

Item	Applies to	Text
A1	CM07-CM10	The seasons read. Each of the 29 panel leagues is read on the season its source labels 2026: the 2026-27 season for the 24 leagues that play across two calendar years, Japan's J1 among them in its first August-to-May season, and the 2027 season for the five that play within one year (Argentina, Brazil, Major League Soccer, Norway and Sweden), which the source labels by the year before. A season is told from its matches' dates, never from a file name.
A2	CM07-CM10	When a league is read. A league is read once the snapshot holds a match of its following season (the source's label 2027), so its test season has ended, and every match of its test season carries a date. This replaces the rule 'as many matches as its 2025-26 season', which fails where a league changes its calendar or its size.
A3	CM07-CM10	The source and its fallback. The source is the transfermarkt-datasets release, whose updates were paused when CM07 to CM10 were registered (it is current to 6 July 2026). If no release holds a league's following season by 30 September 2028, the league's test season is read from Transfermarkt's own match pages, the release's source, in the release's layout. If neither holds it, the league is not read and is named.
A4	CM07-CM10	What a failure means. 2026-27 is the first season under Law 3.3's ten-second exit (IFAB, 2026), which can move a planned stoppage into a later minute, so a failure of CM08 to CM10 is read as the model's or the rule's, not the model's alone. CM09 and CM10 compare two ways of drawing the stoppage minutes and check the fitted seasons' stability more than they test the model. On the one-minute reading CM07 cannot fail, because whole minutes alone lift the concentration above one; CM07a reads it against its null. [stage]'s reader applies A1 to A4; [stage]'s reader is not run.
CM07a	association football	On the team-matches of CM07, the concentration of the gaps between each team's first and last in-play stoppage lies above the 97.5th percentile of its null: 200 samples in which each team keeps its observed first and last stoppage and its other stoppages fall at random between them in distinct whole minutes, each sample spread and fitted as CM07 is ([stage]).

EC14 Register of pre-specified predictions

Table EC52 lists the pre-specified predictions this paper reports, with their outcomes, each quoted in full as it was entered in the register. Its column

‘Bears on’ names the section of the main text each prediction bears on, so each failure sits beside its claim. Of the 39, 28 were committed in commits of their own, two to 19 minutes before their test. The other 11 were numbered when the register was assembled, after their tests existed, and are marked ‘numbered later’. The result tables are not versioned, so the history cannot show that a result was not looked at before its prediction was written. Predictions are quoted from a time when the register said *code* where this paper says sport, *bell* where it says censored normal and *planned share* where it says mass at the limit. Where a prediction names a stage of the software, the name is withheld for review and printed as [stage]. The predictions test different quantities and every outcome is reported, failures included, so no correction for multiplicity is applied. Four predictions on rugby union’s centre, C192, C209, C215 and C216, are printed in this version because the main text reports the rise they test. Three outcomes need a note. C43 was read on two one-way cells and two re-entry cells, football’s and the AFL’s, before the NRL was read, and the NRL’s starters, 63 per cent of whom play the whole match, now break its second half. C04’s own words say the boundary was pre-specified, but it was numbered later, so the history cannot show that it preceded the data. C40 is coded held, although penalty time accounts for only part of the NHL’s shortfall (Section EC4).

Table EC52: The register of pre-specified predictions, each quoted in full as registered, with the section of the main text it bears on; ‘the limits’ is the Discussion’s limitations.

Id	Order	Outcome	Bears on	Prediction
C01	numbered later	held	3.3, the gap	The budget identity holds inside the tolerance of 0.03 in rugby union
C02	numbered later	held	3.3, the gap	Gaelic football joins the identity inside the same tolerance
C03	numbered later	held	3.3, the gap	Hurling joins the identity inside the same tolerance
C04	numbered later	borne out	3.3 and the limits, baseball	Baseball fails the tolerance through its bullpen chains: the boundary was pre-specified as a failure before the arithmetic ran
C05	numbered later	refused	2, the Gaelic panels	A GAA final whose page carries no substitution template is refused whole rather than counted as zero
C50	numbered later	held	3.3, the gap	No football team-match may exceed the budget its season is coded at; where some do, the excess per team-match must sit inside the identity’s tolerance of 0.03 or the rulebook coding of B is not exhaustive and the identity row is restated
C06	numbered later	held	3.3, each month	The identity holds inside its tolerance in every month of the men’s five-substitution family

Table EC52, continued

Id	Order	Outcome	Bears on	Prediction
C40	numbered later	held	2, exposure	The NHL's exposure shortfall is its own penalty time
C43	numbered later	held	3.1, who plays	One-way cells leave most starters to finish and rotational cells do not, as the rulebooks predict
C44	numbered later	failed	3.1 and the limits, who plays	Who finishes a match falls monotonically with the replacement budget per player on the field
C42	numbered later	failed	1, use of the cap	A capped rotational code's utilisation is one strict constant
C174	before, 3 min	held	3.2, the bound	The spend line in its factor form is the censoring identity: in every code-regime with a rule bound (football under three and five, rugby union's internationals, rugby league), the deposited spend equals chi times the bound plus one minus chi times the mean spend of the team-matches that did not exhaust it, within 0.02 of the bound, chi the deposited exhaustion
C151	before, 19 min	held	4 and the limits, the mass	The planned share is the league's, one: with pi, mu and nu free per league and fitted by maximum likelihood on each league's distribution of substitutions per team-match (0 to 5) across the thirty-one five-substitution leagues, the fitted exhaustion share is within 0.03 of the observed in every league, and pi_L spans at least 0.10 from its lowest to its highest league
C152	before, 19 min	failed	4 and the limits, width	The planned share is the league's, two: once pi is the league's, the demand's dispersion nu_L is one constant across the thirty-one leagues (Cochran's Q $p > 0.05$ on the per-league maximum-likelihood estimates with their standard errors)
C188	before, 2 min	failed	4 and the limits, narrowness	The demand is a bell and not a Poisson count: in both free codes the variance-to-mean ratio of the uncensored count (relief pitchers per nine-inning team-game; bench players used with twelve dressed) is below one in every season, its median across seasons below 0.5
C217	before, 3 min	held	4, width	The paper's two readings of the demand's relative dispersion are of one quantity and differ. Fitted by maximum likelihood on the pooled team-match histogram of football's twenty-nine five-substitution leagues with [stage]'s censored-bell likelihood, the planned share and the centre free, the relative dispersion nu is above 0.15, more than one and a half times the 0.09 the eight-code two-moment calibration carries

Table EC52, continued

Id	Order	Outcome	Bears on	Prediction
C218	before, 3 min	failed	4 and the limits, the mass	The eight-code calibration does not identify the dispersion: holding the planned share at 0.56 on rule allotments and zero on rosters and fixing nu at each of 0.09, 0.15, 0.20, 0.26 and 0.32 in turn, with each code's centre solved from its own spend, the eight code-regime exhaustion shares are reproduced within the tolerance of 0.10 set for it at every one of the five values
C219	before, 3 min	failed	4 and the limits, the mass	The calibration stands at the histogram's dispersion: with nu fixed at the inverse-variance pooled league value of [stage] and the planned share re-fitted on the eight pairs, the code-regime exhaustions are reproduced within 0.10 and the root-mean-square residual is at most one and a half times the 0.024 the 0.09 fit achieves
C220	before, 3 min	held	4, width	Intervals for the bell: resampling club-seasons with replacement 200 times within each of football's twenty-nine five-substitution leagues and refitting the censored bell on each resample, the 95 per cent percentile interval of nu excludes 0.09 in at least twenty-six of the twenty-nine leagues
C228	before, 4 min	held	4, width	Follows C218 and C219, which failed: the relative dispersion is not a code-free number, because it is the standard deviation over the centre and the centre depends on where the code's count starts. In basketball the same team-games counted as bench players used with twelve dressed and as all players used give relative dispersions whose ratio equals the ratio of their fitted centres within 0.05, so the two readings carry one standard deviation and two relative dispersions
C229	before, 4 min	failed	4 and the limits, the mass	With the planned share fitted for each code rather than shared across them, the dispersion the league histograms measure serves the codes whose count starts at zero changes: at nu fixed at [stage]'s inverse-variance pooled league value, and the planned share free in each of the five rule-bounded code-regimes (football's two, rugby union's two, rugby league), the fitted exhaustion is within 0.10 of the observed in every one of the five

Table EC52, continued

Id	Order	Outcome	Bears on	Prediction
C230	before, 4 min	held	4, width	The comparable width is the absolute one: the demand's standard deviation, the product of the fitted relative dispersion and the fitted centre, lies within a factor of 2.5 of its median across the five readings taken on each code's own count of changes, football's three-substitution and five-substitution histograms, rugby union's eight-bench sides, basketball's bench players used and baseball's relief pitchers
C244	before, 6 min	held	4, centre and width	The censoring rule the paper states governs a centre and not a width. On planted histograms drawn from the censored-bell likelihood with a known centre and a known standard deviation, at caps 3, 5 and 8 and at centres of 0.6, 0.9, 1.0 and 1.2 times the cap, 200 draws of 4,000 team-matches at each of the twelve cells, refitted with the planned share, the centre and the width free, the median fitted standard deviation is within 15 per cent of the planted one at every one of the twelve cells, including the cells whose centre sits at or beyond its bound
C246	before, 6 min	failed	4 and the limits, width	The width band of C230 carries its uncertainty: resampling team-matches with replacement 400 times within each of the five readings and refitting, every one of the five 95 per cent percentile intervals for the demand's standard deviation contains a value common to all five, so the band is not an artefact of five point estimates
C193	before, 2 min	held	4, the centre	Generalises C191 in the other direction: the centre falls when the bound tightens, in Australian rules. Interchanges per team-game from the AFL feed's sampled games (both sides of each), fitted with the censored bell at the season's cap, 120 in 2014 and 2015, 90 in 2016, 2017 and 2019, 75 in 2022 to 2024, each regime with at least 300 team-games, gives a fitted centre that falls at each tightening: mu under 90 below mu under 120 and mu under 75 below mu under 90, each fall at least 5 interchanges per team-game
C221	before, 3 min	held	4, the centre	A pooled estimate beside the count of leagues: the rise of the demand's centre across rugby union's bound with the allotment by the rule's year (C209), pooled across the five leagues by inverse-variance weighting of the bootstrapped per-league rises, is positive with a 95 per cent interval excluding zero, and its heterogeneity is reported with it

Table EC52, continued

Id	Order	Outcome	Bears on	Prediction
C234	before, 6 min	held	4, the centre	The centre's rise across rugby union's bound is one constant across the five leagues to within a stated width: the between-league standard deviation of the rise, estimated by the method of moments from the bootstrapped per-league rises, has a 95 per cent upper bound below 0.3 replacements per side
C201	before, 2 min	held	5, the line	Generalises C197 across counts within a regime: the k-th change's minute is set by how many changes the team makes in all. In each of football's fourteen leagues with at least 500 team-matches in each regime, under five substitutions, the mean minute of the third substitution among team-matches making exactly three, exactly four and exactly five substitutions falls monotonically with the count in at least twelve of the fourteen leagues, the drop from exactly three to exactly five exceeding 5 minutes in the median league
C202	before, 2 min	held	5, the line	The same law in rugby union within the eight-bench seasons: the mean minute of the seventh replacement's entry among recorded sides using exactly seven exceeds the mean among sides using all eight in at least four of the five leagues, with the median difference above 3 minutes
C236	before, 6 min	held	5, the line	The count law does not turn on the count chosen: under five substitutions the mean minute of the second substitution among team-matches making exactly two, three, four and five substitutions falls monotonically with the count in at least twelve of the fourteen leagues, as C201 found for the third
C238	before, 3 min	held	5, the line's form	The count law has a form and not only a sign, in football. Under five substitutions, over the cells k = 1 to n and n = 1 to 5 pooled across the fourteen leagues, the form is the k-th change's mean minute against k/(n+1), where n is the changes the team made, fitted by weighted least squares over the cells with at least 100 team-matches, the weight the inverse variance of the cell mean: the fit explains at least 0.95 of the weighted variance of the cell means

Table EC52, continued

Id	Order	Outcome	Bears on	Prediction
C239	before, 3 min	held	5, the line's form	The same form serves rugby union. Over the eight-bench sides with the allotment by the rule's year, cells k = 1 to n and n = 1 to 8, the form is the k-th change's mean minute against k/(n+1), where n is the changes the team made, fitted by weighted least squares over the cells with at least 100 team-matches, the weight the inverse variance of the cell mean: the fit explains at least 0.95 of the weighted variance
C240	before, 3 min	held	5, the line's form	One form serves both codes once the minute is read as a share of the match: the intercept and the slope of the k-th change's minute on k/(n+1), each divided by the code's match length, agree between football's five-substitution era and rugby union's eight-bench sides within 0.10 of the match
C241	before, 3 min	refused	women's game, refused	The demand's width replicates in the women's game. From the women's appearance records, substitutions per team-match in the seven leagues under five substitutions, fitted with the censored bell at an allotment of five: the demand's standard deviation, the product of the fitted relative dispersion and centre, lies within a factor of 1.5 of the men's five-substitution value
C242	before, 3 min	refused	women's game, refused	The count law replicates in the women's game: under five substitutions the mean entry minute of the third substitution, read as the match length less the substitute's recorded minutes, falls monotonically over team-matches making exactly three, four and five substitutions in at least five of the seven leagues
C192	before, 2 min	held	4, the centre	Generalises C191 to a second code: the centre rises when the bound loosens in rugby union. In each of the five itsrugby leagues, replacements used per side (bench entries with a recorded minute, on sides that recorded any minute) fitted with the censored bell at the season's bench allotment (seven where the season's sheets list at most seven replacements, eight thereafter), with at least 300 recorded sides in each regime, gives mu under eight above mu under seven in at least four of the five leagues, with the median rise above 0.3 replacements per side; a league whose fitted centre sits beyond its bound in either regime is read as it comes

Table EC52, continued

Id	Order	Outcome	Bears on	Prediction
C209	before, 2 min	held	4, the centre	Re-reads C192 with the allotment set by the rule's year, the first season from which at least fifteen per cent of a league's sides list eight replacements and do so in every season after (the Premiership from its 2009 season, the Top 14 and Pro D2 from 2008, the URC from 2010 and Super Rugby from 2013, seven before): the censored bell's centre of replacements used per recorded side under eight exceeds the centre under seven in at least four of the five leagues, the median rise above 0.3; a regime with fewer than 300 recorded sides is refused for that league
C215	before, 2 min	held	4, the centre	Re-reads C207 with the allotment by the rule's year (the Premiership from its 2009 season, the Top 14 and Pro D2 from 2008, the URC from 2010 and Super Rugby from 2013, seven before): the fitted centre in the first two eight-bench seasons exceeds the centre in the last two seven-bench seasons by more than 0.3 in at least four of the five leagues; a window with fewer than 300 recorded sides is refused for that league
C216	before, 2 min	failed	4 and the limits, the centre	Re-reads C208 with the allotment by the rule's year (the Premiership from its 2009 season, the Top 14 and Pro D2 from 2008, the URC from 2010 and Super Rugby from 2013, seven before): within the eight-bench seasons the fitted centre in the last two seasons minus the centre in the first two is smaller than the step of C215 in at least four of the five leagues

References

Andrews, D. W. K. (2000), 'Inconsistency of the bootstrap when a parameter is on the boundary of the parameter space', *Econometrica* **68**(2), 399–405.

Apesteguia, J. & Palacios-Huerta, I. (2010), 'Psychological pressure in competitive environments: evidence from a randomized natural experiment', *American Economic Review* **100**(5), 2548–2564. Corrigendum: American Economic Review, 101(4), 1636, doi 10.1257/aer.101.4.1636.

Arora, M., Chaganty, N. R. & Sellers, K. F. (2021), 'A flexible regression model for zero- and k-inflated count data', *Journal of Statistical Computation and Simulation* **91**(9), 1815–1845.

Associated Press (2022), 'MLB moving forward with long-delayed 13-pitcher

limit', Wire report, 19 June.
`https://www.baynews9.com/fl/tampa/ap-online/2022/06/19/mlb-moving-forward-with-long-delayed-13-pitcher-limit`

Australian Football League (2026), 'AFL match-centre statistics API', Public statistics interface, read 2026.
`https://api.afl.com.au/`

Avila-Cano, A., Owen, P. D. & Triguero-Ruiz, F. (2023), 'Measuring competitive balance in sports leagues that award bonus points, with an application to rugby union', *European Journal of Operational Research* **309**(2), 939–952.

Baker, R. & Kharrat, T. (2018), 'Event count distributions from renewal processes: fast computation of probabilities', *IMA Journal of Management Mathematics* **29**(4), 415–433.

Becker, G. S. (1968), 'Crime and punishment: an economic approach', *Journal of Political Economy* **76**(2), 169–217.

Bengtsson, H., Ekstrand, J. & Hägglund, M. (2013), 'Muscle injury rates in professional football increase with fixture congestion: an 11-year follow-up of the UEFA Champions League injury study', *British Journal of Sports Medicine* **47**(12), 743–747.

Bhatia, R. & Davis, C. (2000), 'A better bound on the variance', *The American Mathematical Monthly* **107**(4), 353–357.

Black, G. M. & Gabbett, T. J. (2014), 'Match intensity and pacing strategies in rugby league: an examination of whole-game and interchanged players, and winning and losing teams', *Journal of Strength and Conditioning Research* **28**(6), 1507–1516.

Blomquist, S., Newey, W. K., Kumar, A. & Liang, C.-Y. (2021), 'On bunching and identification of the taxable income elasticity', *Journal of Political Economy* **129**(8), 2320–2343.

Botvinik-Nezer, R., Holzmeister, F., Camerer, C. F. et al. (2020), 'Variability in the analysis of a single neuroimaging dataset by many teams', *Nature* **582**(7810), 84–88.

Bradley, V. C., Kuriwaki, S., Isakov, M., Sejdinovic, D., Meng, X.-L. & Flaxman, S. (2021), 'Unrepresentative big surveys significantly overestimated US vaccine uptake', *Nature* **600**(7890), 695–700.

Breznau, N., Rinke, E. M., Wuttke, A. et al. (2022), 'Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty', *Proceedings of the National Academy of Sciences* **119**(44), e2203150119.

Brooks, R., Harris, M. N. & Spencer, C. (2012), 'Inflated ordered outcomes', *Economics Letters* **117**(3), 683–686.

Brown, N. J. L. & Heathers, J. A. J. (2017), 'The GRIM test: a simple technique detects numerous anomalies in the reporting of results in psychology', *Social Psychological and Personality Science* **8**(4), 363–369.

Butler, D., Butler, R. & Singleton, C. (2026), 'Objective calls under the spotlight: referee consistency and behaviour on football's biggest stage', *Journal of Sports Economics* **27**(2), 171–213.

Caley, M. (2024), 'Substitutes, part II: clubs, leagues and managers', Expecting Goals, 8 February.
`https://www.expectinggoals.com/p/substitutes-part-ii-clubs-leagues`

Chen, T. & Chen, L. (2025), 'A study on substitution characteristics and competitive performance of Chinese Super League teams under the five-substitution rule', *PLOS ONE* **20**(5), e0322241.

Clemens, B. (2019), 'The three batter minimum rule barely matters', FanGraphs, 18 December.
`https://blogs.fangraphs.com/the-three-batter-minimum-barely-matters/`

Conrad, S. A. (1976), 'Sales data and the estimation of demand', *Operational Research Quarterly* **27**(1), 123–127.

Cooke, T. & Huber, W. (2026), 'The numerical detective: EDA as methodology for managerial insight', *IMA Journal of Management Mathematics*. Advance access, article dpag021.

Cricsheet (2026), 'Ball-by-ball data for cricket matches', Cricsheet. All matches, in JSON, downloaded 6 September 2026.
`https://cricsheet.org/downloads/`

David, H. A. & Nagaraja, H. N. (2003), *Order Statistics*, 3rd edn, Wiley, Hoboken, NJ.

Deffner, A. & Haeusler, E. (1985), 'A characterization of order statistic point processes that are mixed Poisson processes and mixed sample processes simultaneously', *Journal of Applied Probability* **22**(2), 314–323.

Del Corral, J., Barros, C. P. & Prieto-Rodriguez, J. (2008), 'The determinants of soccer player substitutions: a survival analysis of the Spanish soccer league', *Journal of Sports Economics* **9**(2), 160–172.

Dewenter, R., Gasper, R. & Mackewitz, N. (2026), 'How the new substitution rule affected Bundesliga results', *Sports Economics Review* **15**, 100076.

Donskov, A. S., Humphreys, D. & Dickey, J. P. (2019), 'What is injury in ice hockey: an integrative literature review on injury rates, injury definition, and athlete exposure in men's elite ice hockey', *Sports* **7**(11), 227.

Feigin, P. D. (1979), 'On the characterization of point processes with the order statistic property', *Journal of Applied Probability* **16**(2), 297–304.

Fuller, G. W., Kemp, S. P. T. & Decq, P. (2015), 'The International Rugby Board (IRB) Pitch Side Concussion Assessment trial: a pilot test accuracy study', *British Journal of Sports Medicine* **49**(8), 529–535.

Gabrys, T., Chruscinski, R., Garnys, M., Suva, M. & Švátora, K. (2025), 'The influence of substitution decisions made by national team coaches on final match outcomes at UEFA EURO 2024', *Frontiers in Sports and Active Living* **7**, 1573823.

Gardner, A. J., Iverson, G. L., Stanwell, P., Moore, T., Ellis, J. & Levi, C. R. (2016), 'A video analysis of use of the new 'concussion interchange rule' in the National Rugby League', *International Journal of Sports Medicine* **37**(4), 267–273.

Garicano, L., Palacios-Huerta, I. & Prendergast, C. (2005), 'Favoritism under social pressure', *Review of Economics and Statistics* **87**(2), 208–216.

Gómez, M.-Á., Lago-Peñas, C. & Owen, A. L. (2016), 'The influence of substitutions on elite soccer teams' performance', *International Journal of Performance Analysis in Sport* **16**(2), 553–568.

Greenwood, M. (1946), 'The statistical study of infectious diseases', *Journal of the Royal Statistical Society* **109**(2), 85–110.

Hamada, K. & Tanaka, K.-i. (2021), 'Modelling the order of scoring in team sports', *IMA Journal of Management Mathematics* **32**(3), 283–301.

Harris, M. N. & Zhao, X. (2007), 'A zero-inflated ordered probit model, with an application to modelling tobacco consumption', *Journal of Econometrics* **141**(2), 1073–1099.

Hasija, S., Pinker, E. & Shumsky, R. A. (2010), 'Work expands to fill the time available: capacity estimation and staffing under Parkinson's law', *Manufacturing & Service Operations Management* **12**(1), 1–18.

Heitjan, D. F. & Rubin, D. B. (1991), 'Ignorability and coarse data', *The Annals of Statistics* **19**(4), 2244–2253.

Hills, S. P., Barrett, S., Feltbower, R. G., Barwood, M. J., Radcliffe, J. N., Cooke, C. B., Kilduff, L. P., Cook, C. J. & Russell, M. (2019), 'A match-day analysis of the movement profiles of substitutes from a professional soccer club before and after pitch-entry', *PLOS ONE* **14**(1), e0211563.

Hoenig, T., Edouard, P., Krause, M., Malhan, D., Relógio, A., Junge, A. & Hollander, K. (2022), 'Analysis of more than 20,000 injuries in European professional football by using a citizen science-based approach: an opportunity for epidemiological research?', *Journal of Science and Medicine in Sport* **25**(4), 300–305.

International Cricket Council (2025*a*), 'Men's one-day international playing conditions', International Cricket Council, Dubai. Effective from July 2025; clause 13.9.1, number of overs per bowler.
`https://images.icc-cricket.com/image/upload/prd/d25dbgishkx0kijb4jeu.pdf`

International Cricket Council (2025*b*), 'Men's twenty20 international playing conditions', International Cricket Council, Dubai. Effective from July 2025; clause 13.9.1, number of overs per bowler.
`https://images.icc-cricket.com/image/upload/prd/qfnsie8fz6vhyl1pmcli.pdf`

Jaffe, J. (2022), 'Less is more: relief pitching has dominated this postseason', FanGraphs, 27 October.
`https://blogs.fangraphs.com/less-is-more-relief-pitching-has-dominated-this-postseason/`

Kadane, J. B. (2016), 'Sums of possibly associated Bernoulli variables: the Conway–Maxwell-binomial distribution', *Bayesian Analysis* **11**(2), 403–420.

Kim, S.-H. & Whitt, W. (2014), 'Are call center and hospital arrivals well modeled by nonhomogeneous Poisson processes?', *Manufacturing & Service Operations Management* **16**(3), 464–480.

Kitson, R. (2009), 'ERC decide against changing rules on blood substitutions', The Irish Times, 29 September.
`https://www.irishtimes.com/sport/erc-decide-against-changing-rules-on-blood-substitutions-1.747097`

Kleiber, C. & Zeileis, A. (2016), 'Visualizing count data regressions using rootograms', *The American Statistician* **70**(3), 296–303.

Knodell, A. (2025), 'A brief history of NHL play-by-play data', Puck Over the Glass, 20 November.
`https://puckovertheglass.substack.com/p/a-brief-history-of-nhl-play-by-play`

Kocher, M. G., Lenz, M. V. & Sutter, M. (2012), 'Psychological pressure in competitive environments: new evidence from randomized natural experiments', *Management Science* **58**(8), 1585–1591.

Kocsoy, A. (2025), 'Referee bias in football: actual vs. expected additional time', *Sports Economics Review* **9**, 100047.

Kowal, D. R. & Canale, A. (2020), 'Simultaneous transformation and rounding (STAR) models for integer-valued data', *Electronic Journal of Statistics* **14**(1), 1744–1772.

Lacome, M., Piscione, J., Hager, J.-P. & Carling, C. (2016), 'Analysis of running and technical performance in substitute players in international male rugby union competition', *International Journal of Sports Physiology and Performance* **11**(6), 783–792.

Lazer, D., Kennedy, R., King, G. & Vespignani, A. (2014), 'The parable of Google Flu: traps in big data analysis', *Science* **343**(6176), 1203–1205.

LeagueEyeTest (2021), 'The decline of 80 minute players – NRL round 12 2021 stats and trends', Rugby League Eye Test, 1 June.
`https://www.rugbyleagueeyetest.com/2021/06/01/the-decline-of-80-minute-players-nrl-round-12-2021-stats-and-trends/`

Li, Y., Weber, H. & Link, D. (2026), 'Additional time error in association football is associated with interruption type and goal difference', *Science and Medicine in Football* **10**(1), 59–64.

Liberman, U. (1985), ‘An order statistic characterization of the Poisson renewal process’, *Journal of Applied Probability* **22**(3), 717–722.

Ligue Nationale de Rugby (2025), ‘Carton orange : tout savoir sur la nouvelle règle’, News release, 4 September.
`https://top14.lnr.fr/actualite/carton-orange-tout-savoir-sur-la-nouvelle-regle`

Liu, H., Hopkins, W., Gómez, A. M. & Molinuevo, S. J. (2013), ‘Inter-operator reliability of live football match statistics from OPTA Sportsdata’, *International Journal of Performance Analysis in Sport* **13**(3), 803–821.

Lloyd, E. H. (1952), ‘Least-squares estimation of location and scale parameters using order statistics’, *Biometrika* **39**(1–2), 88–95.

Lorentzen, T. (2017), ‘Allocation of playing time within team sports — a problem for discussion’, *Open Review of Educational Research* **4**(1), 20–32.

Major League Baseball (2020), ‘Official baseball rules, 2020 edition’, Office of the Commissioner of Baseball, New York. Rule 5.10(g), the three-batter minimum for pitchers.

Major League Baseball (2026), ‘Official baseball rules, 2026 edition’, Office of the Commissioner of Baseball, New York. Rule 9.03(c), how to prove a box score.
`https://mktg.mlbstatic.com/mlb/official-information/2026-official-baseball-rules.pdf`

Manasis, V., Ntzoufras, I. & Reade, J. J. (2022), ‘Competitive balance measures and the uncertainty of outcome hypothesis in European football’, *IMA Journal of Management Mathematics* **33**(1), 19–52.

Marylebone Cricket Club (2026), ‘The laws of cricket, 2017 code, 4th edition’, Marylebone Cricket Club, London. Law 17.5, umpire miscounting.
`https://www.lords.org/getattachment/8c96a135-beff-4a46-b5c6-2a02a7245386/Laws-of-Cricket-2017-Code-4th-Edition-(2026).pdf`

McKay, C. D., Tufts, R. J., Shaffer, B. & Meeuwisse, W. H. (2014), ‘The epidemiology of professional ice hockey injuries: a prospective report of six NHL seasons’, *British Journal of Sports Medicine* **48**(1), 57–62.

McKinley, E. (2024), ‘Stoppage time matters: how substitutions and using all minutes played affect player statistics’, American Soccer Analysis, 11 December.
`https://www.americansocceranalysis.com/home/2024/12/8/stoppage-time-matters-how-substitutions-and-using-all-minutes-played-affect-player-statistics`

Meyer, J. & Klatt, S. (2021), ‘Impact of one additional substitution on player load and coaching tactics in elite football’, *Applied Sciences* **11**(16), 7676.

Mosteller, F. (1946), ‘On some useful “inefficient” statistics’, *Annals of Mathematical Statistics* **17**(4), 377–408.

Mukherjee, S., Huang, Y., Neidhardt, J., Uzzi, B. & Contractor, N. (2019), ‘Prior shared success predicts victory in team competitions’, *Nature Human Behaviour* **3**(1), 74–81. Author correction (acknowledgements only): Nature Human Behaviour, 3(4), 406, doi 10.1038/s41562-019-0581-y.

Nahmias, S. (1994), ‘Demand estimation in lost sales inventory systems’, *Naval Research Logistics* **41**(6), 739–757.

National Football League (2026), ‘2026 official playing rules of the national football league’, NFL Football Operations, New York. Rule 5, Section 2, Article 3, legal substitutions.
`https://operations.nfl.com/rules-officiating/2026-nfl-rulebook/`

National Hockey League (2026), ‘NHL web and statistics APIs’, Public statistics interfaces, shift charts, read 2026.
`https://api-web.nhle.com/v1/`

Ntzoufras, I., Palaskas, V. & Drikos, S. (2021), ‘Bayesian models for prediction of the set-difference in volleyball’, *IMA Journal of Management Mathematics* **32**(4), 491–518.

Nuijten, M. B., Hartgerink, C. H. J., van Assen, M. A. L. M., Epskamp, S. & Wicherts, J. M. (2016), ‘The prevalence of statistical reporting errors in psychology (1985–2013)’, *Behavior Research Methods* **48**(4), 1205–1226.

Pappalardo, L., Cintia, P., Rossi, A., Massucco, E., Ferragina, P., Pedreschi, D. & Giannotti, F. (2019), ‘A public data set of spatio-temporal match events in soccer competitions’, *Scientific Data* **6**, 236.

Premier League (2021), 'Football Association and Premier League statement', News release, 29 January, read 16 October 2026.
https://www.premierleague.com/news/2002209

Premier League (2023), 'What's new in 2023/24: applications of the laws of the game', News release, 9 August.
https://www.premierleague.com/en/news/3617054

Puri, P. S. (1982), 'On the characterization of point processes with the order statistic property without the moment condition', *Journal of Applied Probability* **19**(1), 39–51.

Raftery, M., Tucker, R. & Falvey, É. C. (2021), 'Getting tough on concussion: how welfare-driven law change may improve player safety—a Rugby Union experience', *British Journal of Sports Medicine* **55**(10), 527–529.

Ribeiro, C. F. B., Siqueira, L. S., Pinto, D. P. & Silva, C. D. (2020), 'The three and six-substitution rules in football: a preliminary comparative analysis in quantitative replacing, game statistics, win rate and winning probability', *Motriz: Revista de Educação Física* **26**, e10200188.

Roberts, S. P., Stokes, K. A., Williams, S., West, S. W., Kemp, S., Cross, M., Moore, I. S., Bitchell, C. L., Mathema, P., Quarrie, K., McDonald, W., Fortington, L., Rubio del Castillo, E., Readhead, C., Sewry, N., Falvey, É. & Tucker, R. (2024), 'Injury in starting and replacement players from five professional men's rugby unions', *Sports Medicine* **54**(8), 2157–2167.

Sargan, J. D. (1958), 'The estimation of economic relationships using instrumental variables', *Econometrica* **26**(3), 393–415.

Scarf, P., Khare, A. & Alotaibi, N. (2022), 'On skill and chance in sport', *IMA Journal of Management Mathematics* **33**(1), 53–73.

Schmook, N. (2016), 'Power in a spin as rotation stats emerge', AFL.com.au; undated, written after round 9 of the 2016 season.
https://www.afl.com.au/news/78213/power-in-a-spin-as-rotation-stats-emerge

Scholtus, S. (2011), 'Algorithms for correcting sign errors and rounding errors in business survey data', *Journal of Official Statistics* **27**(3), 467–490.
https://www.scb.se/contentassets/ca21efb41fee47d293bbee5bf7be7fb3/algorithms-for-correcting-sign-errors-and-rounding-errors-in-business-survey-data.pdf

Schuirmann, D. J. (1987), 'A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability', *Journal of Pharmacokinetics and Biopharmaceutics* **15**(6), 657–680.

Segar, D. (2024), 'How substitute trends compare across Europe's top five leagues', Opta Analyst, 28 March, read 16 October 2026. `https://theanalyst.com/2024/03/how-substitute-trends-compare-across-europe-top-five-leagues/`

Self, S. G. & Liang, K.-Y. (1987), 'Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions', *Journal of the American Statistical Association* **82**(398), 605–610.

Serevin, J. (2026), 'Top 14 – Pro D2 : comment le retour à huit remplacements va changer les fins de match', Rugbyscope, 4 August. `https://www.rugbyscope.fr/top-14-pro-d2-comment-le-retour-a-huit-remplacements-va-changer-les-fins-de-match/`

Sharp, J. C. M., Murray, G. D. & Macleod, D. A. D. (2001), 'A unique insight into the incidence of rugby injuries using referee replacement reports', *British Journal of Sports Medicine* **35**(1), 34–37.

Siegle, M. & Lames, M. (2012), 'Game interruptions in elite soccer', *Journal of Sports Sciences* **30**(7), 619–624.

Silberzahn, R., Uhlmann, E. L., Martin, D. P. et al. (2018), 'Many analysts, one data set: making transparent how variations in analytic choices affect results', *Advances in Methods and Practices in Psychological Science* **1**(3), 337–356.

Singh, A., Scarf, P. & Baker, R. (2023), 'A unified theory for bivariate scores in possessive ball-sports: the case of handball', *European Journal of Operational Research* **304**(3), 1099–1112.

Stats Perform (n.d.), 'Opta event definitions', Accessed 27 September 2026. `https://www.statsperform.com/opta-event-definitions/`

Sun, R., Wang, C., Qin, Z. & Han, C. (2024), 'Temporal features of goals, substitutions, and fouls in football games in the five major European league from 2018 to 2021', *Heliyon* **10**(5), e27014.

The Associated Press (2026), 'Pérez throws one-hitter for 11th MLB complete-game shutout this year as Marlins beat Braves', TSN, via The

Canadian Press, 25 September.
https://www.tsn.ca/mlb/article/perez-throws-one-hitter-for-11th-mlb-complete-game-shutout-this-year-as-marlins-beat-braves/

The International Football Association Board (2017), 'Guidelines for return substitutes', Approved at the 131st Annual General Meeting, 3 March; Laws of the Game 2026/27, online edition.
https://www.theifab.com/laws/latest/guidelines-for-return-substitutes/

The International Football Association Board (2020), 'Five substitutes option temporarily allowed for competition organisers', News release, 8 May.
https://www.theifab.com/news/five-substitutes-option-temporarily-allowed-for-competition-organisers/

The International Football Association Board (2022), 'The IFAB permanently approves five-substitute option in top-level competitions', News release, 13 June; Laws of the Game 2022/23, Law 3.2.
https://www.theifab.com/news/the-ifab-permanently-approves-five-substitute-option-in-top-level-competitions/

The International Football Association Board (2024), 'Additional permanent concussion substitutions protocol', Laws of the Game, in force from 1 July 2024; approved at the 138th Annual General Meeting, 2 March 2024.
https://www.theifab.com/laws/latest/additional-permanent-concussion-substitutions-protocol/

The International Football Association Board (2026*a*), 'Additional permanent concussion substitutions protocol', Laws of the Game 2026/27, Notes and modifications, read 16 October 2026.
https://theifab.com/laws/latest/additional-permanent-concussion-substitutions-protocol

The International Football Association Board (2026*b*), 'The IFAB introduces further measures to improve match flow and player behaviour', News release, 28 February; in force from 1 July 2026.
https://www.theifab.com/news/the-ifab-introduces-further-measures-to-improve-match-flow-and-player-behaviour/

Turan, Ç., Harbili, E., Çetin, O. & Tokul, E. (2026), 'Impact of the revised stoppage-time regulation on the temporal and locomotor demands of pro-

fessional football', *International Journal of Sports Science & Coaching* (online first, 23 September).

Volf, P. (2009), 'A random point process model for the score in sport matches', *IMA Journal of Management Mathematics* **20**(2), 121–131.

Wilson, P. (2015), 'The misuse of the Vuong test for non-nested models to test for zero-inflation', *Economics Letters* **127**, 51–53.

Winkelmann, R. (1995), 'Duration dependence and dispersion in count-data models', *Journal of Business & Economic Statistics* **13**(4), 467–474.

World Rugby (2020), 'Rugby world cup 2019: statistical analysis', World Rugby Game Analysis.
`https://www.world.rugby/news/691225/rugby-world-cup-2019-statistical-analysis`

World Rugby (2023), 'Laws of the game: rugby union', World Rugby, Dublin. Law 3, team: 3.19 and 3.20, a front-row player suspended or sent off; 3.33, tactically replaced players rejoining.

World Rugby (2026), '2026-07 law updates', World Rugby Passport, Laws of the Game, What's new; read 15 October 2026.
`https://passport.world.rugby/laws-of-the-game/whats-new/2026-07-law-updates/`

Xiao, Z. & Zhang, H. (2024), 'More substitutions changed team substitution strategy? An analysis of the FIFA World Cup 2002–2022', *BMC Sports Science, Medicine and Rehabilitation* **16**, 165.

Zhang, H., Mathes, J., Garnica Caparrós, M. & Memmert, D. (2025), 'Assessing the impact of substitution rule changes in European football leagues', *International Journal of Sports Science & Coaching* **20**(3), 1107–1124.